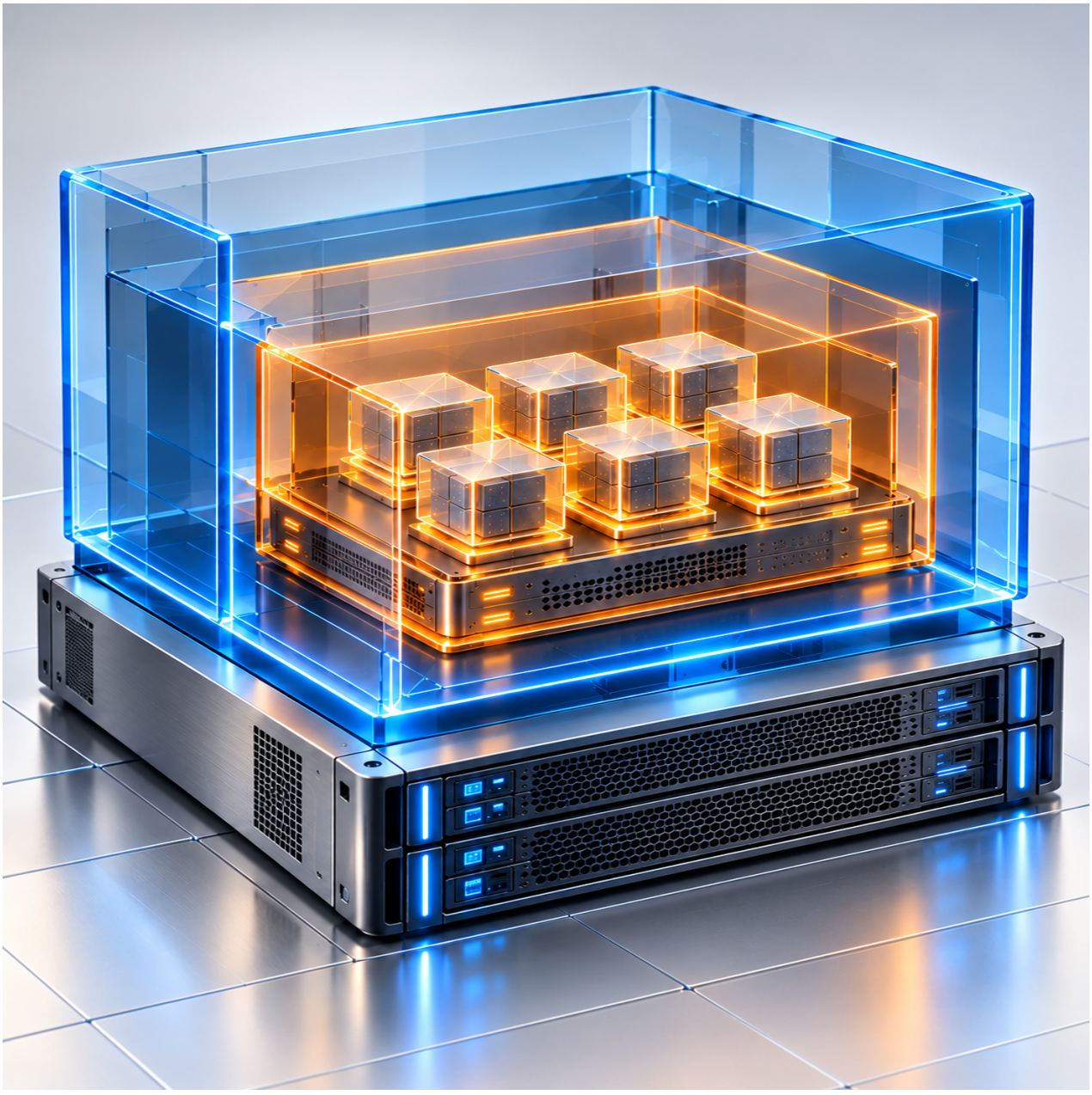
Kiformatou

DOCUMENTATION TECHNIQUE

Mise en place d’une infrastructure virtualisée

Procédures



Version : Windows 11 • **Date** : 29/10/2026 • **Auteur(s)** : Nathan Yang

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Présentation du projet

Cette documentation vise à accompagner le personnel informatique à mettre en place l'infrastructure de l'école Kiformatou. Le projet est limité par une configuration de 32 Go de RAM et 475 Go de stockage.

Architecture

L'infrastructure respectera l'architecture tant au niveau du plan d'adressage IP et organisation des appareils :

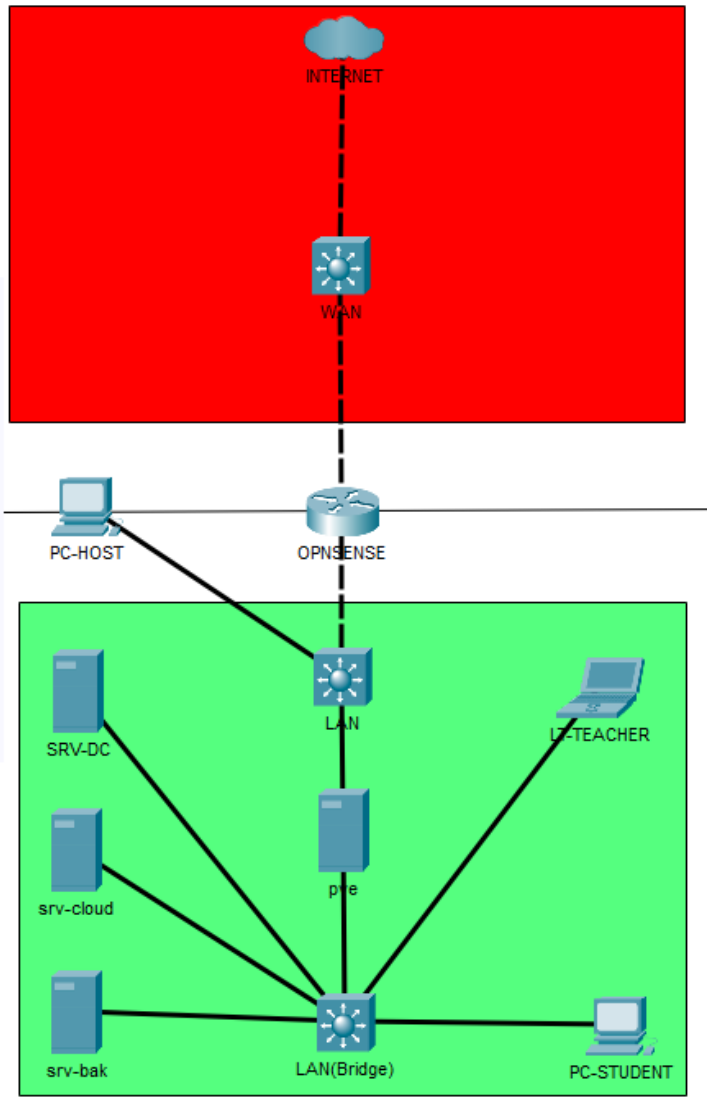
```
# LAN KIFORMATOU

## GENERAL
- @opnsense(WAN)     = DHCP
- @opnsense(LAN)     = 192.168.0.1/24
- @pve               = 192.168.0.2/24
- @PC-HOST           = 192.168.0.100/24

## VLAN 10 (SERVERS)
- hosts = 4
- @OPNSENSE(LAN)     = 192.168.10.1/29
- @SRV-DC            = 192.168.10.2/29
- @srv-cloud         = 192.168.10.3/29
- @srv-bak           = 192.168.10.4/29

## VLAN 20 (TEACHERS)
- hosts = 4
- @opnsense(LAN.20)  = 192.168.20.1/29
- @LT-TEACHER        = DHCP(192.168.20.2/29 - 192.168.20.5/29)

## VLAN 30 (STUDENTS)
- hosts = 96
- @pnsense(LAN.30)   = 192.168.30.1/25
- @PC-STUDENT        = DHCP(192.168.30.2/25 - 192.168.30.97/25)
```



Préparation de Hyper-V

Nous allons utiliser Hyper-V, un hyperviseur de type 1 pour simuler l'infrastructure de Kiformatou. Nous allons l'installer et le configurer afin de s'assurer que les VM puissent communiquer entre eux.

1. Activer Hyper-V

Sur Windows 10/11 Pro ou Entreprise, ouvrez PowerShell en administrateur, puis exécutez cette commande pour activer l'ensemble Hyper-V, y compris les outils de gestion et le module PowerShell :

```
Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V-All -All -NoRestart
```

Après l'installation, redémarrez :

```
Restart-Computer
```

Après le redémarrage, vérifiez les composants :

```
Get-WindowsOptionalFeature -Online |
    Where-Object FeatureName -Like "*Hyper-V*" |
    Format-Table FeatureName, State
```

Assurez-vous d'avoir l'affichage suivant où toutes les fonctionalités sont à **Enabled**:

```
FeatureName                                 State
-----------                                 -----
Microsoft-Hyper-V-All                     Enabled
Microsoft-Hyper-V                         Enabled
Microsoft-Hyper-V-Tools-All               Enabled
Microsoft-Hyper-V-Management-PowerShell   Enabled
Microsoft-Hyper-V-Hypervisor              Enabled
Microsoft-Hyper-V-Services                Enabled
Microsoft-Hyper-V-Management-Clients      Enabled
```

3. Créer les vNICS

Nous voulons créer un switch interne pour le LAN Kiformatou et un switch externe pour l'accès Internet. Exécutez les commandes suivantes pour les créer :

S'il n'y a pas de doublons, exécutez ces commandes pour créer les vSwitches de l'infrastructure sur l'hôte :

```
New-VMSwitch `
    -Name "KIFORMATOU" `
    -SwitchType Internal `


Rename-VMSwitch -Name "KIFORMATOU" -NewName "KIFORMATOU"


New-VMSwitch `
    -Name "INTERNET" `
    -NetAdapterName "Ethernet" `
    -AllowManagementOS $true
```

Vérifez ensuite :

```
Get-VMSwitch
```

Vous devriez obtenir une affichage similaire :

```
Name           SwitchType NetAdapterInterfaceDescription
----           ---------- ------------------------------
Default Switch Internal
KIFORMATOU     Internal
INTERNET       External   Realtek Gaming 2.5GbE Family Controller
```

(WIP : Vous pouvez également copier le script du dépôt de travail pour cette section)

Création de la VM OPNsense (REVOIR)

La VM OPNsense fera office de routeur/pare-feu pour relier le LAN Kiformatou à Internet, la création se fera en PowerShell. N'oubliez pas de prendre l'ISO de OPNsense v26.7 à cette adresse https://opnsense.org/download/

1. Créer la VM

Vous pouvez exécuter le script suivant pour créer la VM en adaptant le chemin absolu :

```
# Configuration
$OPNsense = [PSCustomObject]@{
    Name          = "opnsense"
    Generation    = 2
    CPUCount      = 2
    VRAMBytes     = 4GB
    Path          = "V:\Kiformatou\opnsense"
    VHDPath       = "V:\Kiformatou\opnsense\opnsense.vhdx"
    VHDSizeBytes  = 16GB
    SwitchNameWAN = "INTERNET"
    SwitchNameLAN = "KIFORMATOU"
    ISOPath       = "V:\ISO\opnsense-26.7.iso"
}

# Create the VM directory
New-Item -ItemType Directory -Path $OPNsense.Path -Force | Out-Null

# Create a Generation 2 VM with 4 GB RAM and a 64 GB virtual disk
New-VM -Name $OPNsense.Name `
    -Generation $OPNsense.Generation `
    -MemoryStartupBytes $OPNsense.VRAMBytes `
    -Path $OPNsense.Path `
    -NewVHDPath $OPNsense.VHDPath `
    -NewVHDSizeBytes $OPNsense.VHDSizeBytes `
    -SwitchName $OPNsense.SwitchNameWAN

# Assign 2 virtual processors and disable secure boot
Set-VMProcessor -VMName $OPNsense.Name -Count $OPNsense.CPUCount
```

```
Set-VMFirmware -VMName $OPNsense.Name -EnableSecureBoot Off


# Attach the installation ISO and boot from it
$dvdDrive = Add-VMDvdDrive -VMName $OPNsense.Name -
Path $OPNsense.ISOPath -Passthru
Set-VMFirmware -VMName $OPNsense.Name -FirstBootDevice $dvdDrive


# Disable dynamic memory
Set-VMMemory -VMName $OPNsense.Name -DynamicMemoryEnabled $false -
StartupBytes '@OPNsense'.VRAMBytes


# Add a second vNIC for LAN
Add-VMNetworkAdapter -VMName $OPNsense.Name `
    -Name $OPNsense.SwitchNameLAN `
    -SwitchName $OPNsense.SwitchNameLAN


# Active Trunk for VLANs 10, 20 and 30
Set-VMNetworkAdapterVlan -VMName $OPNsense.Name `
  -VMNetworkAdapterName $OPNsense.SwitchNameLAN `
  -Trunk -AllowedVlanIdList "10,20,30" -NativeVlanId 0



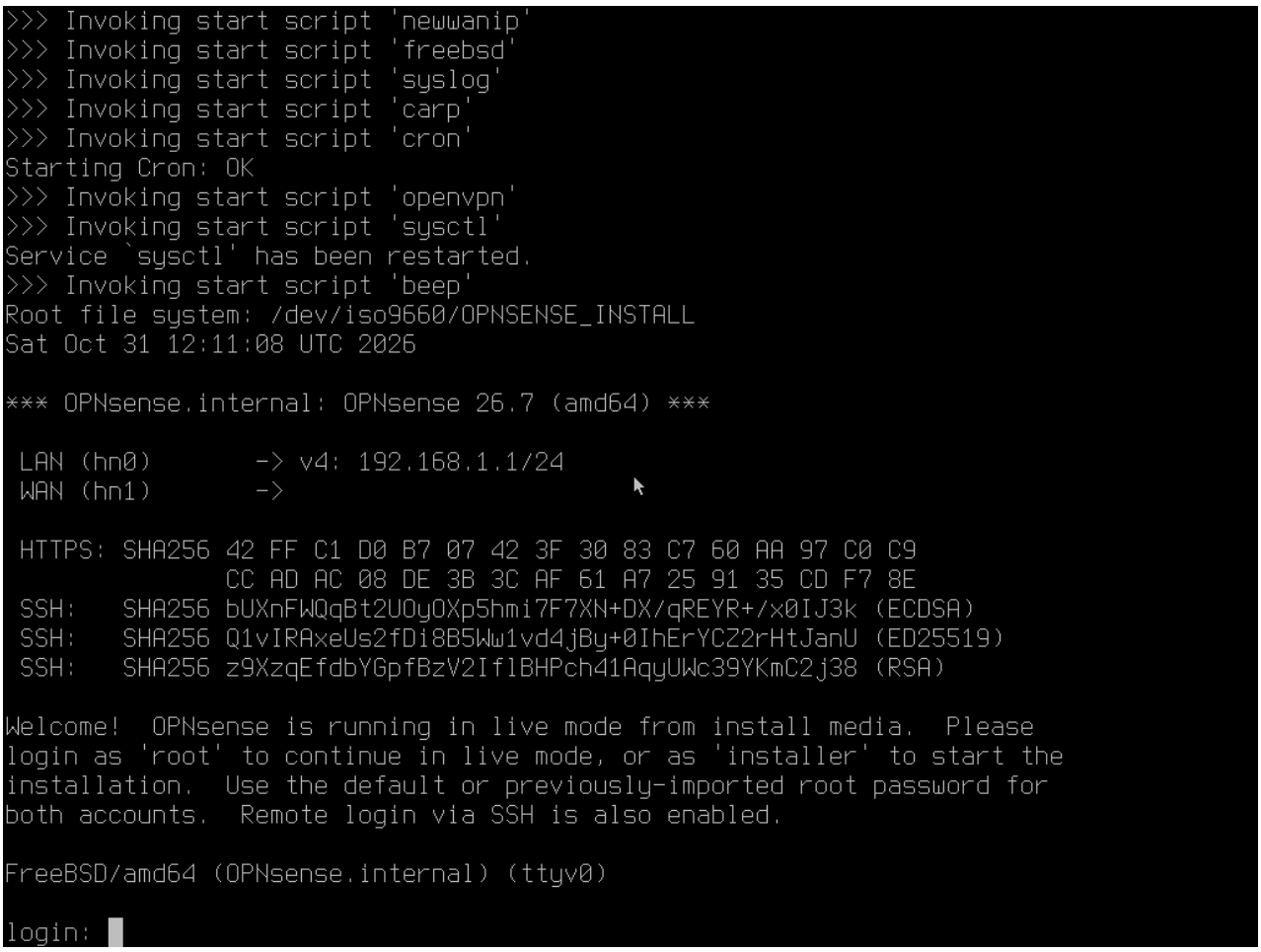
# Display the VM status
Get-VM -Name $OPNsense.Name
Get-VMNetworkAdapterVlan -VMName $OPNsense.Name
```

2. Installer la VM

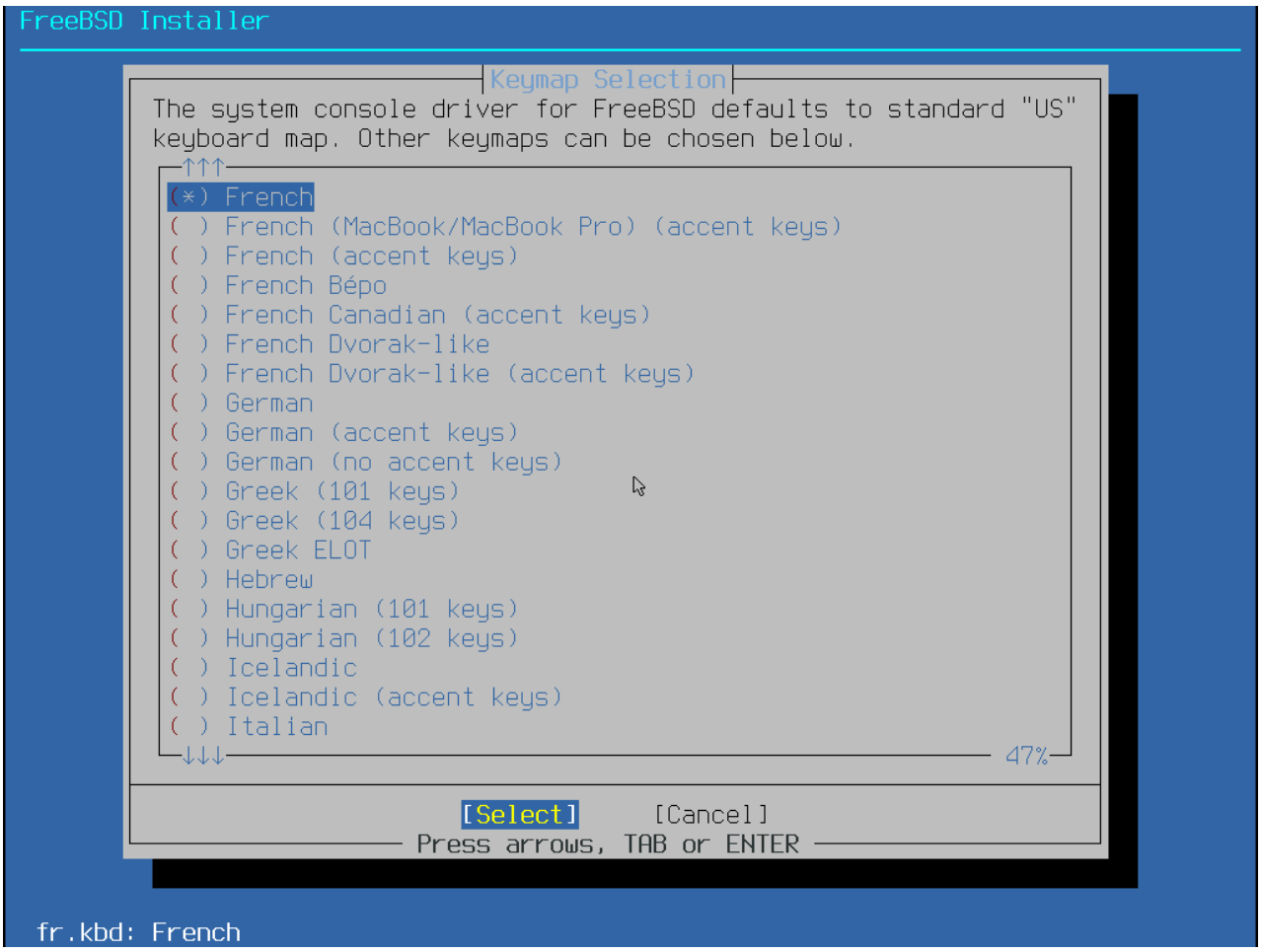
Démarrez la VM avec :

```
Get-VM
Start-VM -Name "opnsense"
```

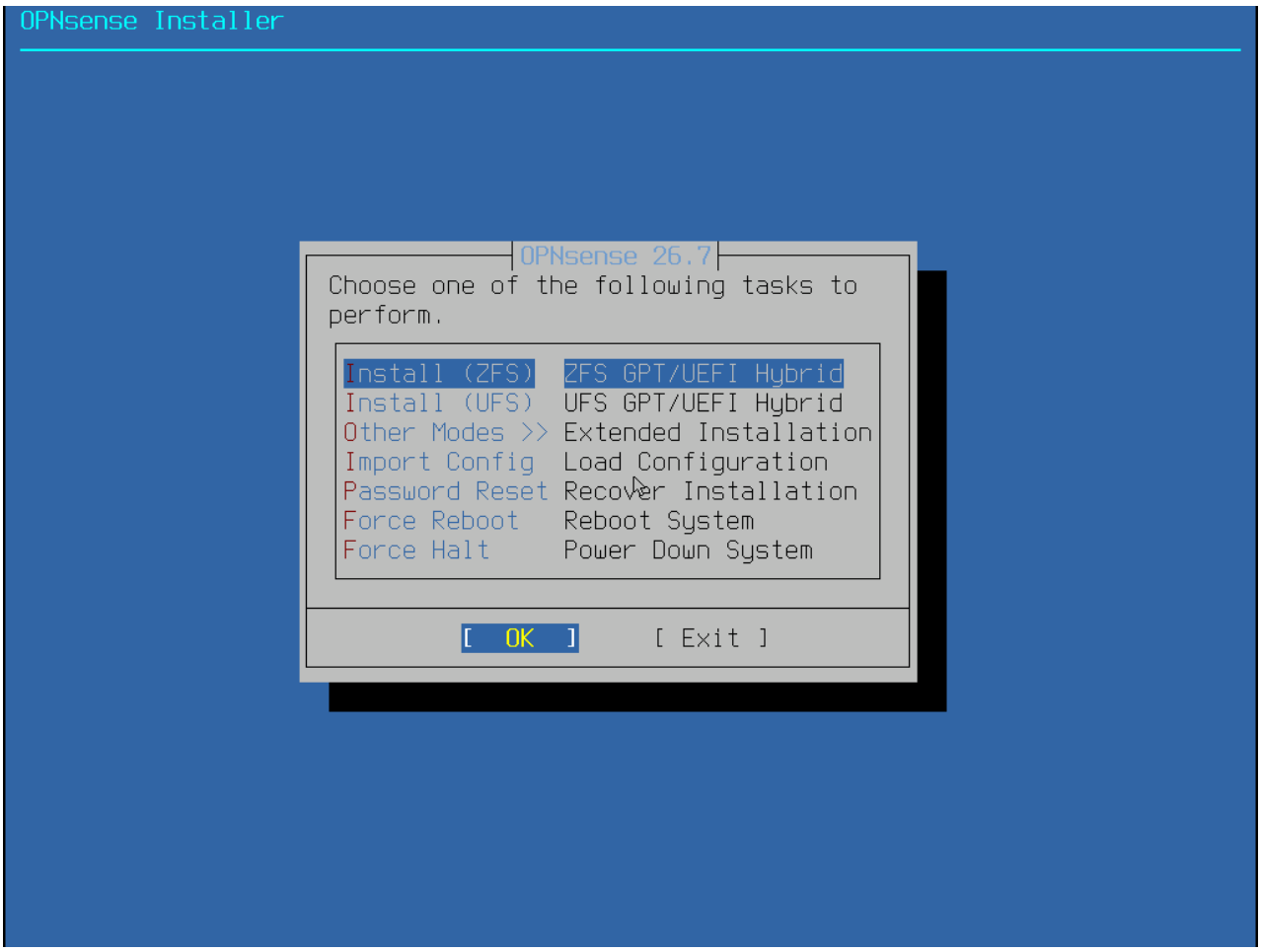
Puis attendez jusqu'à avoir cette affichage :



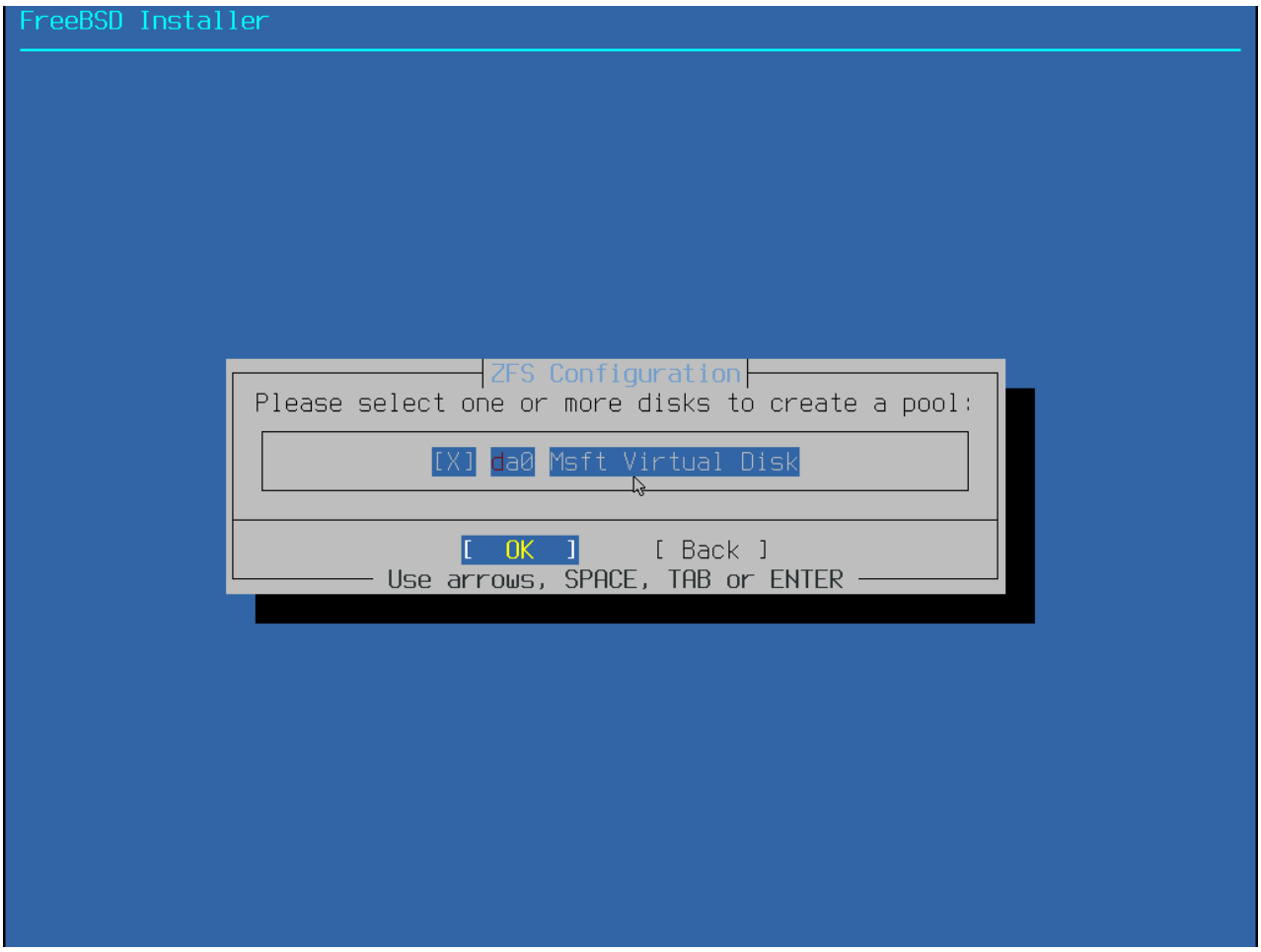
Rentrez l'identifiant **installer** et le mot de passe **opnsense** (ATTENTION : le clavier est en qwerty).



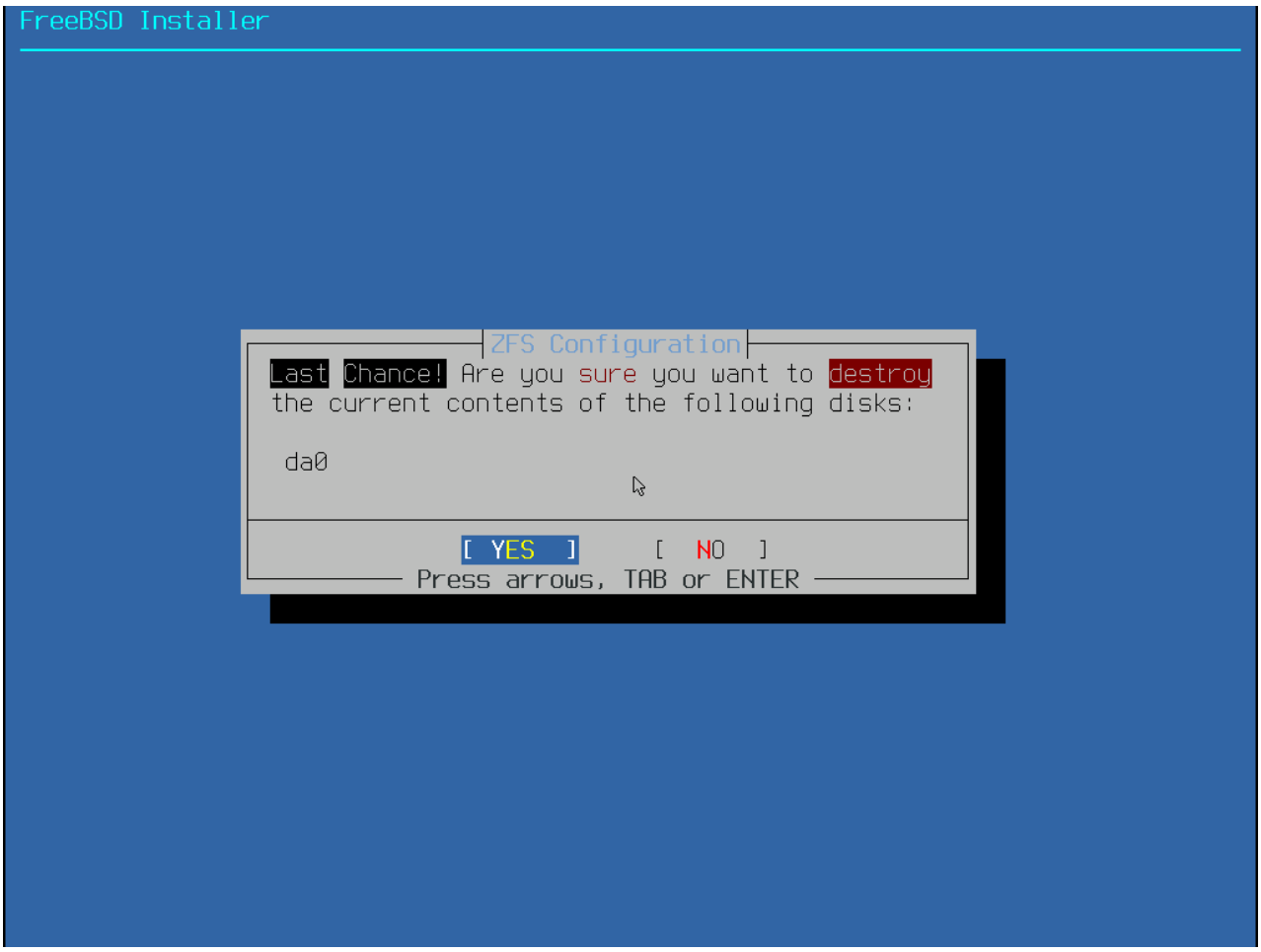
Choisissez **French**



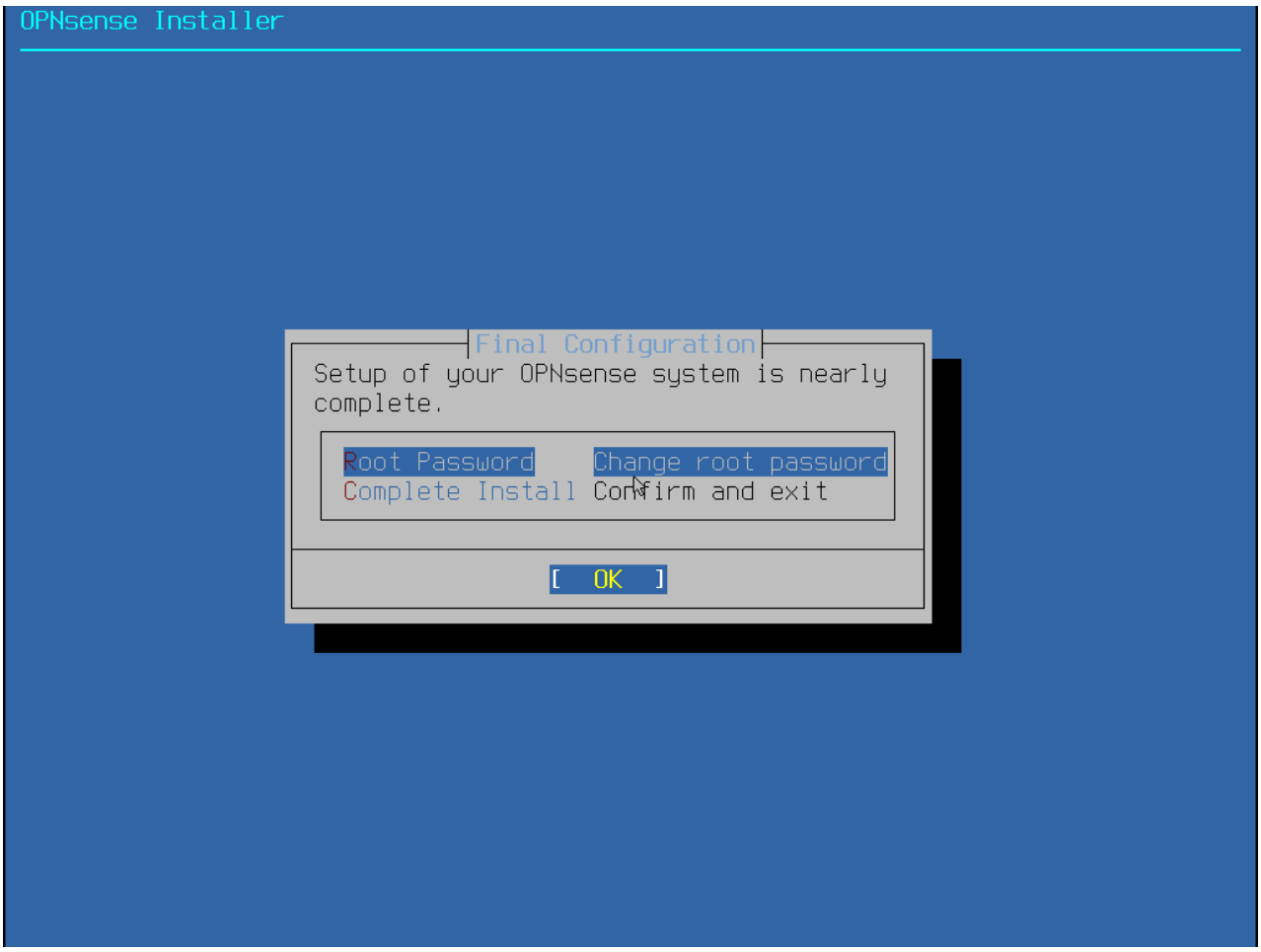
Continuez avec l'option par défaut **Install (ZFS)**

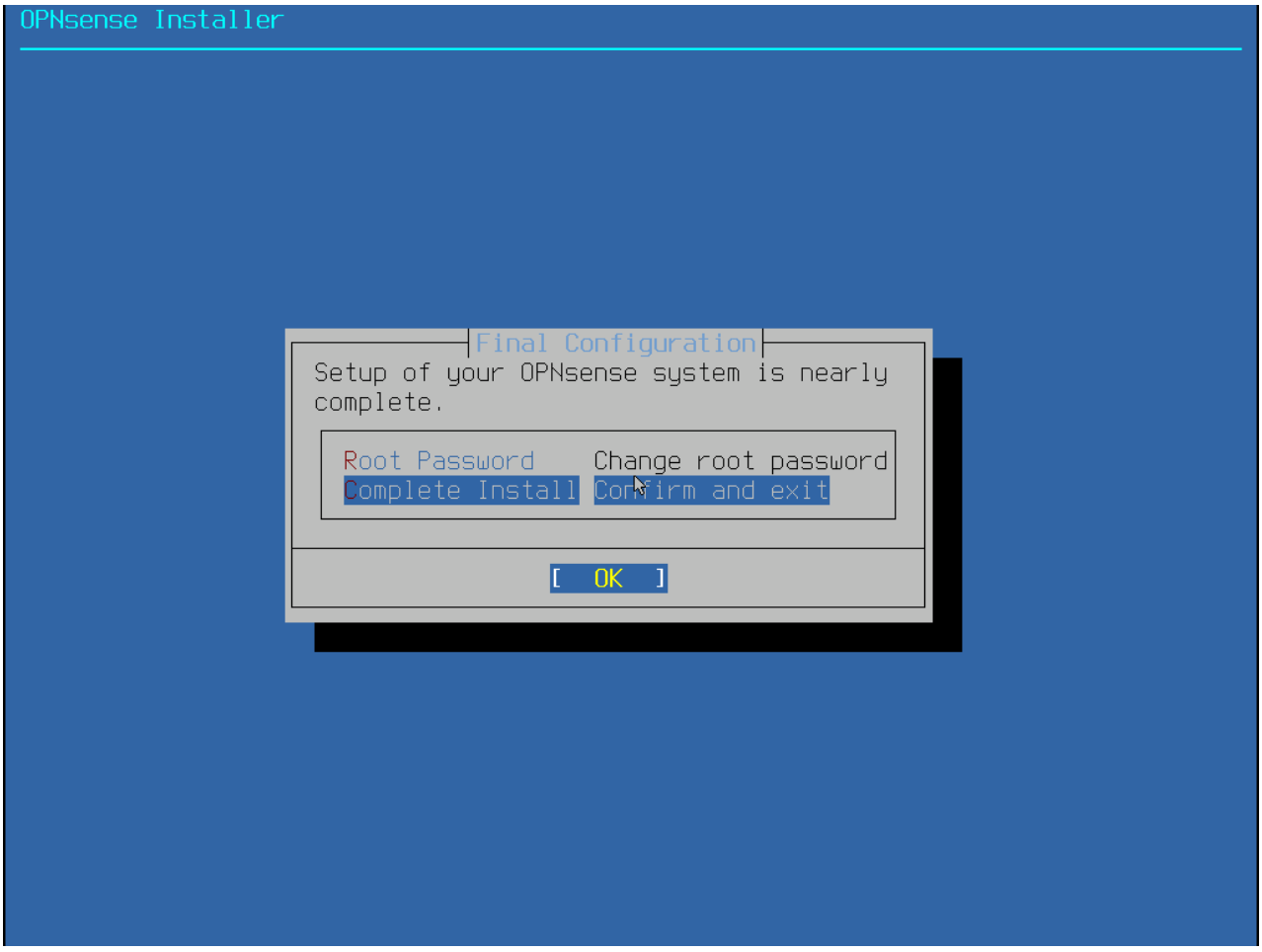


Sélectionnez l'unique disque et continuez

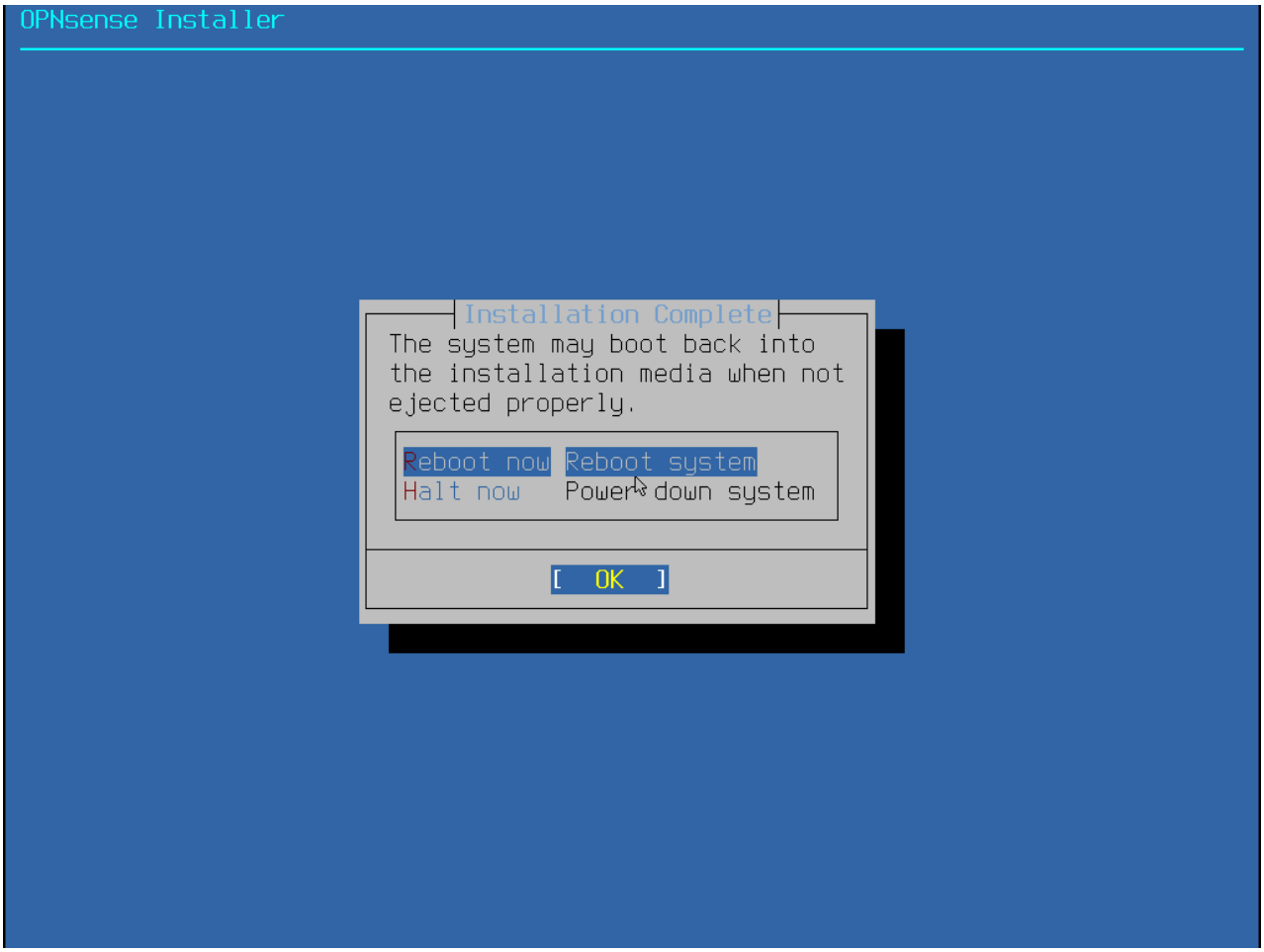

FreeBSD Installer
ZFS Configuration
Last Chance! Are you sure you want to destroy
the current contents of the following disks:
da0
[YES]
[NO]
Press arrows, TAB or ENTER

Confirmez


OPNsense Installer
Final Configuration
Setup of your OPNsense system is nearly complete.
Root Password Change root password
Complete Install Confirm and exit
[OK]


OPNsense Installer
Final Configuration
Setup of your OPNsense system is nearly complete.
Root Password Change root password
Complete Install Confirm and exit
[OK]

Changez le mot de passe du root et terminez l'installation



Redémarrez

3. Configurer la VM

Connectez-vous avec vos identifiants

3.1. Configurer les interfaces réseau

```
FreeBSD/amd64 (OPNsense.internal) (ttyv0)

login: root
Password:
-------------------------------------------------
|       Hello, this is OPNsense 26.7            |
|                                               |
|  Website:     https://opnsense.org/           |
|  Handbook:    https://docs.opnsense.org/      |
|  Forums:      https://forum.opnsense.org/     |
|  Code:        https://github.com/opnsense     |
|  Reddit:      https://reddit.com/r/opnsense   |
-------------------------------------------------

*** OPNsense.internal: OPNsense 26.7 (amd64) ***

 LAN (hn0)       -> v4: 192.168.1.1/24
 WAN (hn1)       ->

 HTTPS: SHA256 71 BC C5 00 CA 87 7A 5A 1C 81 CD CA 26 80 10 27
               43 8C 40 48 C9 97 58 1B 99 7A D9 45 D6 6A 70 CB

  0) Logout                              7) Ping host
  1) Assign interfaces                   8) Shell
  2) Set interface IP address            9) pfTop
  3) Reset the root password            10) Firewall log
  4) Reset to factory defaults          11) Reload all services
  5) Power off system                   12) Update from console
  6) Reboot system                      13) Restore a backup

Enter an option:
```

Tapez **1**

Tapez **N** à l'option "Do you want to configure LAGGS now ?"

Tapez **N** à l'option "Do you want to configure VLANs now ?"

Vous devrez choisir les interfaces pour le WAN et le LAN, c'est nos vSwitches nommés INTERNET et KIFORMATOU. Exécutez la commande suivante dans PowerShell Administrateur pour les identifier :

```
Get-VMNetworkAdapter -VMName "opnsense" |

    Select-Object VMName, Name, MacAddress
```

Vous devrez tomber sur l'affichage suivant (à adapter selon vos adresses MAC) :

```
VMName   Name            MacAddress

------   ----            ----------

opnsense Network Adapter 00155D00B72F

opnsense KIFORMATOU      00155D00B730
```

Tapez ensuite l'interface qui concerne chaque adresse MAC puis confirmez :

```
  5) Power off system                  12) Update from console
  6) Reboot system                     13) Restore a backup

Enter an option: 1

Do you want to configure LAGGs now? [y/N]: N
Do you want to configure VLANs now? [y/N]: N

Valid interfaces are:

hn0             00:15:5d:00:b7:2f Hyper-V Network Interface
hn1             00:15:5d:00:b7:30 Hyper-V Network Interface

If you do not know the names of your interfaces, you may choose to use
auto-detection. In that case, disconnect all interfaces now before
hitting 'a' to initiate auto detection.

Enter the WAN interface name or 'a' for auto-detection: hn0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(or nothing if finished): hn1

Enter the Optional interface 1 name or 'a' for auto-detection
(or nothing if finished):

The interfaces will be assigned as follows:

WAN  -> hn0
LAN  -> hn1

Do you want to proceed? [y/N]:
```

Nous allons ensuite modifier la configuration IP de chaque interface sachant qu'on veut du DHCP sur notre WAN et du statique sur le LAN. Tapez 2 puis configurez d'abord le LAN :

```
Available interfaces:

1 - LAN (hn1 - static, idassoc6)
2 - WAN (hn0 - dhcp, dhcp6)

Enter the number of the interface to configure: 1

Configure IPv4 address LAN interface via DHCP? [y/N] n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.0.1


Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via WAN tracking? [Y/n] n
Configure IPv6 address LAN interface via DHCP6? [y/N] n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? [y/N] n
```

Configurez ensuite la vNIC “INTERNET” en DHCP :

```
Enter an option: 2

Available interfaces:

1 - LAN (hn1 - static)
2 - WAN (hn0 - dhcp, dhcp6)

Enter the number of the interface to configure: 2

Configure IPv4 address WAN interface via DHCP? [Y/n] y

Configure IPv6 address WAN interface via DHCP6? [Y/n] n

Enter the new WAN IPv6 address. Press <ENTER> for none:
>

Do you want to change the web GUI protocol from HTTPS to HTTP? [y/N] n
Do you want to generate a new self-signed web GUI certificate? [y/N] n
Restore web GUI access defaults? [y/N] n

Writing configuration...done.
Generating /etc/resolv.conf...done.
Generating /etc/hosts...done.
Configuring WAN interface...done.
Setting up routes for wan...done.
Starting Dnsmasq...done.
Starting Unbound DNS...done.
Configuring firewall.......done.
Starting Dnsmasq...done.
Starting Unbound DNS...done.
```

Nous pouvons ainsi accéder à l'interface web de OPNsense.

3.2. Accèder à l'interface Web

Avant d'accéder à l'interface Web de OPNsense, assurez-vous d'avoir la configuration IP suivante sur la vSwitch **"KIFORMATOU"** pour votre PC hôte :

```
Get-NetIPConfiguration -InterfaceAlias "vEthernet (KIFORMATOU)"


InterfaceAlias       : vEthernet (KIFORMATOU)
InterfaceIndex       : 13
InterfaceDescription : Hyper-V Virtual Ethernet Adapter #2
NetProfile.Name      : Réseau 5
IPv4Address          : 192.168.0.100
IPv6DefaultGateway   :
IPv4DefaultGateway   : 192.168.0.1
DNSServer            : 1.1.1.1
                       8.8.8.8
```

Si ce n'est pas le cas, exécutez les commandes suivantes avec **PowerShell Administrateur**:

```
$interfaceAlias = "Ethernet"

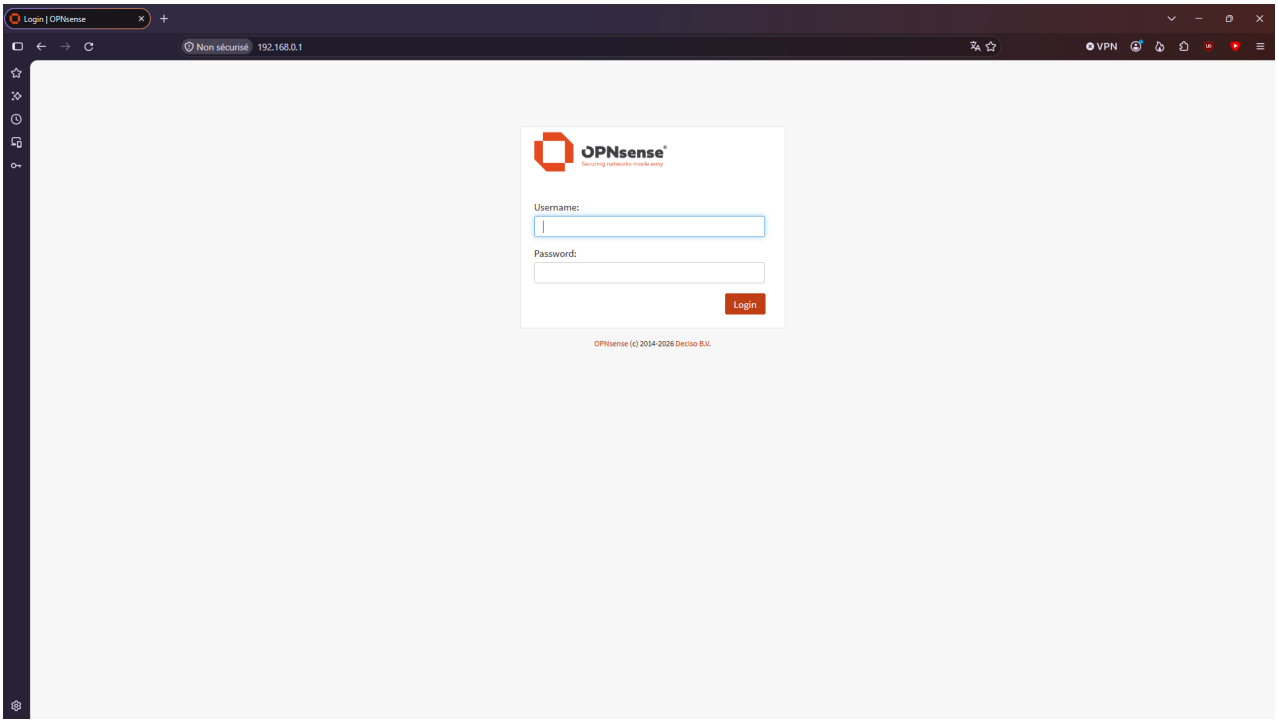

# Disable DHCP
Set-NetIPInterface -InterfaceAlias $interfaceAlias -AddressFamily IPv4 -
Dhcp Disabled


# Set the IPv4 address, subnet mask (/24 = 255.255.255.0), and default ga
teway
New-NetIPAddress -InterfaceAlias $interfaceAlias `
    -IPAddress "192.168.0.100" `
    -PrefixLength 24 `
    -DefaultGateway "192.168.0.1"

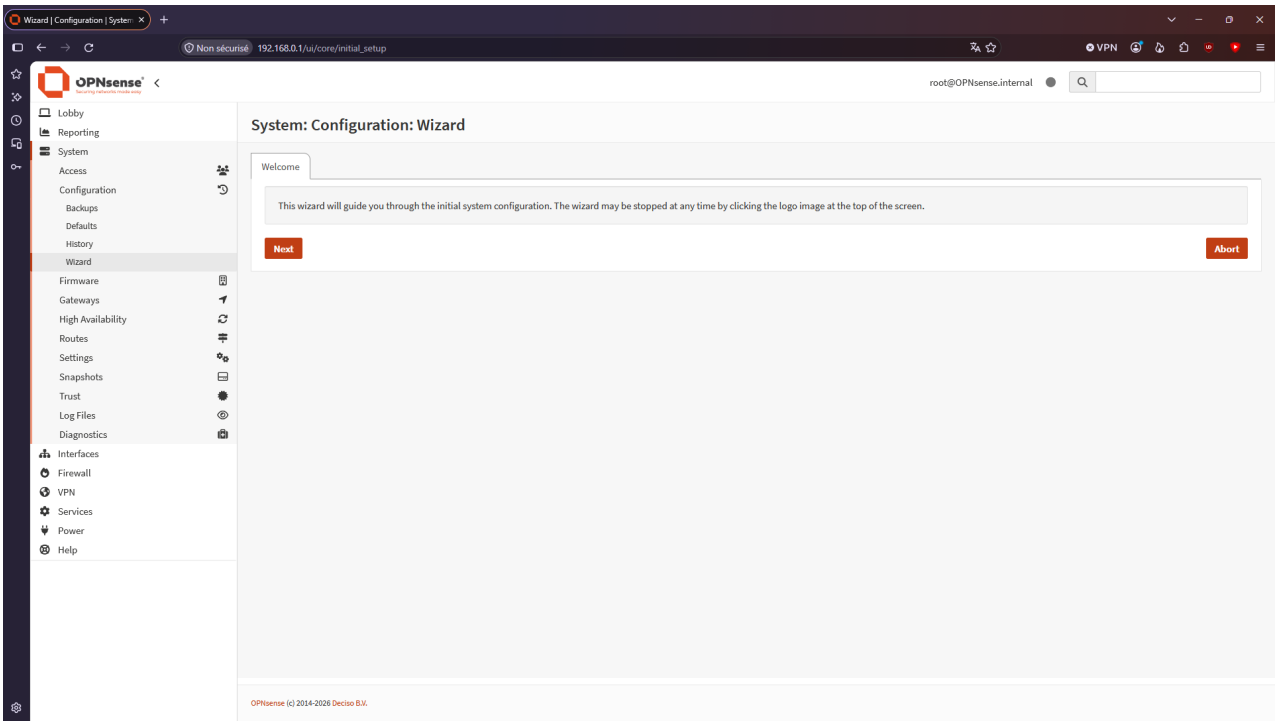

# Set DNS servers
Set-DnsClientServerAddress -InterfaceAlias $interfaceAlias `
    -ServerAddresses ("1.1.1.1", "8.8.8.8")

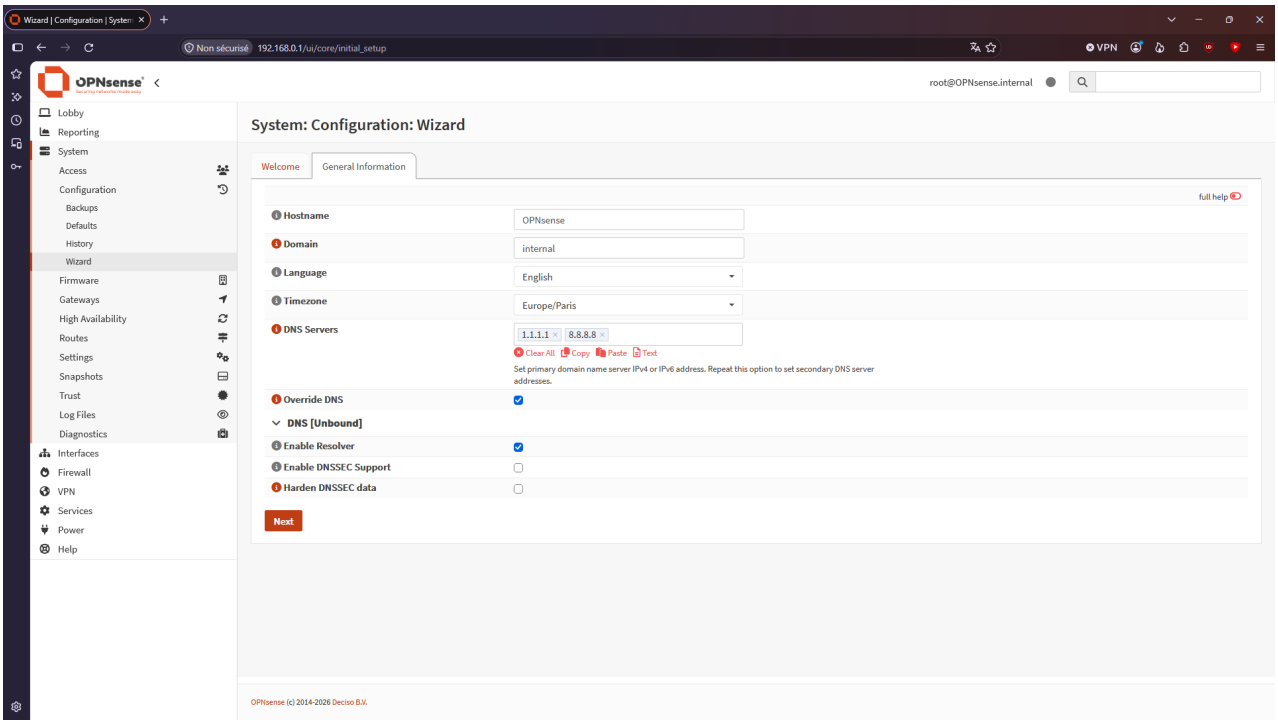

# Verify the network configuration
Get-NetIPConfiguration -InterfaceAlias $interfaceAlias
```

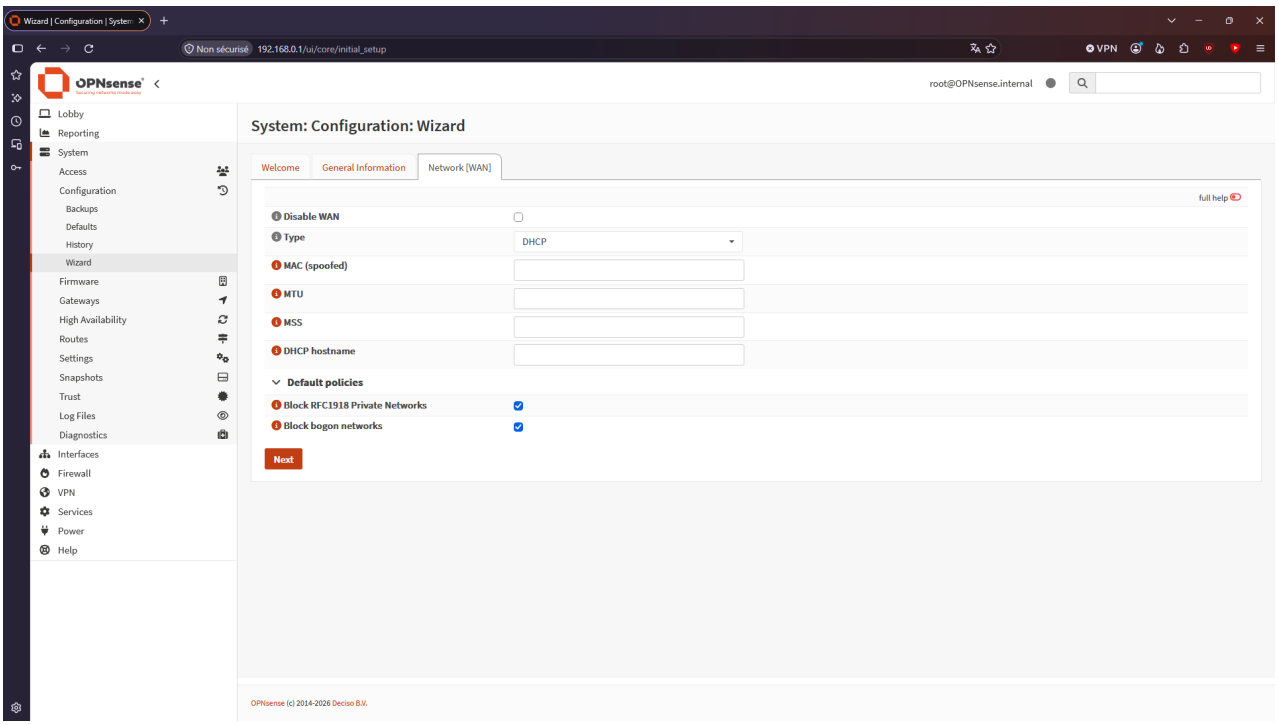
Connectez-vous ensuite avec le navigateur web de votre choix à l'adresse https://192.168.0.1. Vous devrez tomber sur la fenêtre suivante :

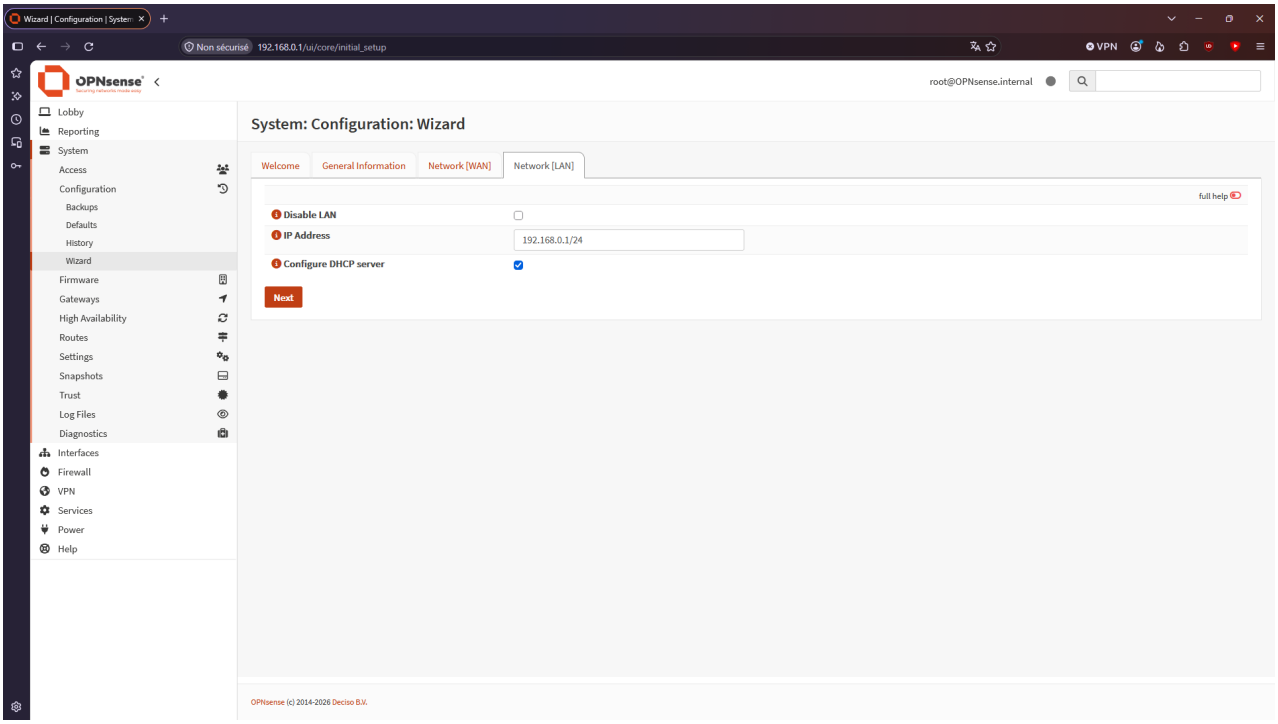


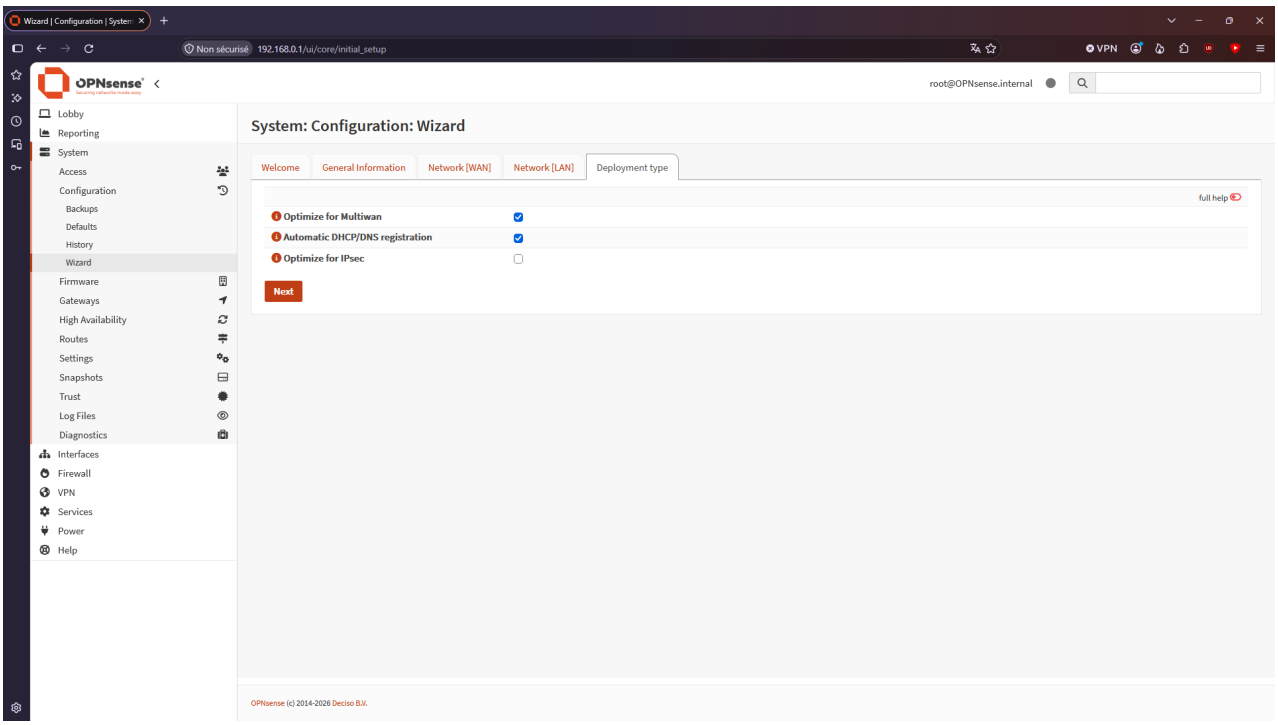
Rentrez vos identifiants pour le root puis suivez la configuration initale à l'arrivée :

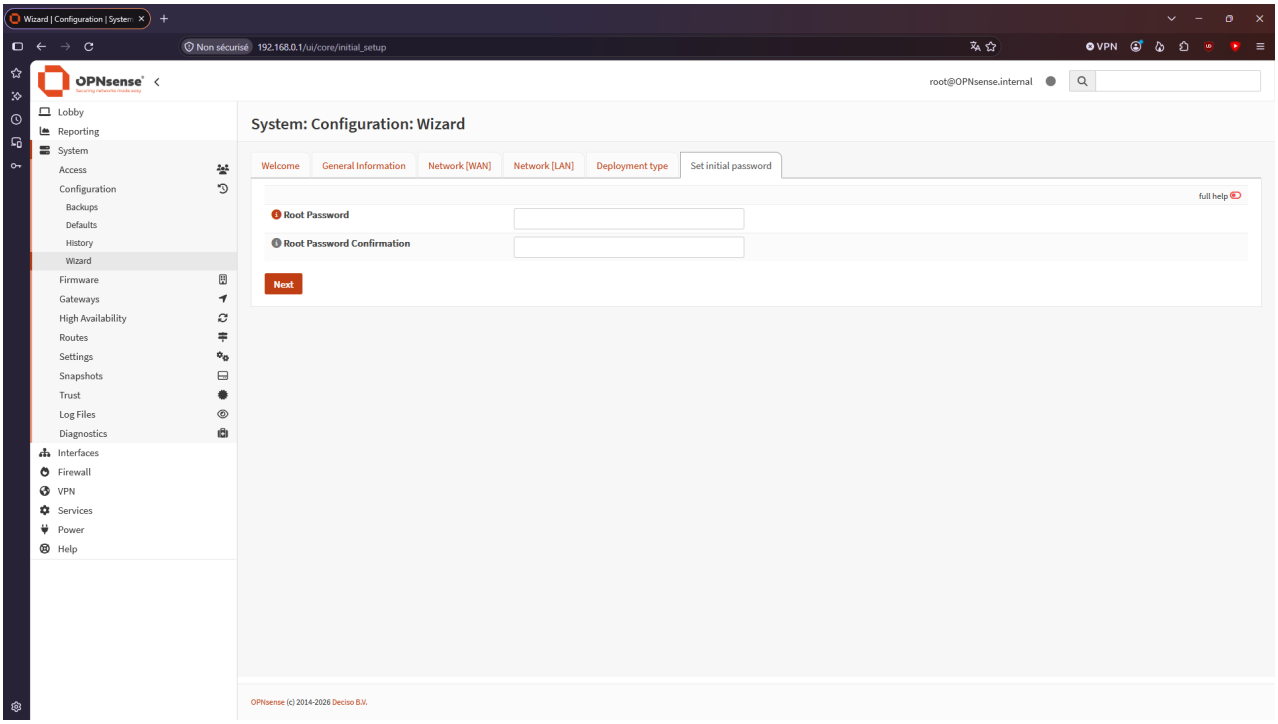


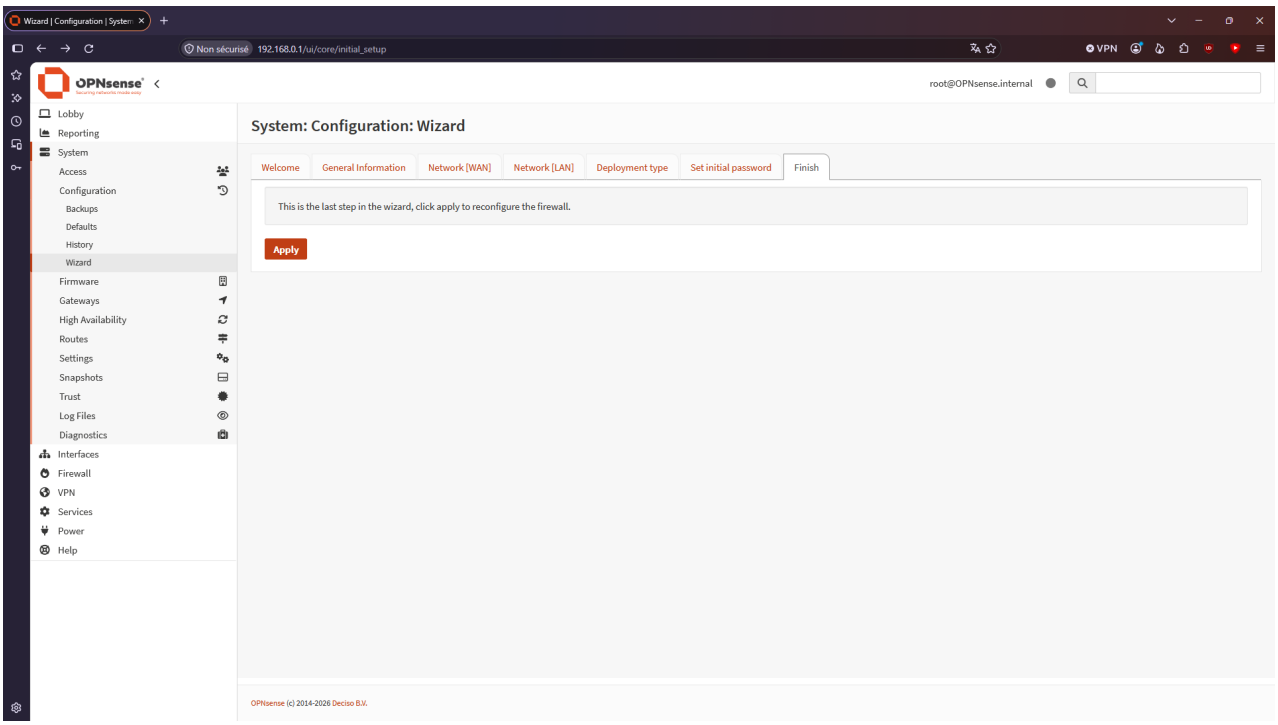


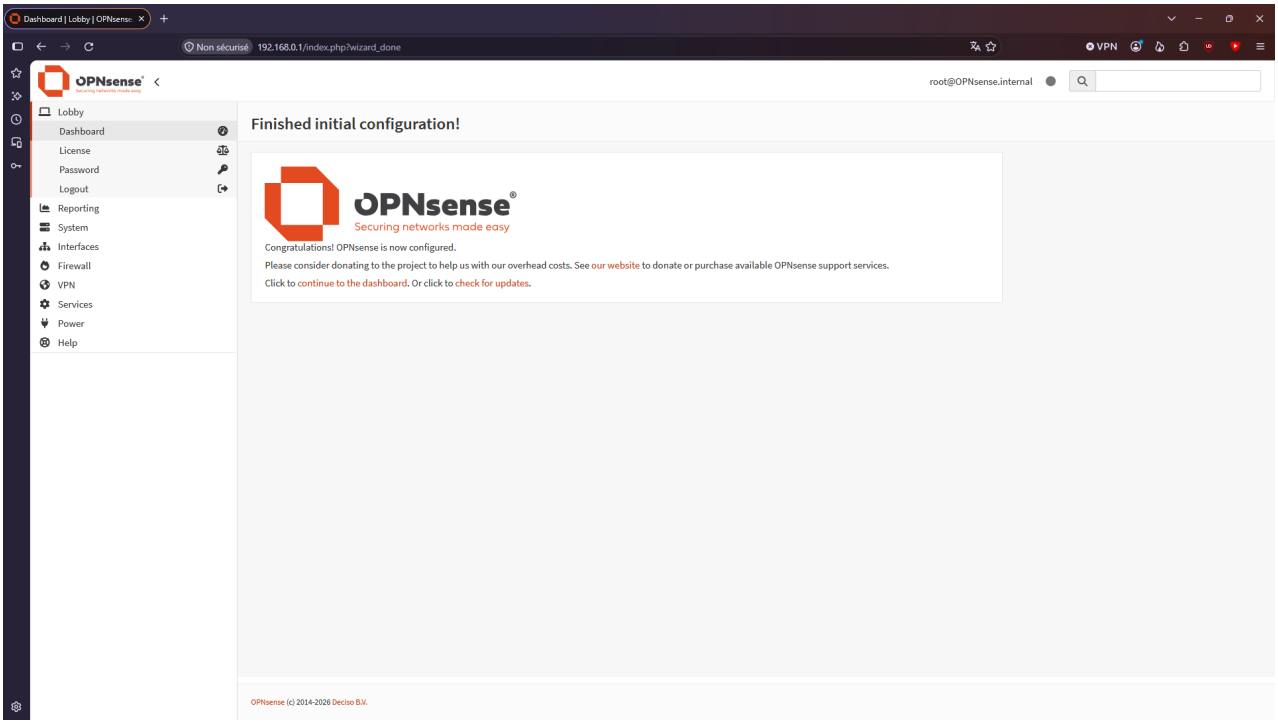

Wizard | Configuration | System
Non sécurisé 192.168.0.1/ui/core/initial_setup
VPN
OPNsense
root@OPNsense.internal
Lobby
Reporting
System
Access
Configuration
Backups
Defaults
History
Wizard
Firmware
Gateways
High Availability
Routes
Settings
Snapshots
Trust
Log Files
Diagnostics
Interfaces
Firewall
VPN
Services
Power
Help
System: Configuration: Wizard
Welcome
General Information
Network [WAN]
full help
Disable WAN
Type
DHCP
MAC (spoofed)
MTU
MSS
DHCP hostname
Default policies
Block RFC1918 Private Networks
Block bogon networks
Next
OPNsense (c) 2014-2026 Deciso B.V.


Wizard | Configuration | System
Non sécurisé 192.168.0.1/ui/core/initial_setup
VPN
OPNsense
root@OPNsense.internal
Lobby
Reporting
System
Access
Configuration
Backups
Defaults
History
Wizard
Firmware
Gateways
High Availability
Routes
Settings
Snapshots
Trust
Log Files
Diagnostics
Interfaces
Firewall
VPN
Services
Power
Help
System: Configuration: Wizard
Welcome
General Information
Network [WAN]
Network [LAN]
full help
Disable LAN
IP Address
192.168.0.1/24
Configure DHCP server
Next
OPNsense (c) 2014-2026 Deciso B.V.

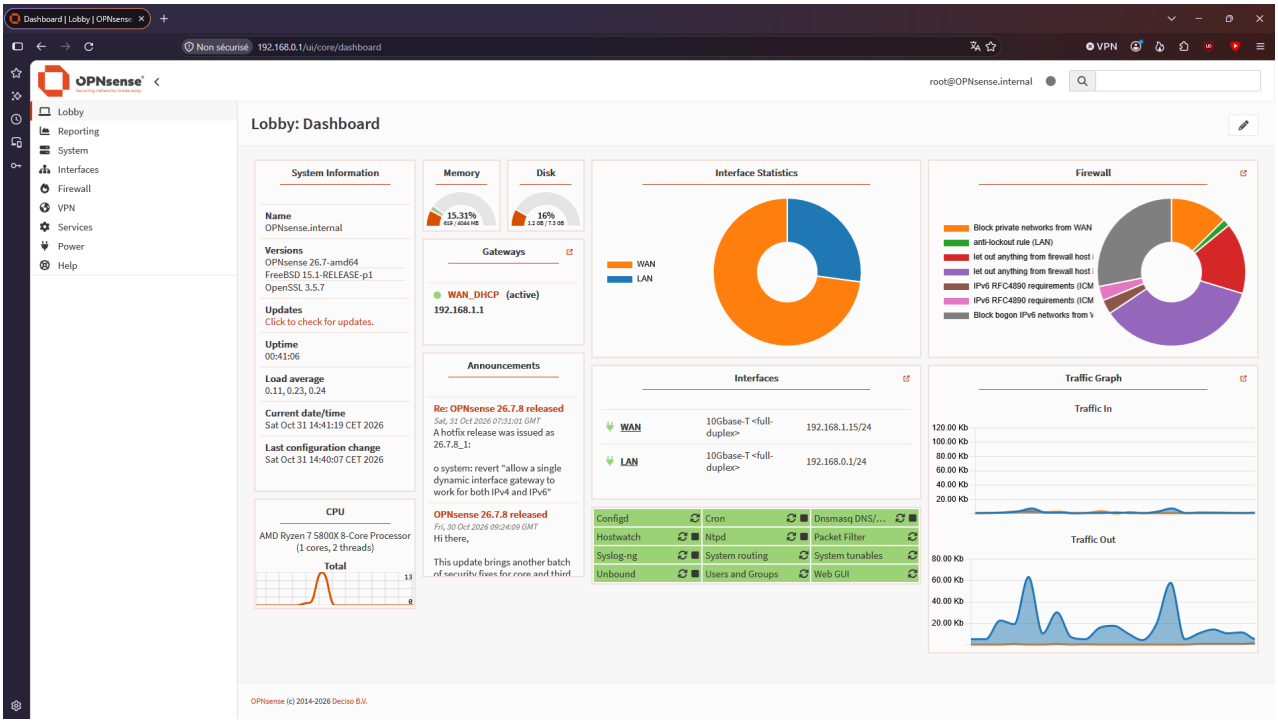

Wizard | Configuration | System
Non sécurisé 192.168.0.1/ui/core/initial_setup
VPN
OPNsense
root@OPNsense.internal
Lobby
Reporting
System
Access
Configuration
Backups
Defaults
History
Wizard
Firmware
Gateways
High Availability
Routes
Settings
Snapshots
Trust
Log Files
Diagnostics
Interfaces
Firewall
VPN
Services
Power
Help
System: Configuration: Wizard
Welcome
General Information
Network [WAN]
Network [LAN]
Deployment type
full help
Optimize for Multiwan
Automatic DHCP/DNS registration
Optimize for IPsec
Next
OPNsense (c) 2014-2026 Deciso B.V.


Wizard | Configuration | System
Non sécurisé 192.168.0.1/ui/core/initial_setup
VPN
OPNsense
root@OPNsense.internal
Lobby
Reporting
System
Access
Configuration
Backups
Defaults
History
Wizard
Firmware
Gateways
High Availability
Routes
Settings
Snapshots
Trust
Log Files
Diagnostics
Interfaces
Firewall
VPN
Services
Power
Help
System: Configuration: Wizard
Welcome
General Information
Network [WAN]
Network [LAN]
Deployment type
Set initial password
full help
Root Password
Root Password Confirmation
Next
OPNsense (c) 2014-2026 Deciso B.V.

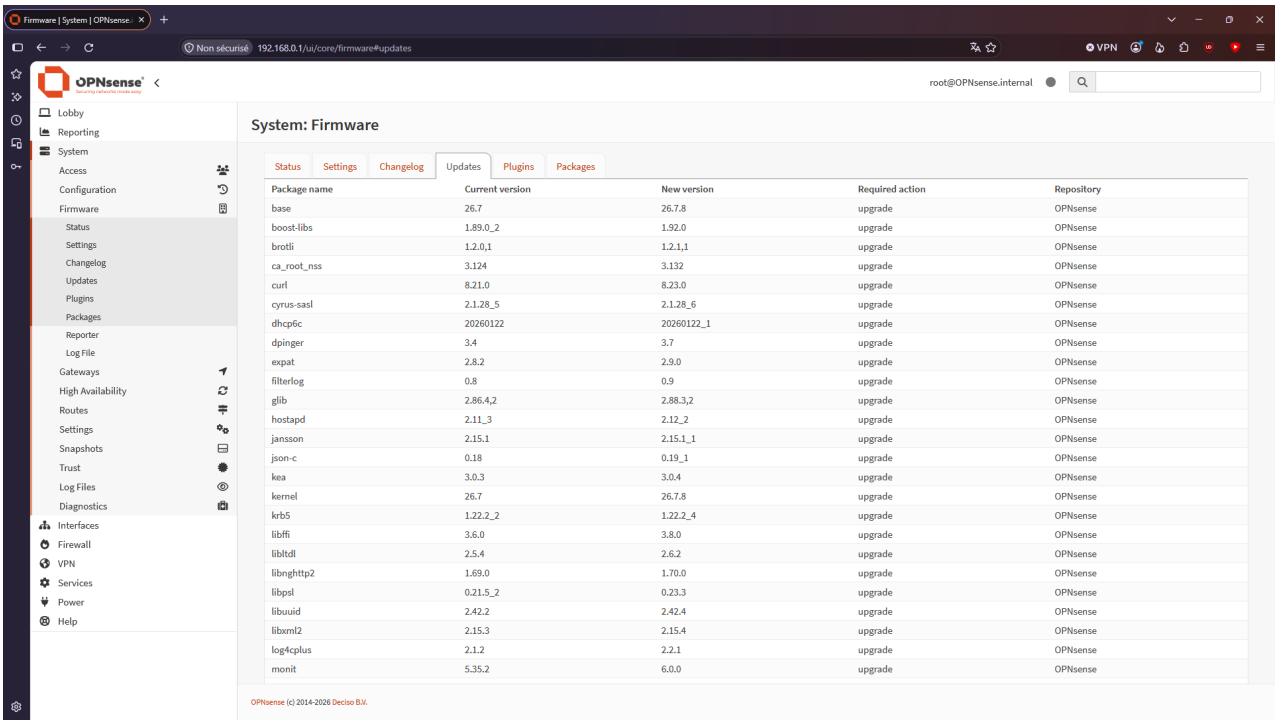




Terminez la configuration initale et revenir dans **Lobby/Dashboard** pour confirmez les informations.



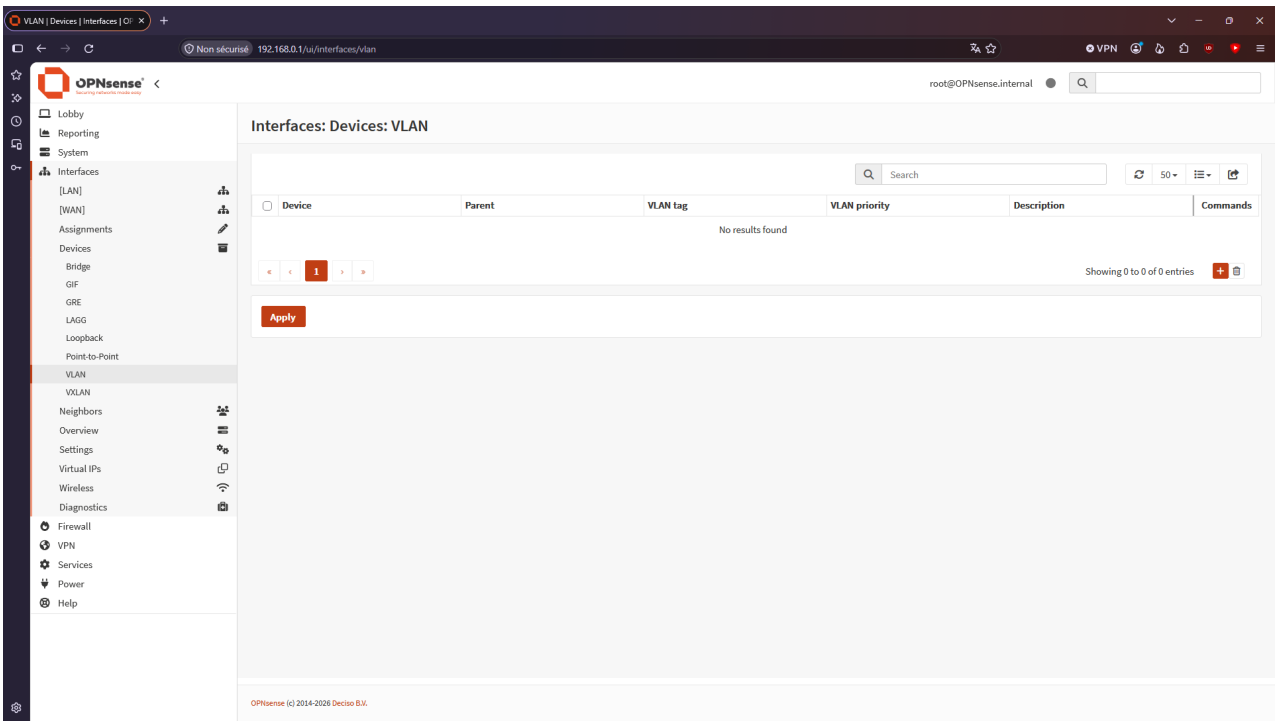
Mettez à jour OPNsense en allant dans System/Firmware/Updates



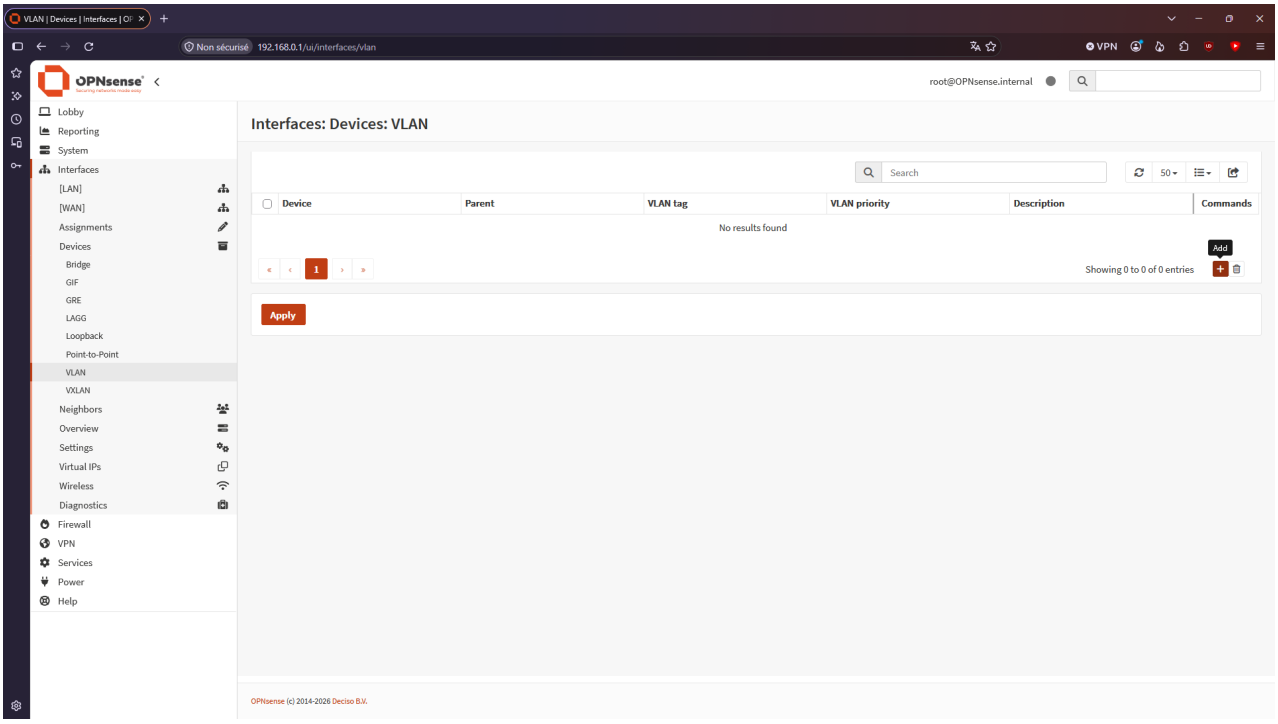
Appliquez les MAJ. Vous avez donc terminé l'installation et la configuration de OPNsense. Vous pouvez également créer les VLANs (optionnel) et des serveurs DHCP. N'oubliez pas d'exporter la VM pour la conserver comme template du projet.

3.3. Créer les VLAN

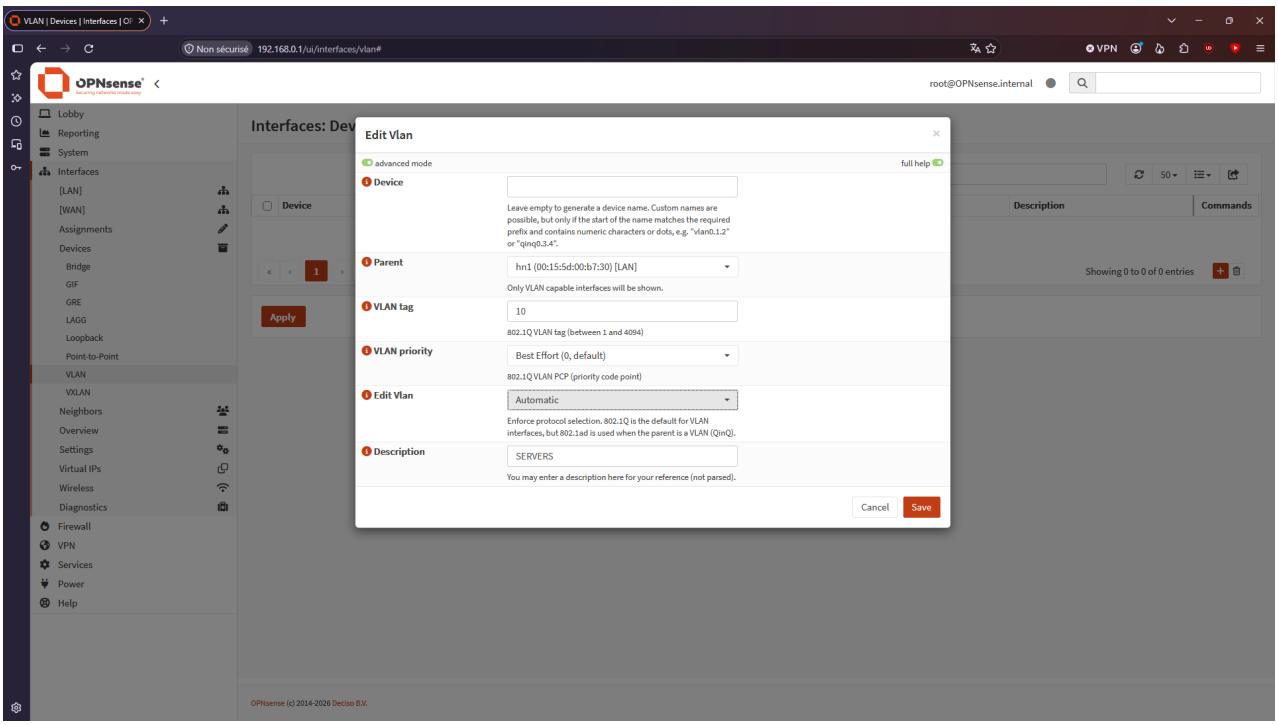
Nous allons créer les VLAN 10 (SERVERS), 20 (TEACHERS) et 30 (STUDENTS) pour notre infrastructure. Allez dans Interfaces/VLAN sur l'interface web de OPNsense :

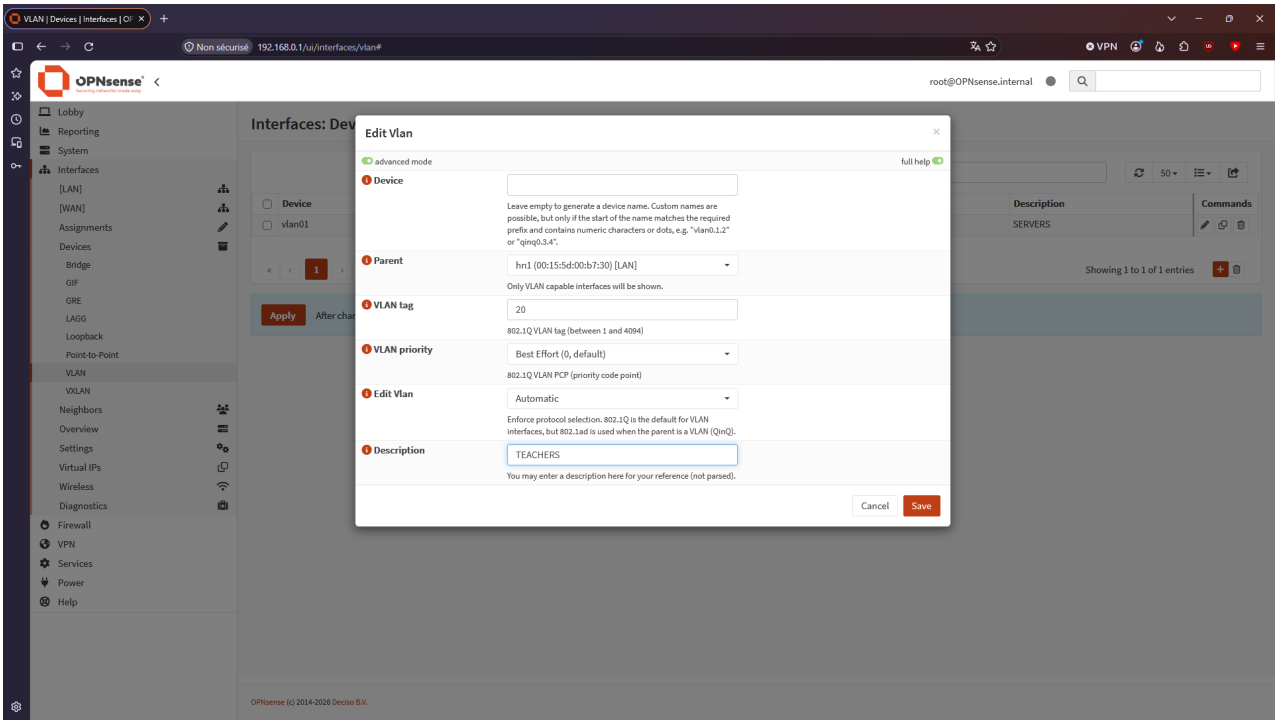


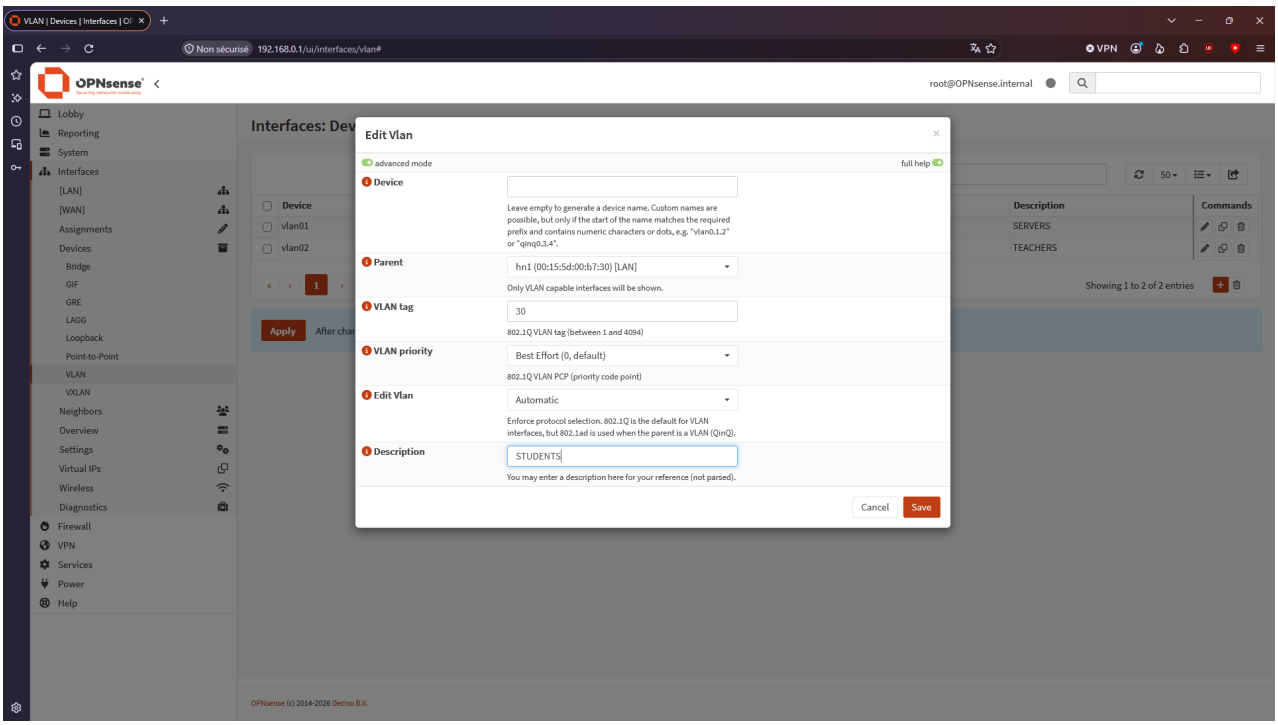
Cliquez sur le bouton **+ (Add)** pour ajouter un VLAN :

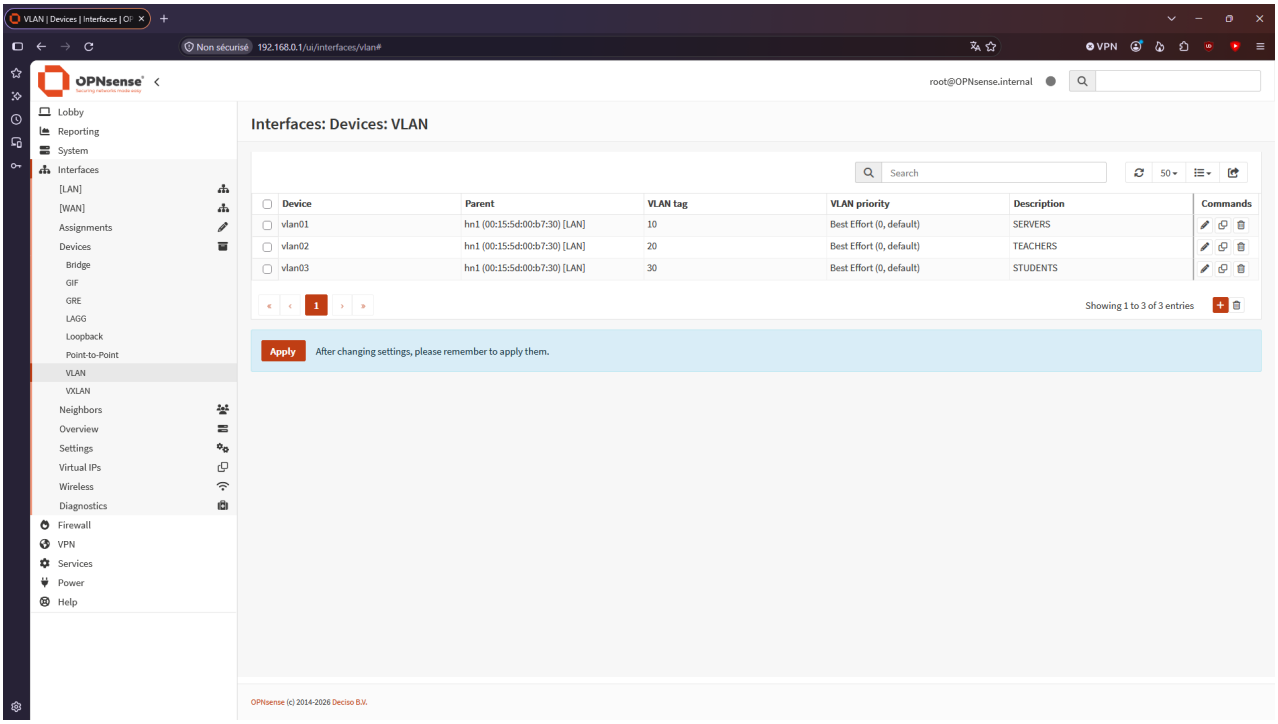


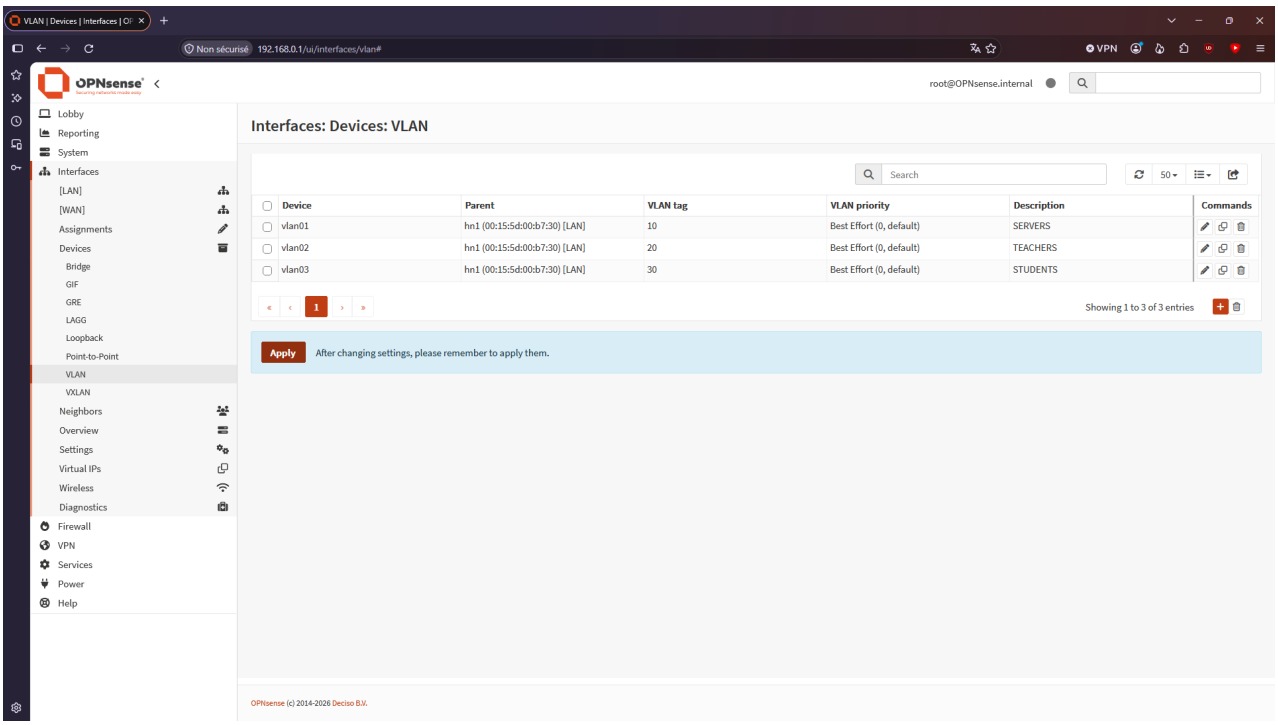
Ajoutez les paramètres suivants pour chaque création de VLAN puis cliquez sur Save :


Edit Vlan
advanced mode
full help
Device
Leave empty to generate a device name. Custom names are possible, but only if the start of the name matches the required prefix and contains numeric characters or dots, e.g. "vlan0.1.2" or "qinq0.3.4".
Parent
hn1 (00:15:5d:00:b7:30) [LAN]
Only VLAN capable interfaces will be shown.
VLAN tag
10
802.1Q VLAN tag (between 1 and 4094)
VLAN priority
Best Effort (0, default)
802.1Q VLAN PCP (priority code point)
Edit Vlan
Automatic
Enforce protocol selection. 802.1Q is the default for VLAN interfaces, but 802.1ad is used when the parent is a VLAN (QinQ).
Description
SERVERS
You may enter a description here for your reference (not parsed).
Cancel
Save

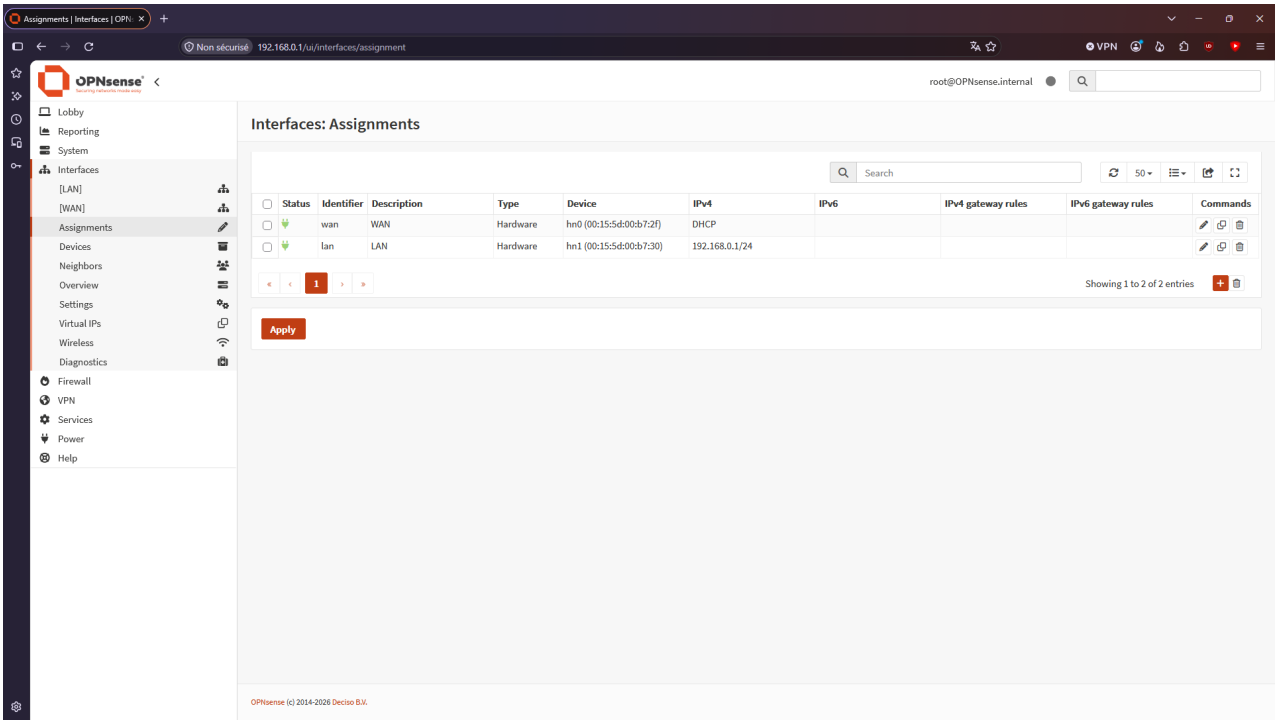

Edit Vlan
advanced mode
full help
Device
Leave empty to generate a device name. Custom names are possible, but only if the start of the name matches the required prefix and contains numeric characters or dots, e.g. "vlan0.1.2" or "qinq0.3.4".
Parent
hn1 (00:15:5d:00:b7:30) [LAN]
Only VLAN capable interfaces will be shown.
VLAN tag
20
802.1Q VLAN tag (between 1 and 4094)
VLAN priority
Best Effort (0, default)
802.1Q VLAN PCP (priority code point)
Edit Vlan
Automatic
Enforce protocol selection. 802.1Q is the default for VLAN interfaces, but 802.1ad is used when the parent is a VLAN (QinQ).
Description
TEACHERS
You may enter a description here for your reference (not parsed).
Cancel
Save


Edit Vlan
advanced mode
full help
Device
Leave empty to generate a device name. Custom names are possible, but only if the start of the name matches the required prefix and contains numeric characters or dots, e.g. "vlan0.1.2" or "qinq0.3.4".
Parent
hn1 (00:15:5d:00:b7:30) [LAN]
Only VLAN capable interfaces will be shown.
VLAN tag
30
802.1Q VLAN tag (between 1 and 4094)
VLAN priority
Best Effort (0, default)
802.1Q VLAN PCP (priority code point)
Edit Vlan
Automatic
Enforce protocol selection. 802.1Q is the default for VLAN interfaces, but 802.1ad is used when the parent is a VLAN (QinQ).
Description
STUDENTS
You may enter a description here for your reference (not parsed).
Cancel
Save

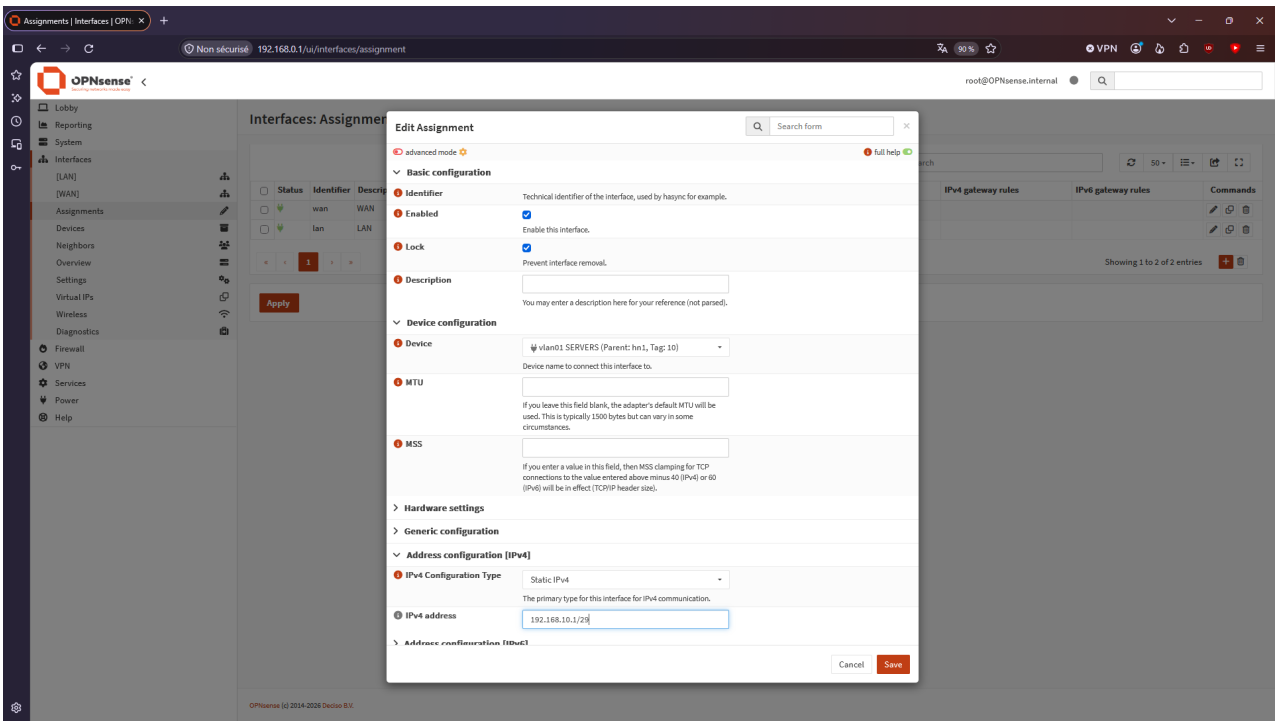

Interfaces: Devices: VLAN
Search
Device
Parent
VLAN tag
VLAN priority
Description
Commands
vlan01
hn1 (00:15:5d:00:b7:30) [LAN]
10
Best Effort (0, default)
SERVERS
vlan02
hn1 (00:15:5d:00:b7:30) [LAN]
20
Best Effort (0, default)
TEACHERS
vlan03
hn1 (00:15:5d:00:b7:30) [LAN]
30
Best Effort (0, default)
STUDENTS
Showing 1 to 3 of 3 entries
Apply
After changing settings, please remember to apply them.

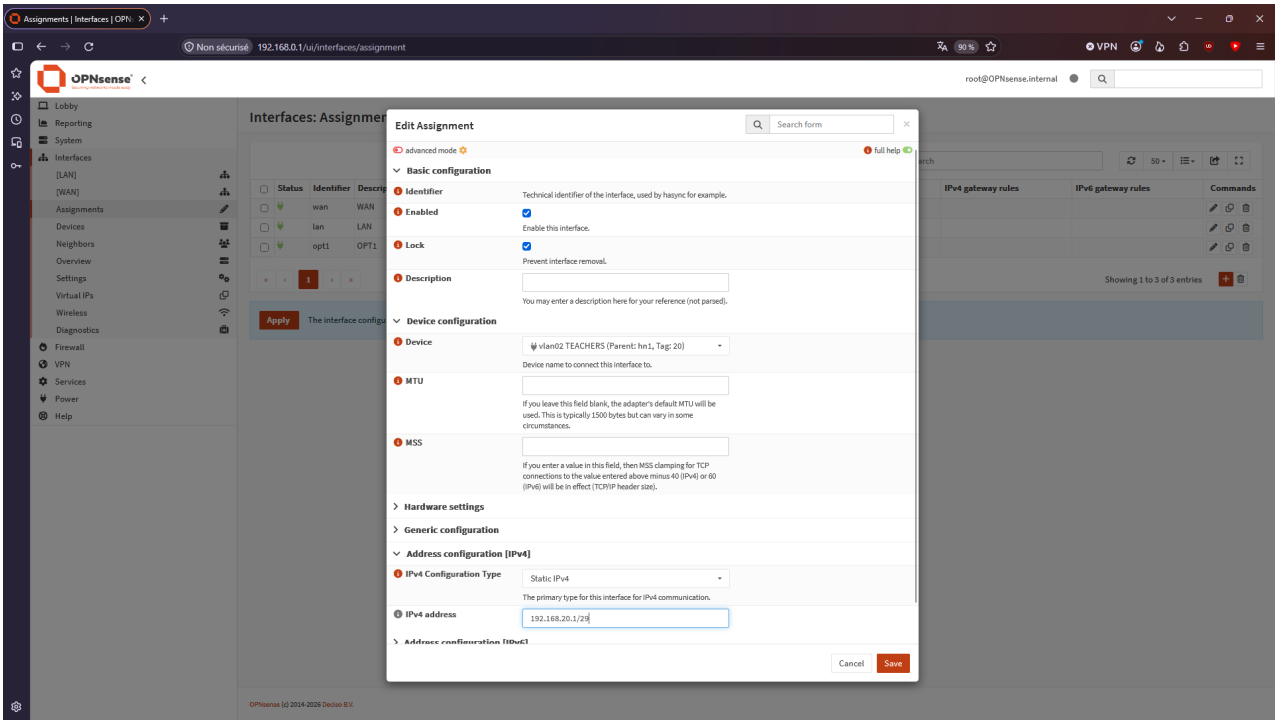


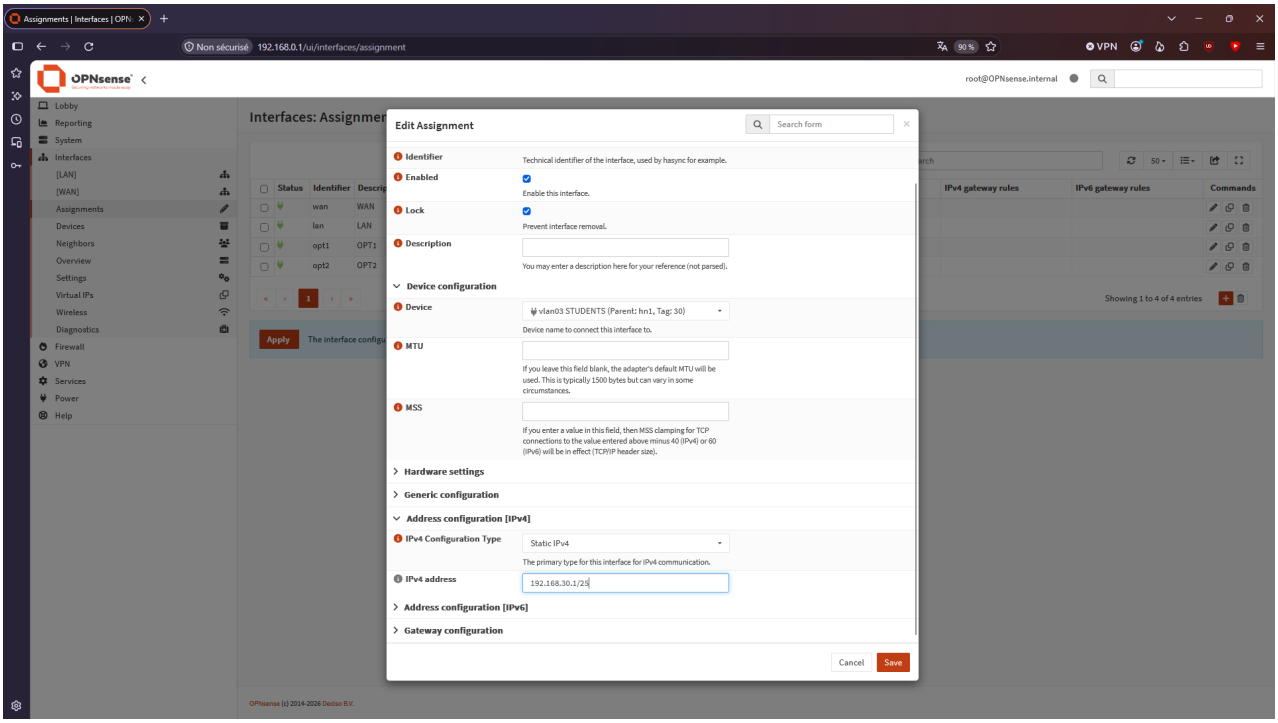
Appliquez les modifications en cliquant sur **Apply**. Allez ensuite dans Interfaces/Assignments :

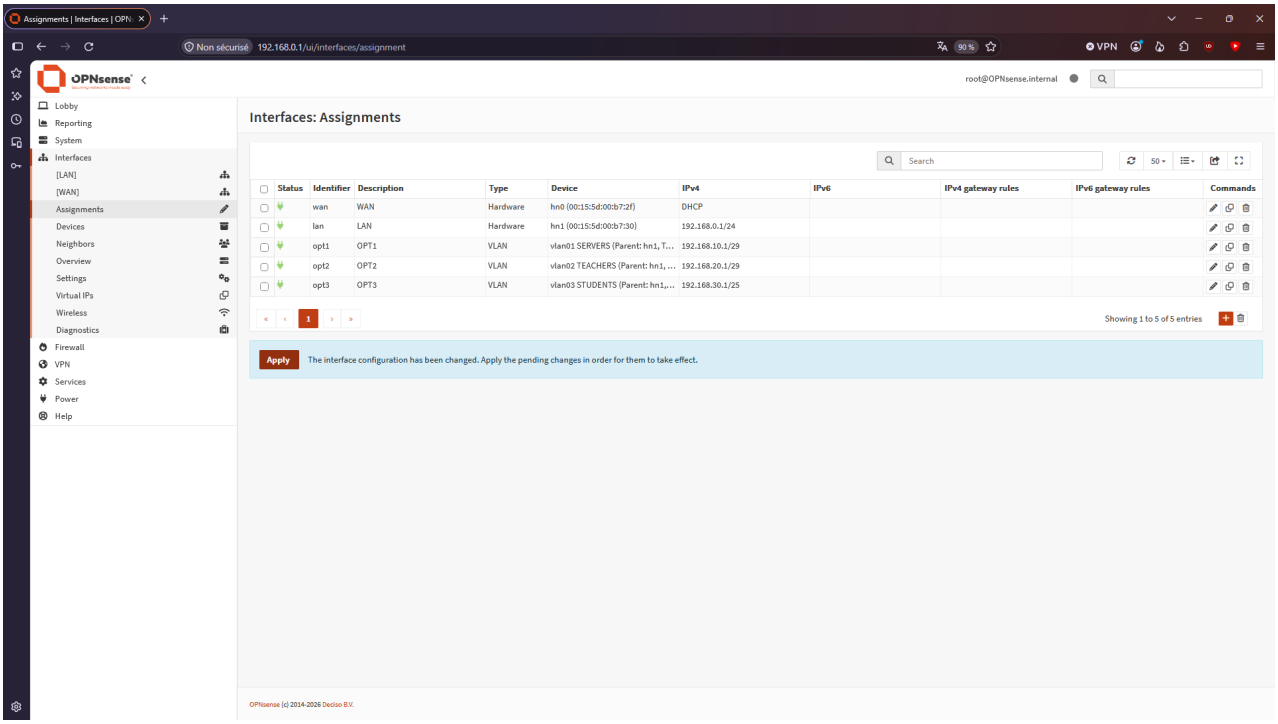


Affectez et activer les interfaces pour chaque VLAN avec une adresse statique :

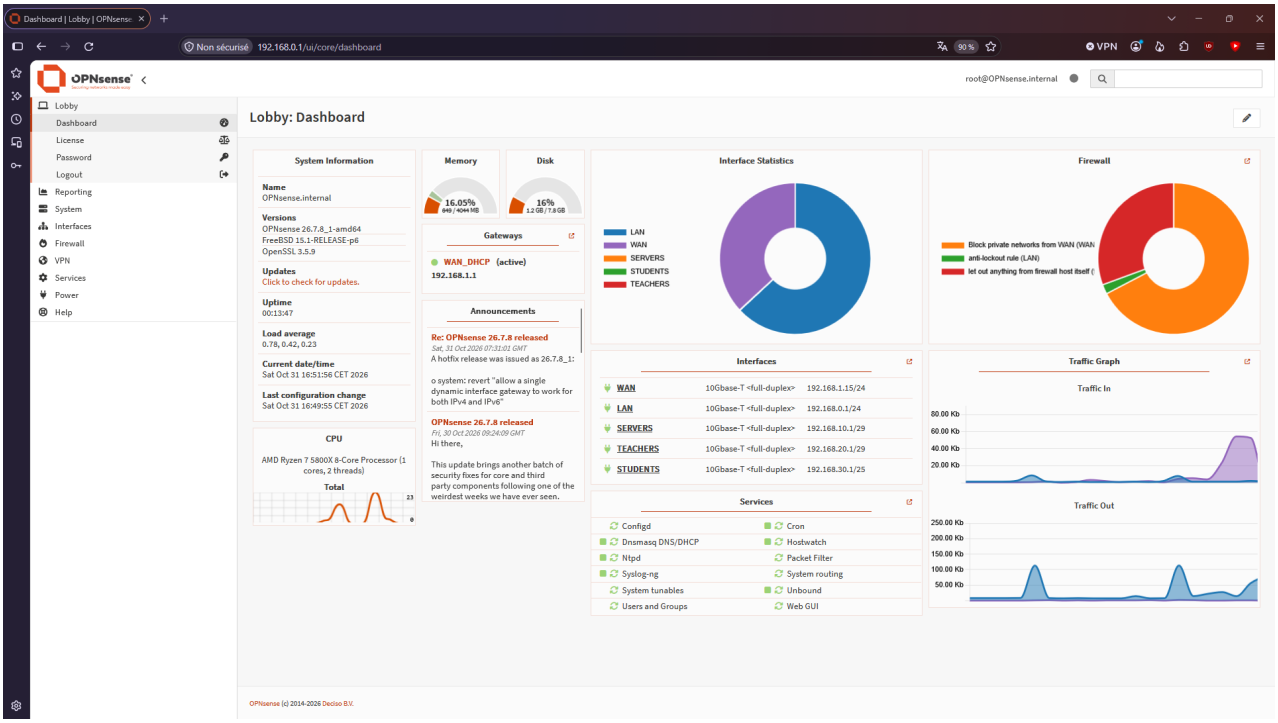

Edit Assignment
Search form
advanced mode
full help
Basic configuration
Identifier
Technical identifier of the interface, used by hasync for example.
Enabled
Enable this interface.
Lock
Prevent interface removal.
Description
You may enter a description here for your reference (not parsed).
Device configuration
Device
vlan01 SERVERS (Parent: hn1, Tag: 10)
Device name to connect this interface to.
MTU
If you leave this field blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.
MSS
If you enter a value in this field, then MSS clamping for TCP connections to the value entered above minus 40 (IPv4) or 60 (IPv6) will be in effect (TCP/IP header size).
Hardware settings
Generic configuration
Address configuration [IPv4]
IPv4 Configuration Type
Static IPv4
The primary type for this interface for IPv4 communication.
IPv4 address
192.168.10.1/29
Cancel
Save


Edit Assignment
Search form
advanced mode
full help
Basic configuration
Identifier
Technical identifier of the interface, used by hasync for example.
Enabled
Enable this interface.
Lock
Prevent interface removal.
Description
You may enter a description here for your reference (not parsed).
Device configuration
Device
vlan02 TEACHERS (Parent: hn1, Tag: 20)
Device name to connect this interface to.
MTU
If you leave this field blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.
MSS
If you enter a value in this field, then MSS clamping for TCP connections to the value entered above minus 40 (IPv4) or 60 (IPv6) will be in effect (TCP/IP header size).
Hardware settings
Generic configuration
Address configuration [IPv4]
IPv4 Configuration Type
Static IPv4
The primary type for this interface for IPv4 communication.
IPv4 address
192.168.20.1/29
Cancel
Save





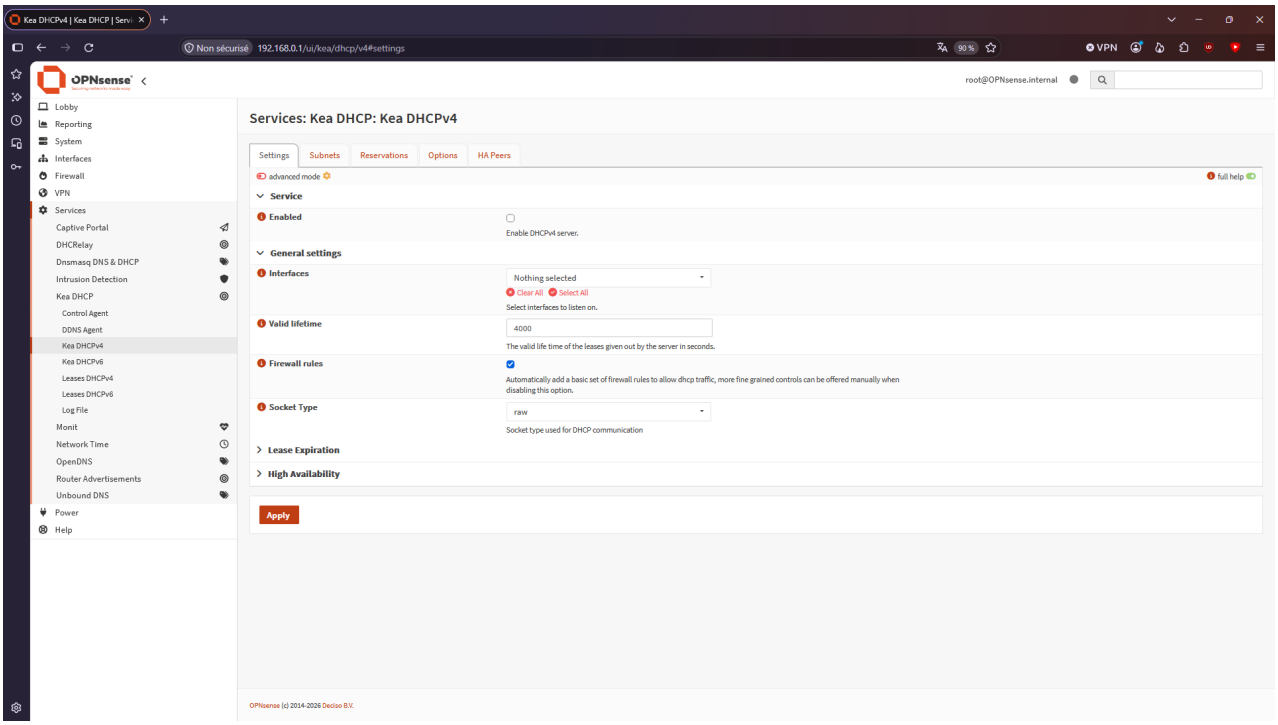
Appliquez ensuite les modifications (vous pouvez également ajoutez une description pour chaque VLAN comme SERVERS). En retournant dans **Lobby/Dashbord**, vous devez retrouver les interfaces de chaque VLAN :



Nous avons terminé avec la création des VLAN.

3.4. Créer les serveurs DHCP

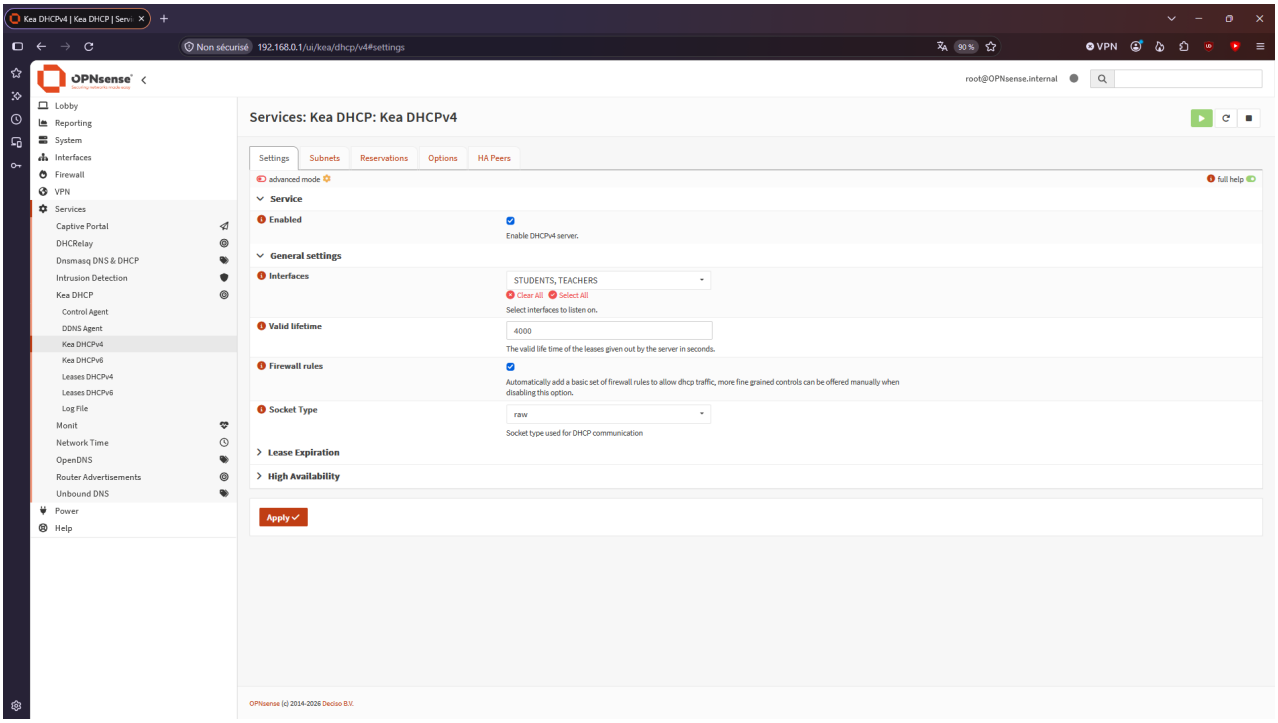
Nous allons après créer les serveurs DHCP pour les VLAN 20 et 30. Le VLAN 10 contient uniquement des serveurs avec une adresse IPv4 statique. Allez dans Services/Kea DHCP/Kea DHCPv4 :



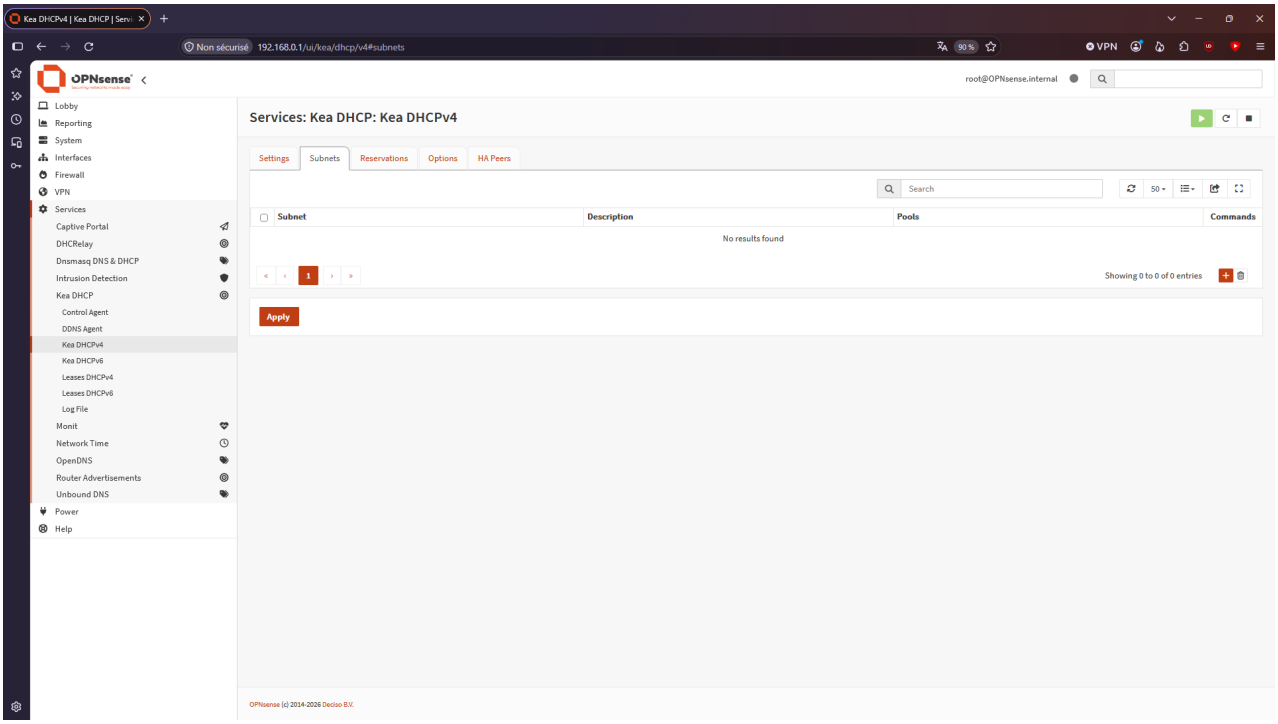
Vous allez procéder comme ceci :

1) Cochez **Enabled**

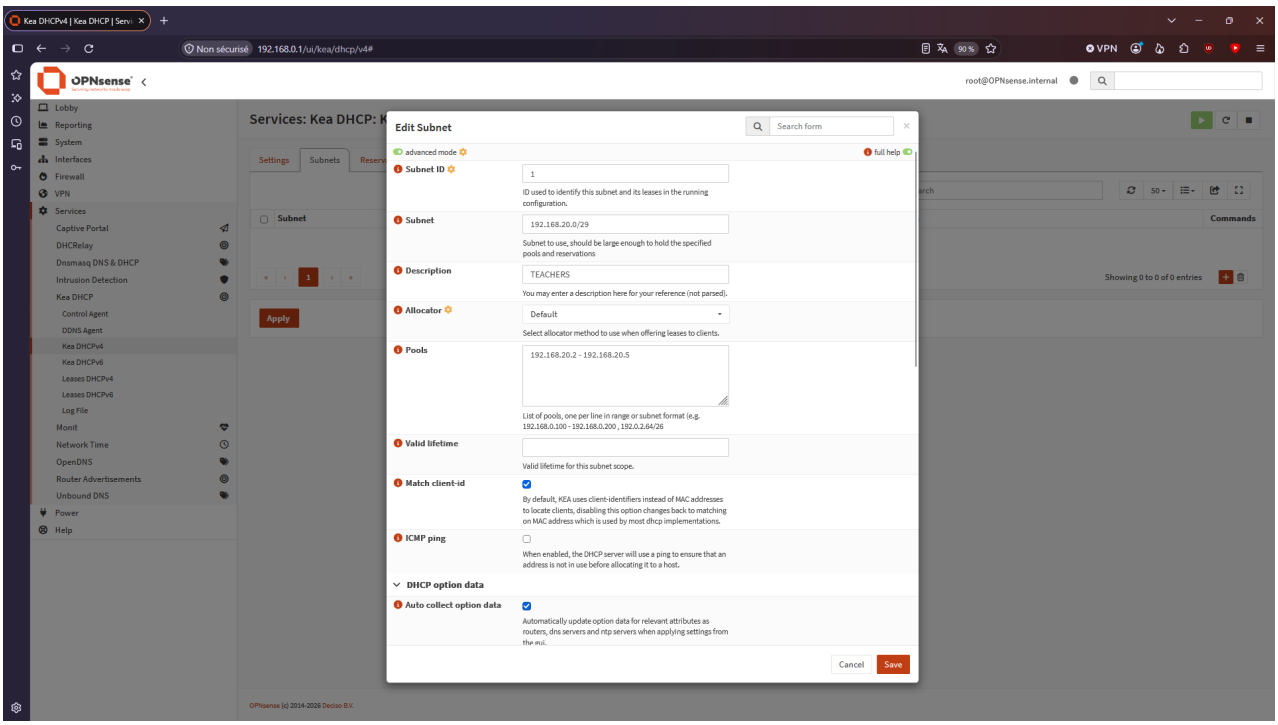
2) Sélectionnez les interface **20 (TEACHERS)** et **30 (STUDENTS)**
3) Activez **Firewall rules** pour autoriser les échanges DHCP
4) Appliquez

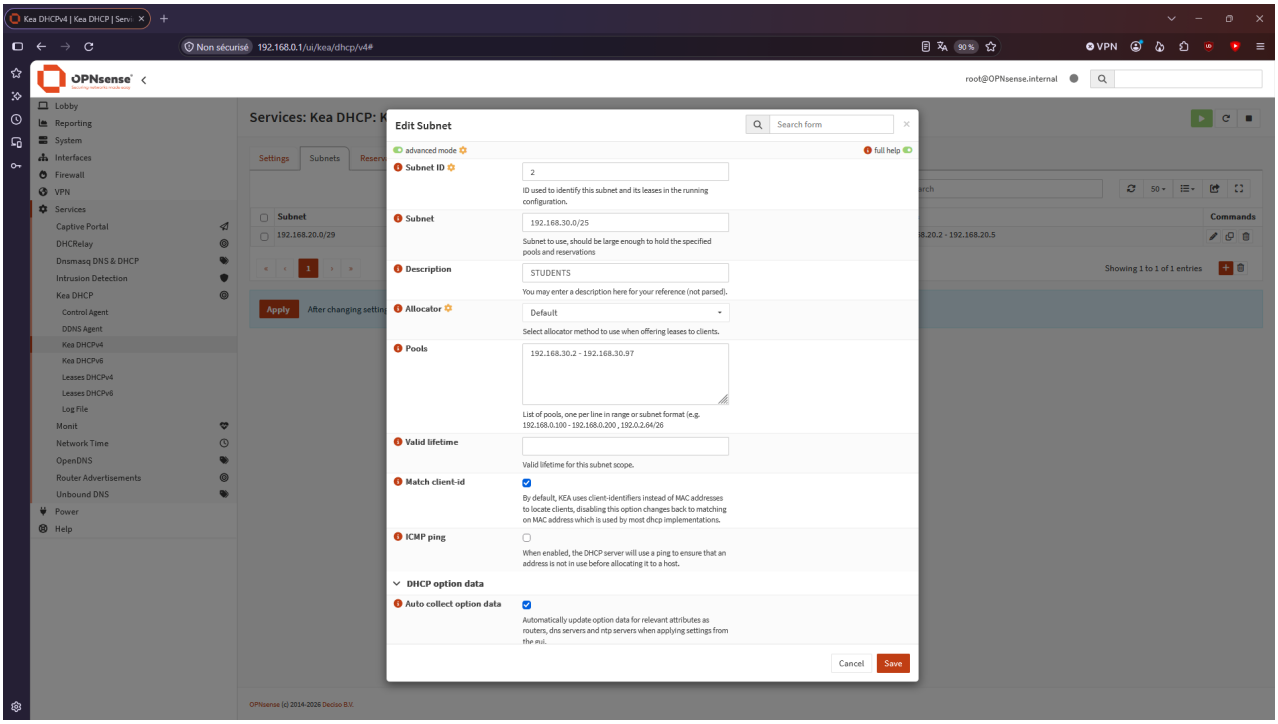


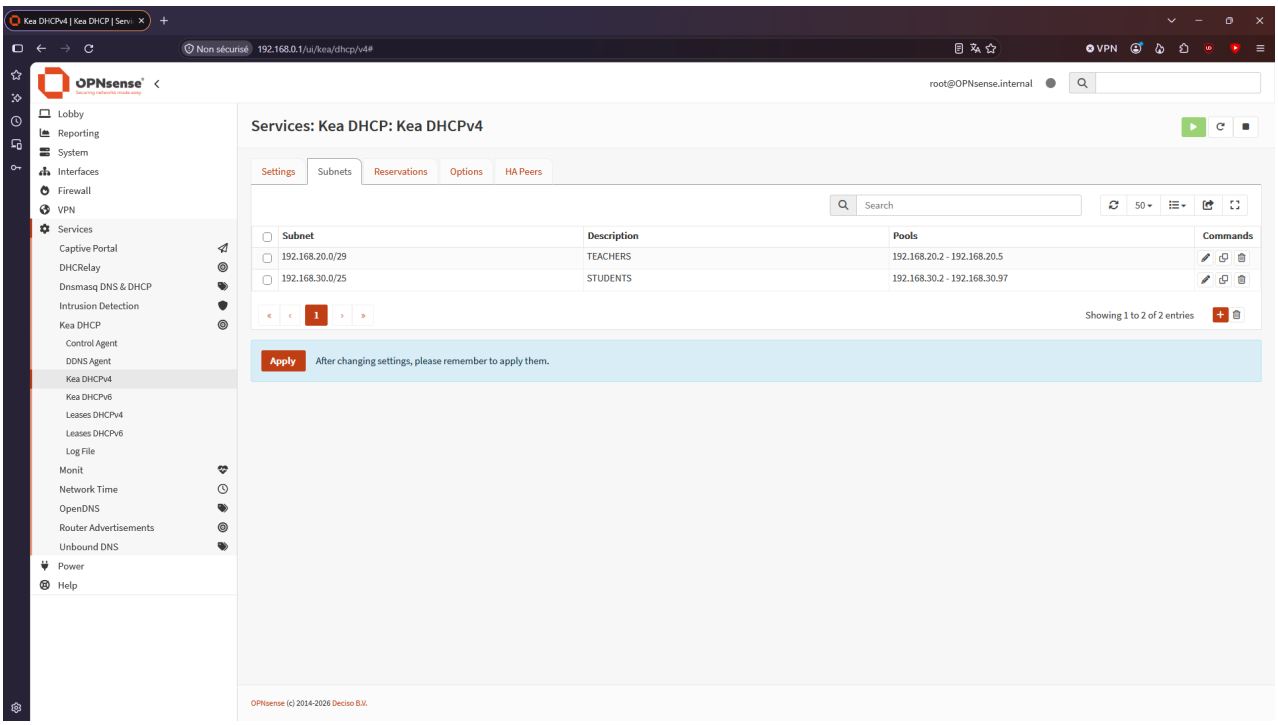
Allez ensuite dans **Subnets** :



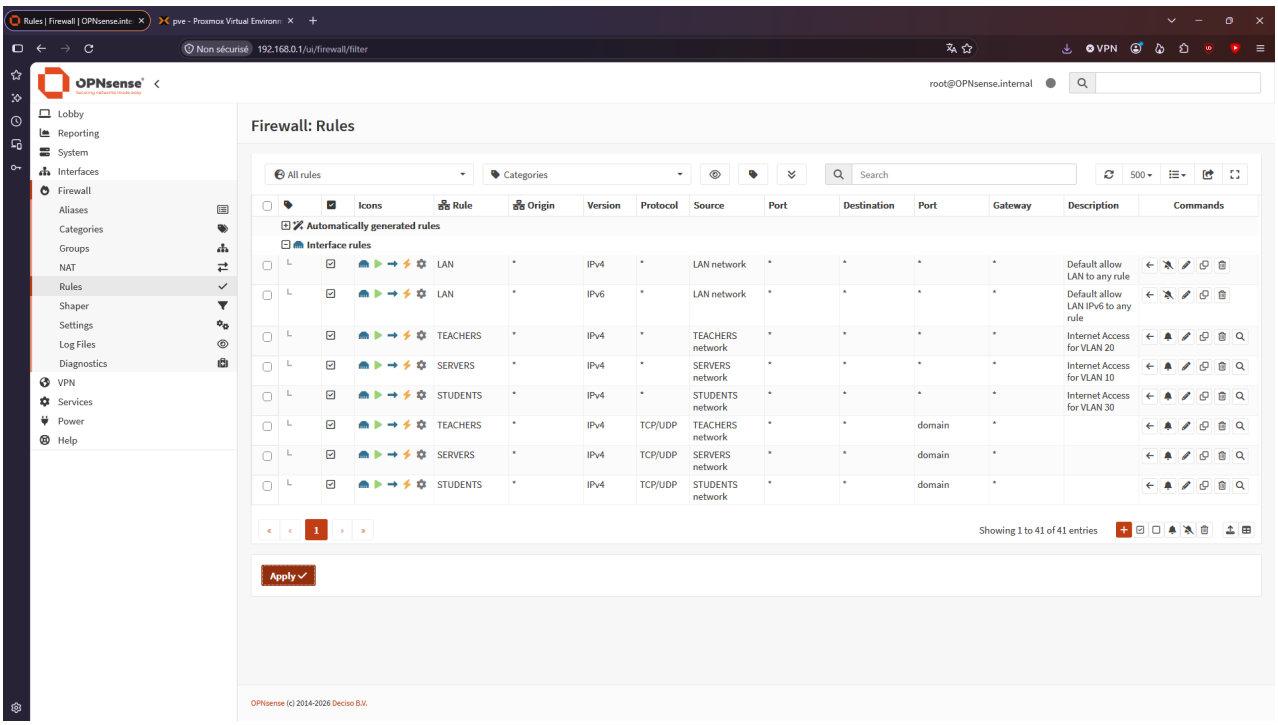
Ajoutez les entrées suivantes pour chaque VLAN :


Edit Subnet
Search form
advanced mode
full help
Subnet ID
1
ID used to identify this subnet and its leases in the running configuration.
Subnet
192.168.20.0/29
Subnet to use, should be large enough to hold the specified pools and reservations
Description
TEACHERS
You may enter a description here for your reference (not parsed).
Allocator
Default
Select allocator method to use when offering leases to clients.
Pools
192.168.20.2 - 192.168.20.5
List of pools, one per line in range or subnet format (e.g. 192.168.0.100 - 192.168.0.200 , 192.0.2.64/26
Valid lifetime
Valid lifetime for this subnet scope.
Match client-id
By default, KEA uses client-identifiers instead of MAC addresses to locate clients, disabling this option changes back to matching on MAC address which is used by most dhcp implementations.
ICMP ping
When enabled, the DHCP server will use a ping to ensure that an address is not in use before allocating it to a host.
DHCP option data
Auto collect option data
Automatically update option data for relevant attributes as routers, dns servers and ntp servers when applying settings from
Cancel
Save

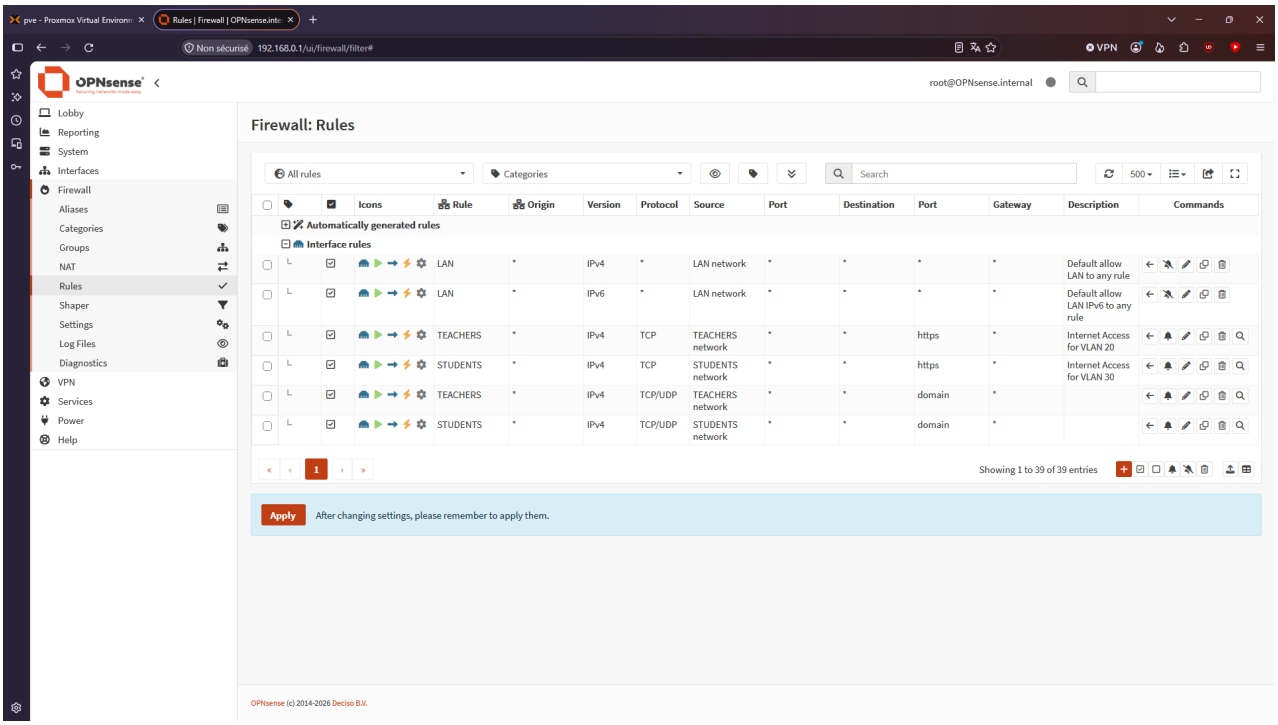

Edit Subnet
Search form
advanced mode
full help
Subnet ID
2
ID used to identify this subnet and its leases in the running configuration.
Subnet
192.168.30.0/25
Subnet to use, should be large enough to hold the specified pools and reservations
Description
STUDENTS
You may enter a description here for your reference (not parsed).
Allocator
Default
Select allocator method to use when offering leases to clients.
Pools
192.168.30.2 - 192.168.30.97
List of pools, one per line in range or subnet format (e.g. 192.168.0.100 - 192.168.0.200 , 192.0.2.64/26
Valid lifetime
Valid lifetime for this subnet scope.
Match client-id
By default, KEA uses client-identifiers instead of MAC addresses to locate clients, disabling this option changes back to matching on MAC address which is used by most dhcp implementations.
ICMP ping
When enabled, the DHCP server will use a ping to ensure that an address is not in use before allocating it to a host.
DHCP option data
Auto collect option data
Automatically update option data for relevant attributes as routers, dns servers and ntp servers when applying settings from
Cancel
Save



Appliquez les modifications et vous aurez vos serveurs DHCP pour les VLAN 20 et 30. Vous pouvez vérifiez le DHCP sur un poste client.


Rules | Firewall | OPNsense.int
pve - Proxmox Virtual Environ
Non sécurisé 192.168.0.1/ui/firewall/filter
VPN
OPNsense
root@OPNsense.internal
Lobby
Reporting
System
Interfaces
Firewall
Aliases
Categories
Groups
NAT
Rules
Shaper
Settings
Log Files
Diagnostics
VPN
Services
Power
Help
Firewall: Rules
All rules
Categories
Search
500
Icons
Rule
Origin
Version
Protocol
Source
Port
Destination
Port
Gateway
Description
Commands
Automatically generated rules
Interface rules
LAN
IPv4
LAN network
Default allow LAN to any rule
LAN
IPv6
LAN network
Default allow LAN IPv6 to any rule
TEACHERS
IPv4
TEACHERS network
Internet Access for VLAN 20
SERVERS
IPv4
SERVERS network
Internet Access for VLAN 10
STUDENTS
IPv4
STUDENTS network
Internet Access for VLAN 30
TEACHERS
IPv4
TCP/UDP
TEACHERS network
domain
SERVERS
IPv4
TCP/UDP
SERVERS network
domain
STUDENTS
IPv4
TCP/UDP
STUDENTS network
domain
1
Showing 1 to 41 of 41 entries
Apply
OPNsense (c) 2014-2026 Deciso B.V.

3.5. Paramétrer le NAT pour l'accès Internet (WIP)



Création de Proxmox VE (REVOIR)

Proxmox VE est une plateforme open source qui permet d'exéctuer plusieurs systèmes isolés sur une machine physique. Nous allons l'utiliser pour faire de la virtualisation imbriquée en regroupant les machines de l'infrastructure.

1. Créer la VM

Nous allons mettre en place Proxmox VE comme VM de Hyper-V. Commencez par récupérer l'ISO de Proxmox VE version 8.4 à cette adresse https://www.proxmox.com/en/downloads/proxmox-virtual-environment. Puis exécutez le script suivant en respectant les paramètres :

```powershell
# Configuration
$PVE = [PSCustomObject]@{
    Name          = "pve"
    Generation    = 2
    CPUCount      = 6
    VRAMBytes     = 20GB
    Path          = "V:\Kiformatou\pve"
    VHDPath       = "V:\Kiformatou\pve\pve.vhdx"
    VHDSizeBytes  = 300GB
    SwitchName    = "KIFORMATOU"
    ISOPath       = "V:\ISO\pve_8.4-1.iso"
```

```
}

# Create the VM directory
New-Item -ItemType Directory -Path $PVE.Path -Force | Out-Null

# Create a Generation 2 VM with 20 GB vRAM and a 300 GB vHD
New-VM -Name $PVE.Name `
    -Generation $PVE.Generation `
    -MemoryStartupBytes $PVE.VRAMBytes `
    -Path $PVE.Path `
    -NewVHDPath $PVE.VHDPath `
    -NewVHDSizeBytes $PVE.VHDSizeBytes `
    -SwitchName $PVE.SwitchName

# Assign 6 vCPU
Set-VMProcessor -VMName $PVE.Name -Count $PVE.CPUCount

# Disable secure boot
Set-VMFirmware -VMName $PVE.Name -EnableSecureBoot Off

# Activate exposed virtualization
Set-VMProcessor -VMName $PVE.Name -ExposeVirtualizationExtensions $true

# Allow tagged VLANs 10 and 20
Get-VMNetworkAdapter -VMName $PVE.Name |
    Set-VMNetworkAdapterVlan -Trunk `
        -AllowedVlanIdList "0,10,20,30" `
        -NativeVlanId 0

# Allow traffic from nested VM MAC addresses
Get-VMNetworkAdapter -VMName $PVE.Name |
    Set-VMNetworkAdapter -MacAddressSpoofing On


# Attach the installation ISO and boot from it
$dvdDrive = Add-VMDvdDrive -VMName $PVE.Name -Path $PVE.ISOPath -Passthru
```

```
Set-VMFirmware -VMName $PVE.Name -FirstBootDevice $dvdDrive


# Disable dynamic memory
Set-VMMemory -VMName $PVE.Name -DynamicMemoryEnabled $false -
StartupBytes $PVE.VRAMBytes


# Display the VM status
Get-VM -Name $PVE.Name
Get-VMProcessor -VMName $PVE.Name |
    Select-Object VMName, ExposeVirtualizationExtensions
```

Puis, démarrez :

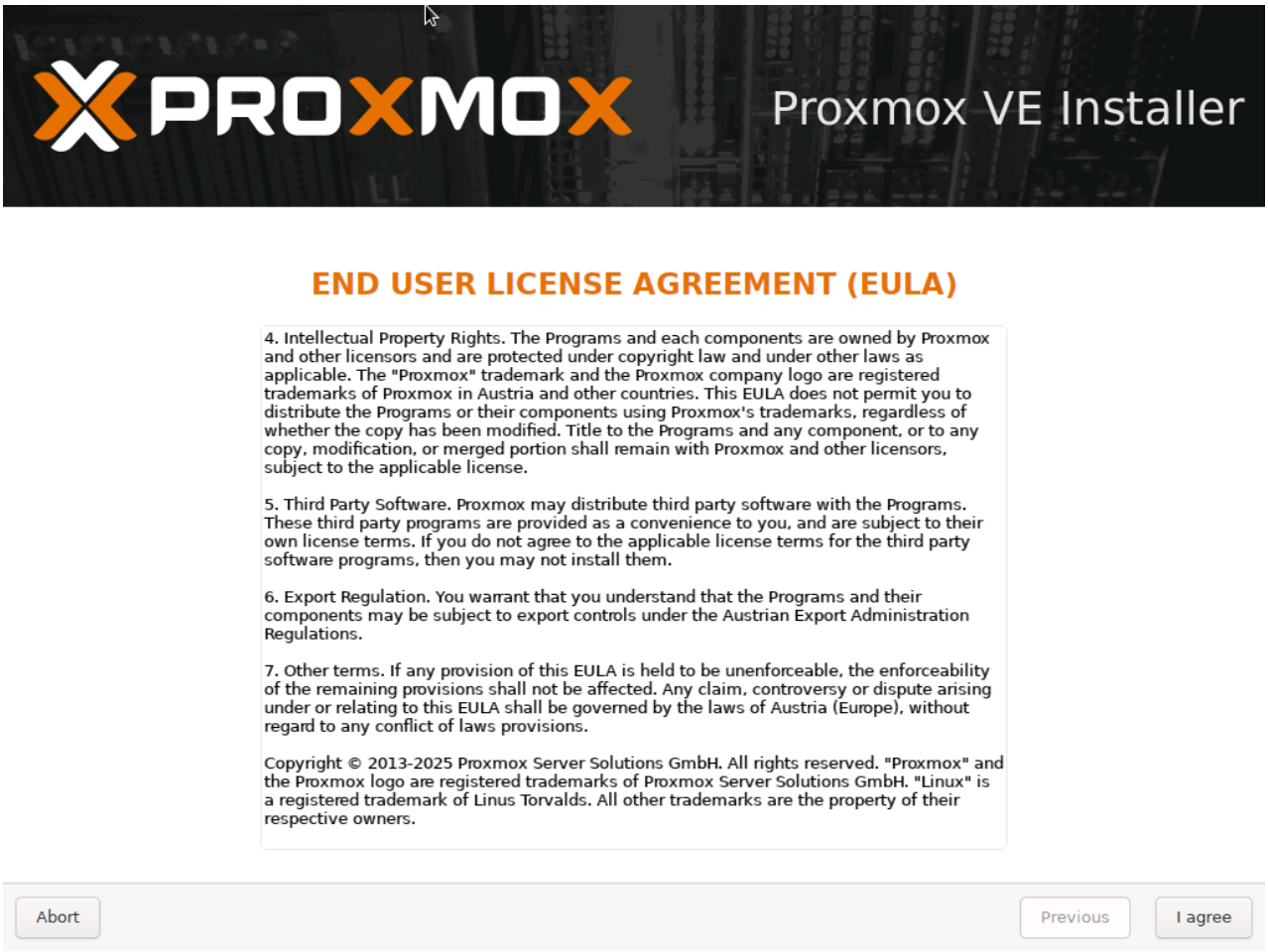
```
Start-VM -Name "pve"
vmconnect.exe localhost "pve"
```

2. Installer la VM

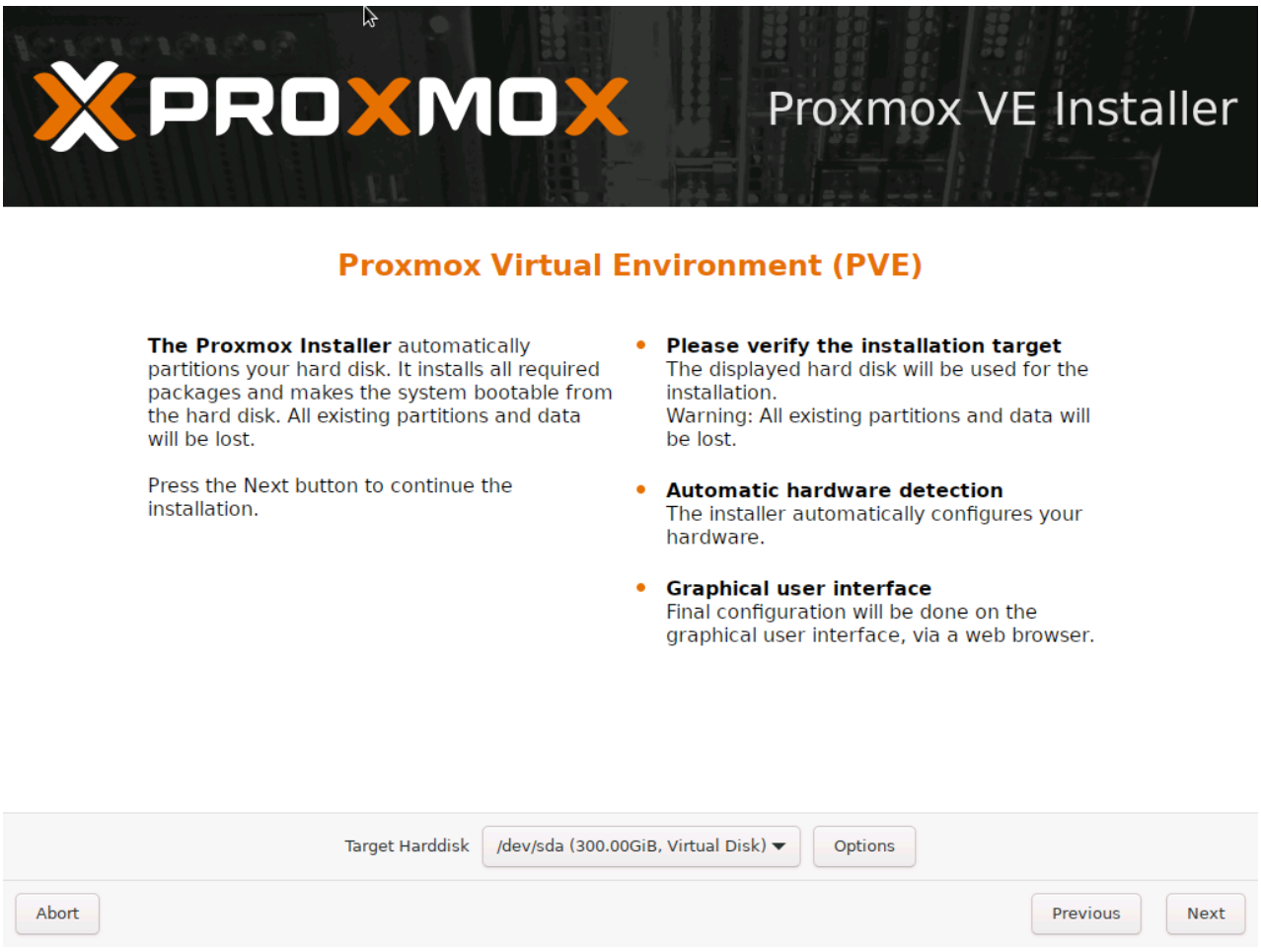
Après démarrage de la VM, choisissez **Install Proxmox VE (Graphical)** :



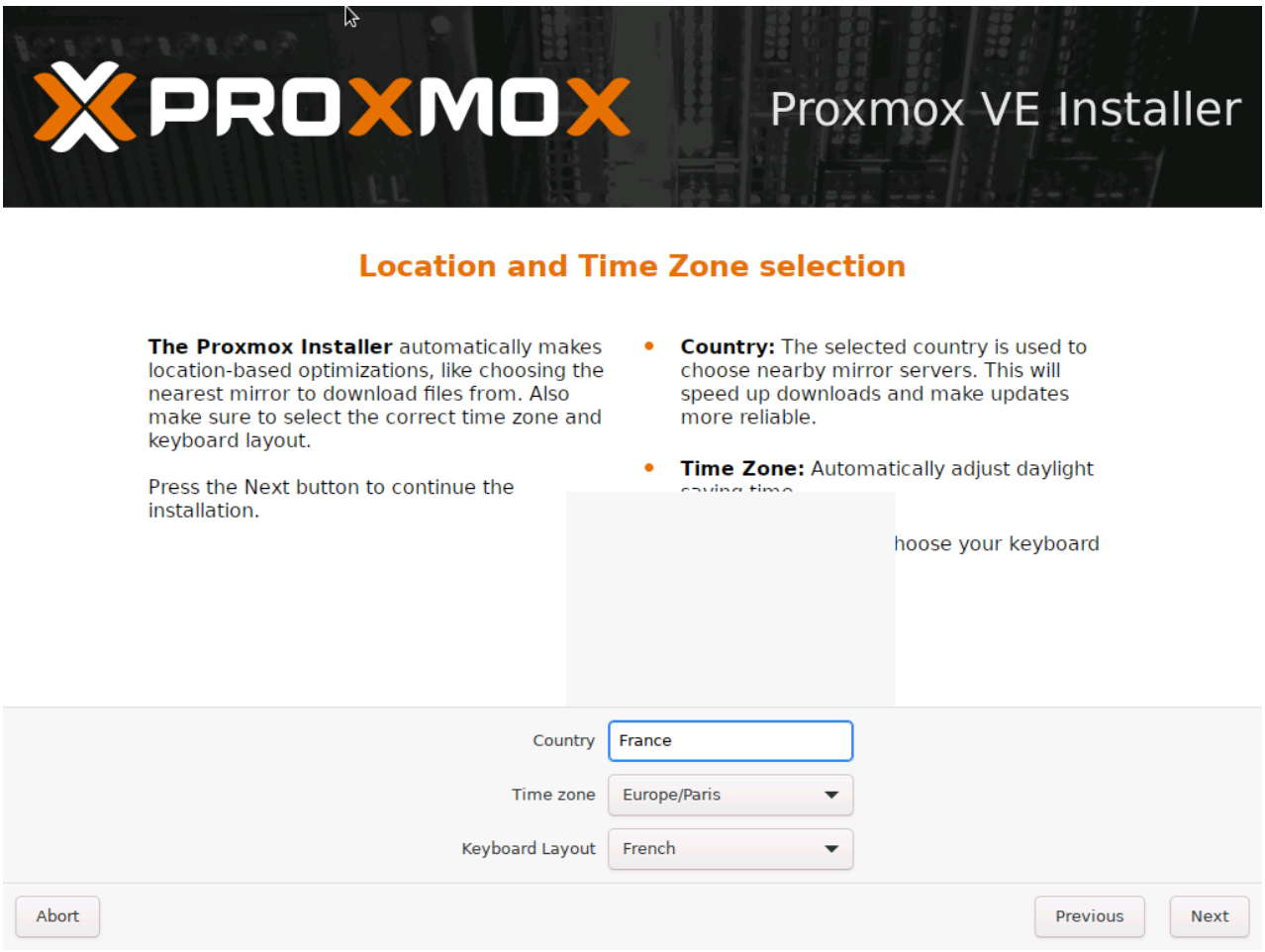
Acceptez **EULA :**



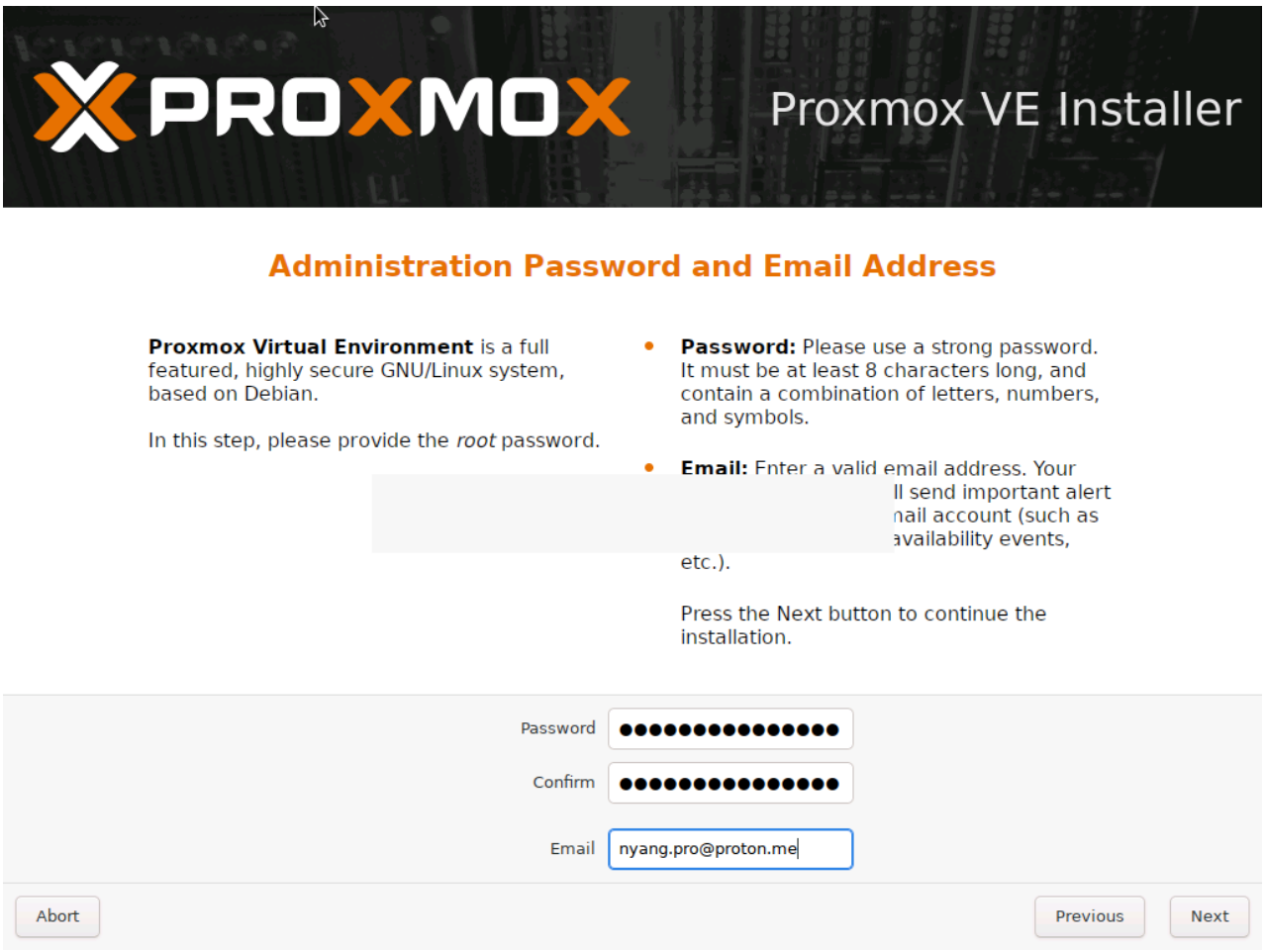
Cliquez sur **Next**



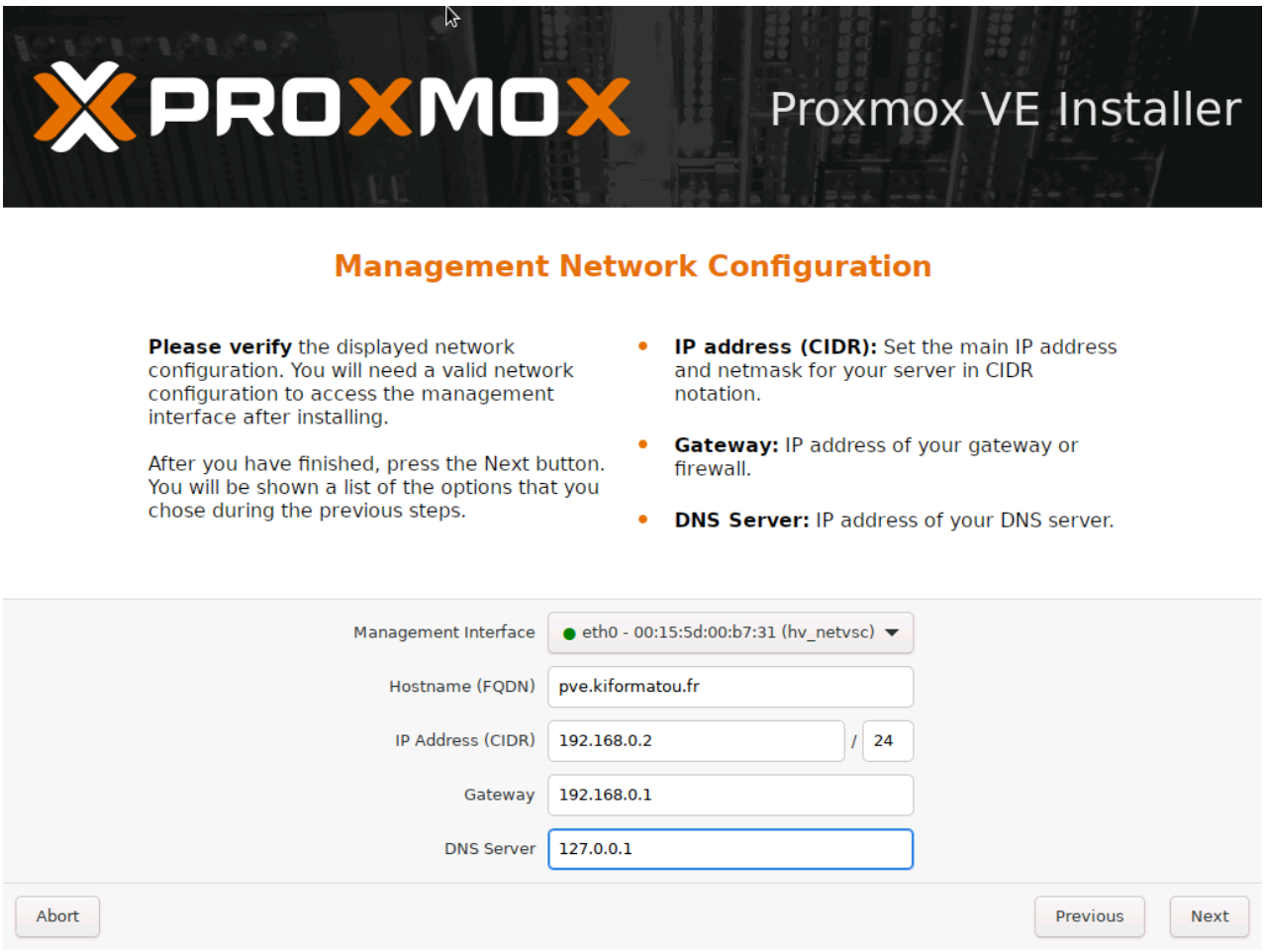
Tapez **France**, les paramètres restants se mettrons à jour automatiquement :



Tapez un mot de passe votre choix puis un email :



Entrez les paramètres suivants pour la configuration réseau :



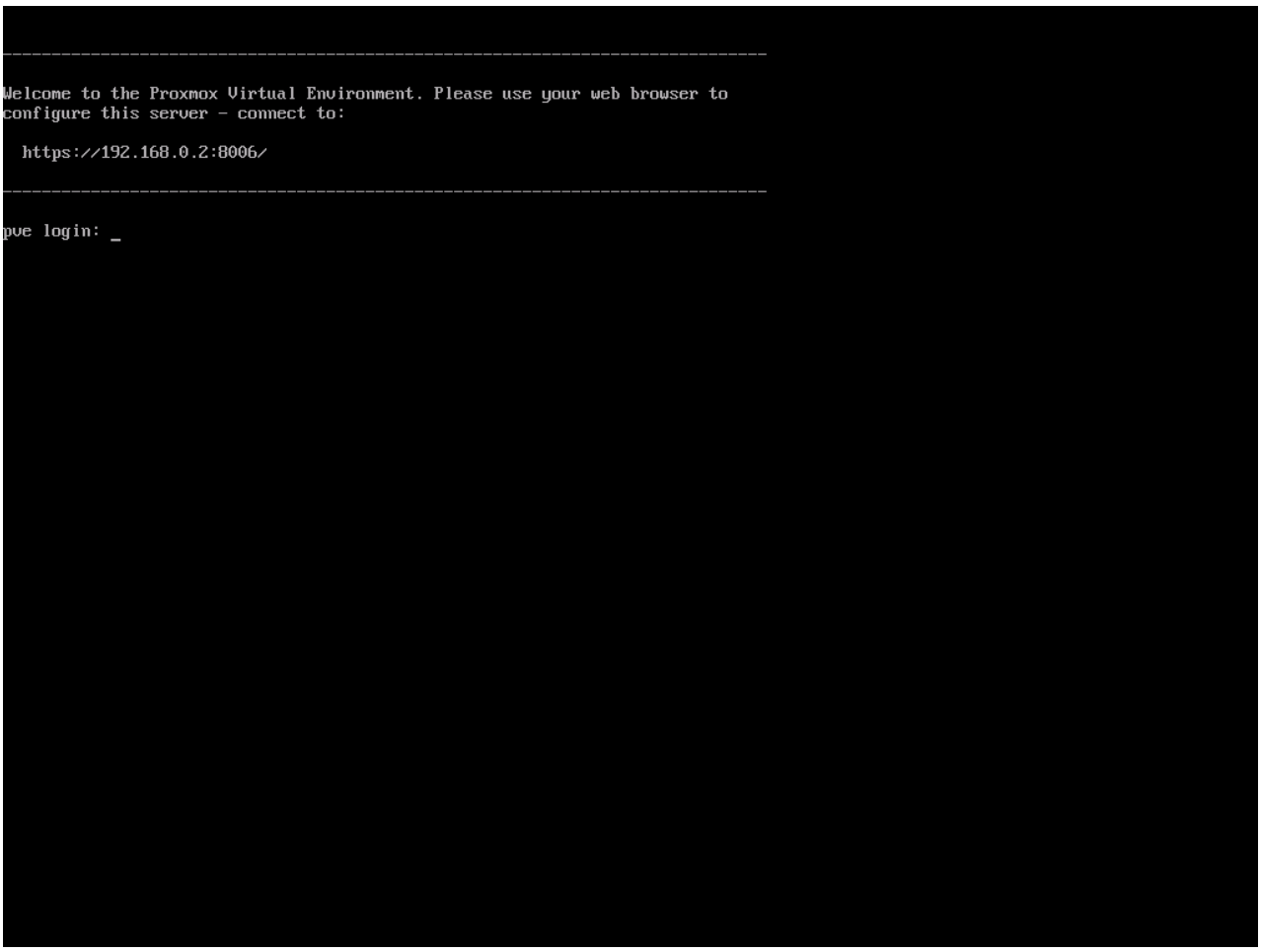
Vérifiez vos informations et débutez l'installation :



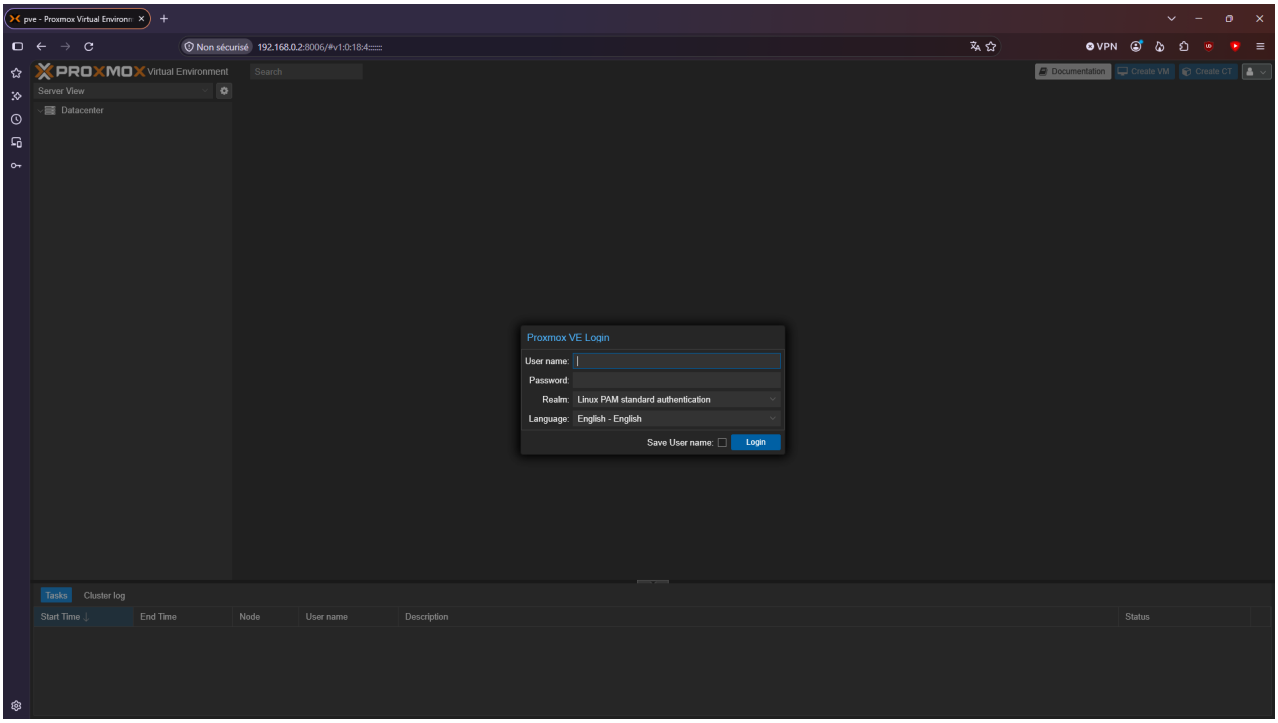
3. Configurer la VM

3.1. Mettre à jour Proxmox

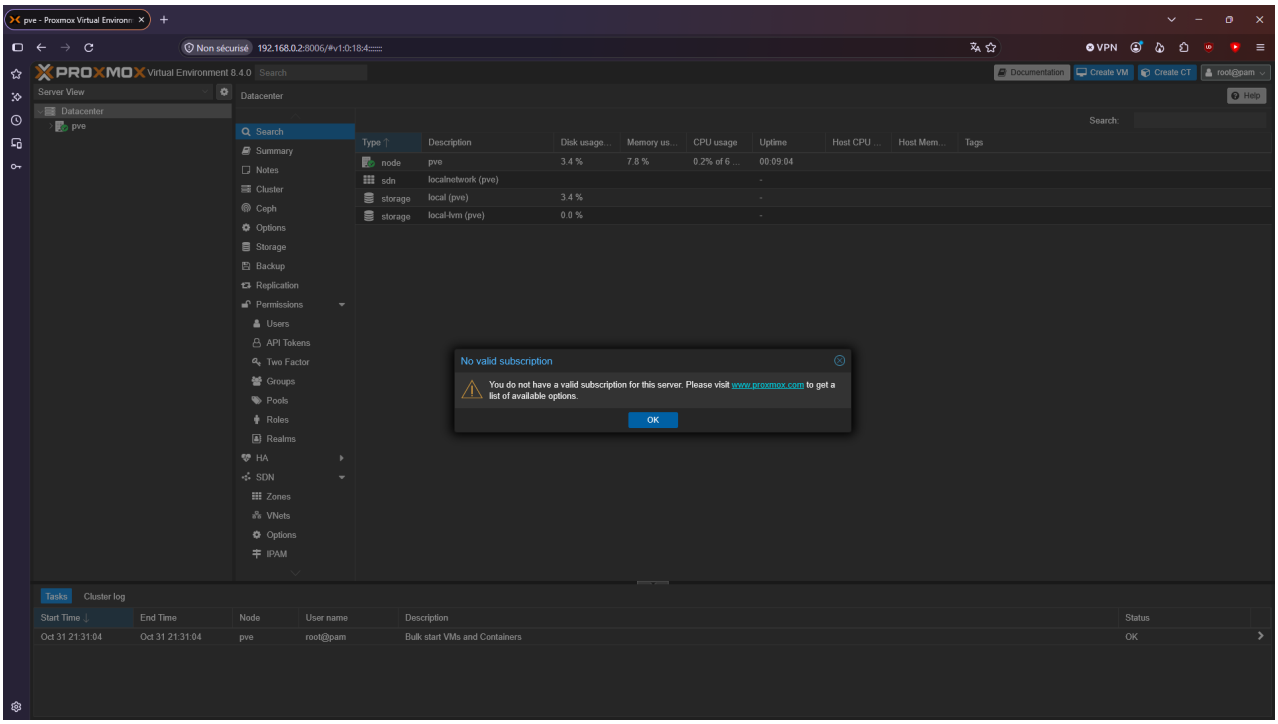
Après avoir terminé l'installation, la VM redémarrera. Vous devez tomber sur l'affichage suivant :



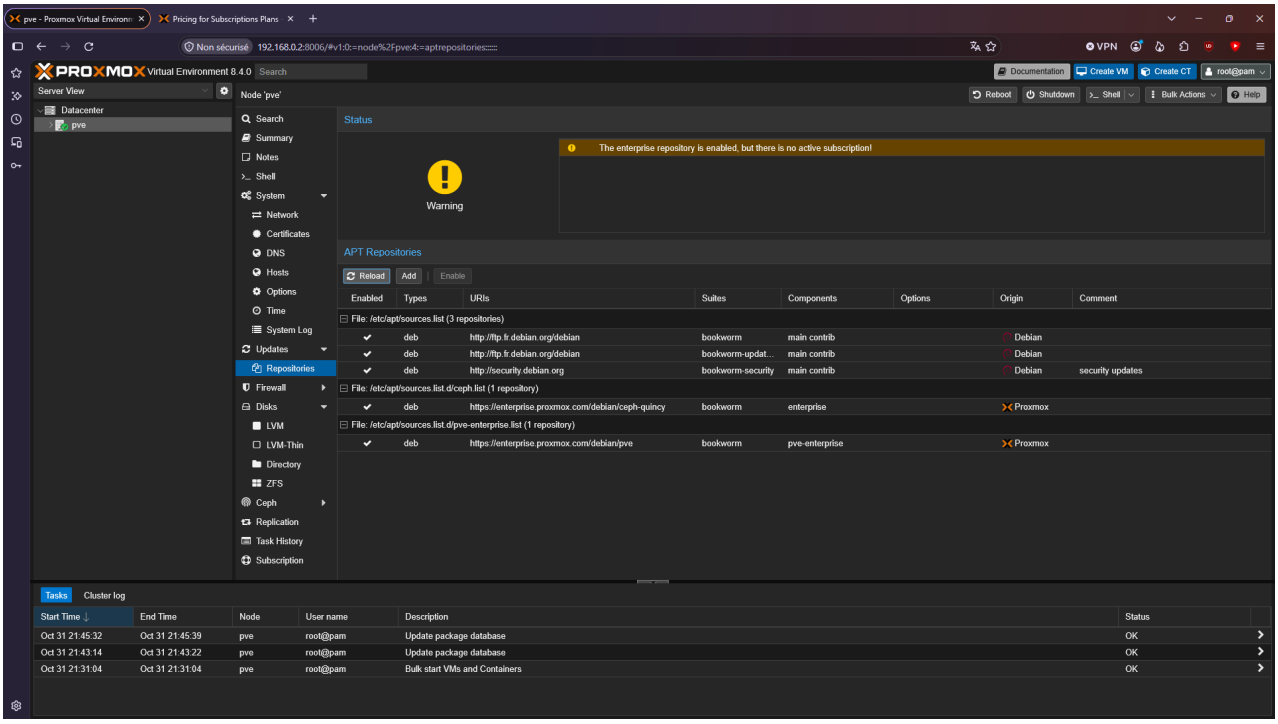
Accéder à l'interface Web de Proxmox VE en allant à l'adresse https://192.168.0.2:8006 :



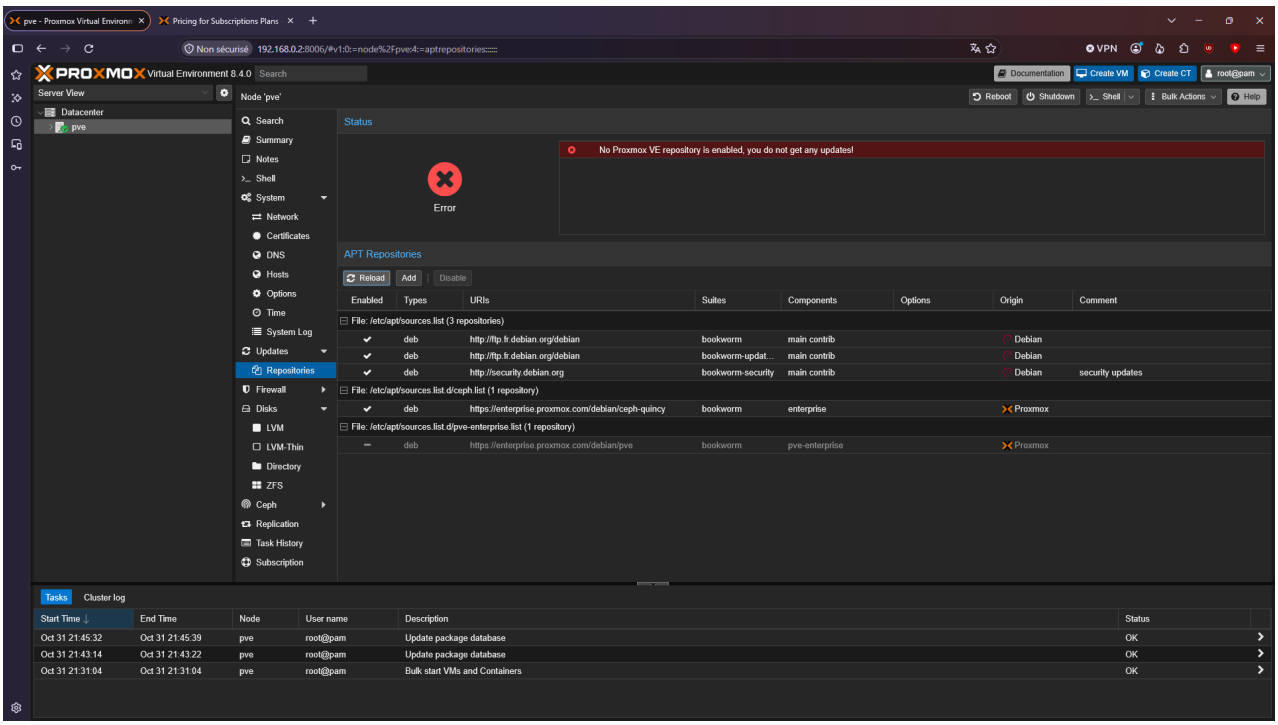
Entre vos identifiants pour **root**, vous serez ensuite sur le tableau de bord de Proxmox :



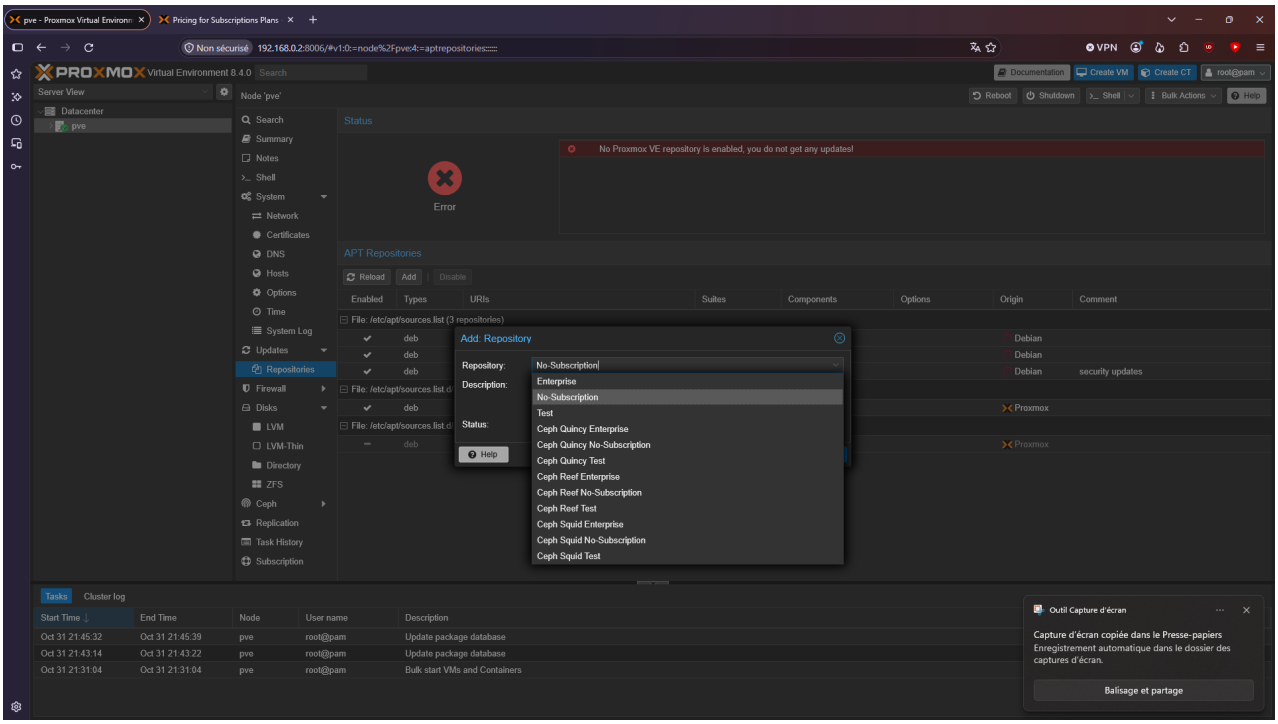
Commencez par mettre à jour Proxmox en cliquant sur votre noeud **pve -> Updates -> Repositories** :



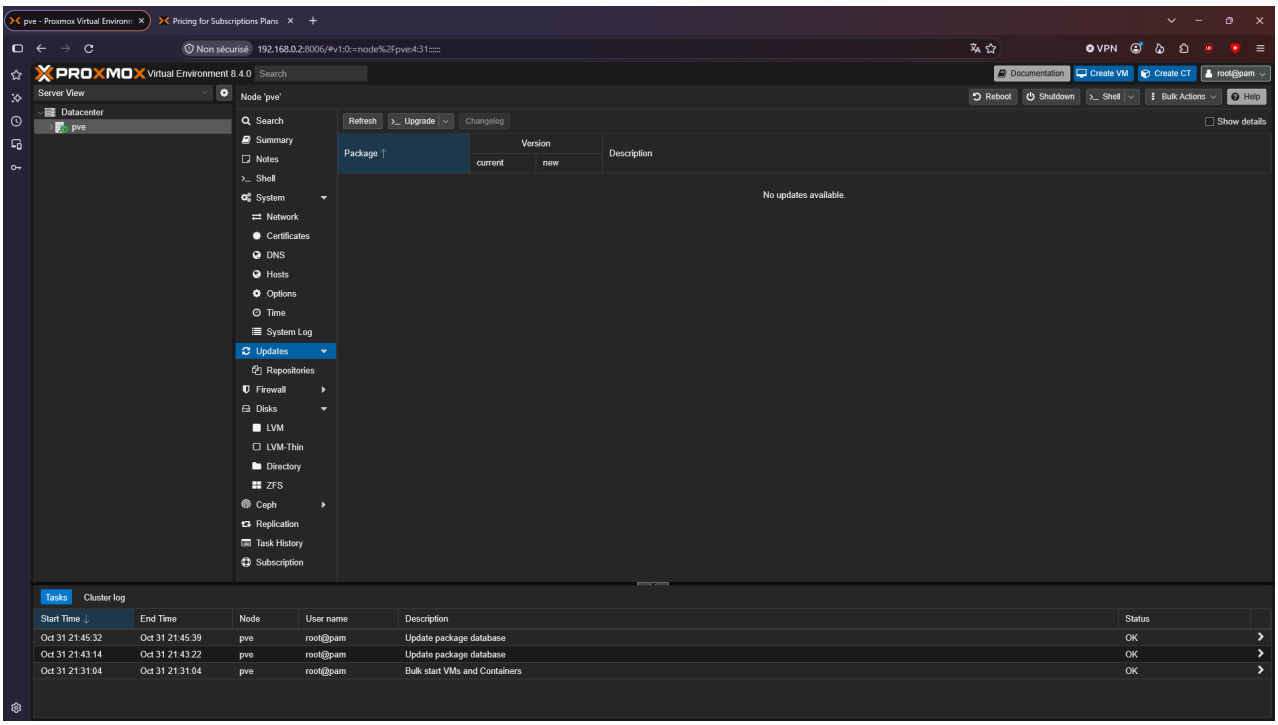
Désactivez les composants **enterprise** et **pve-enterprise** :



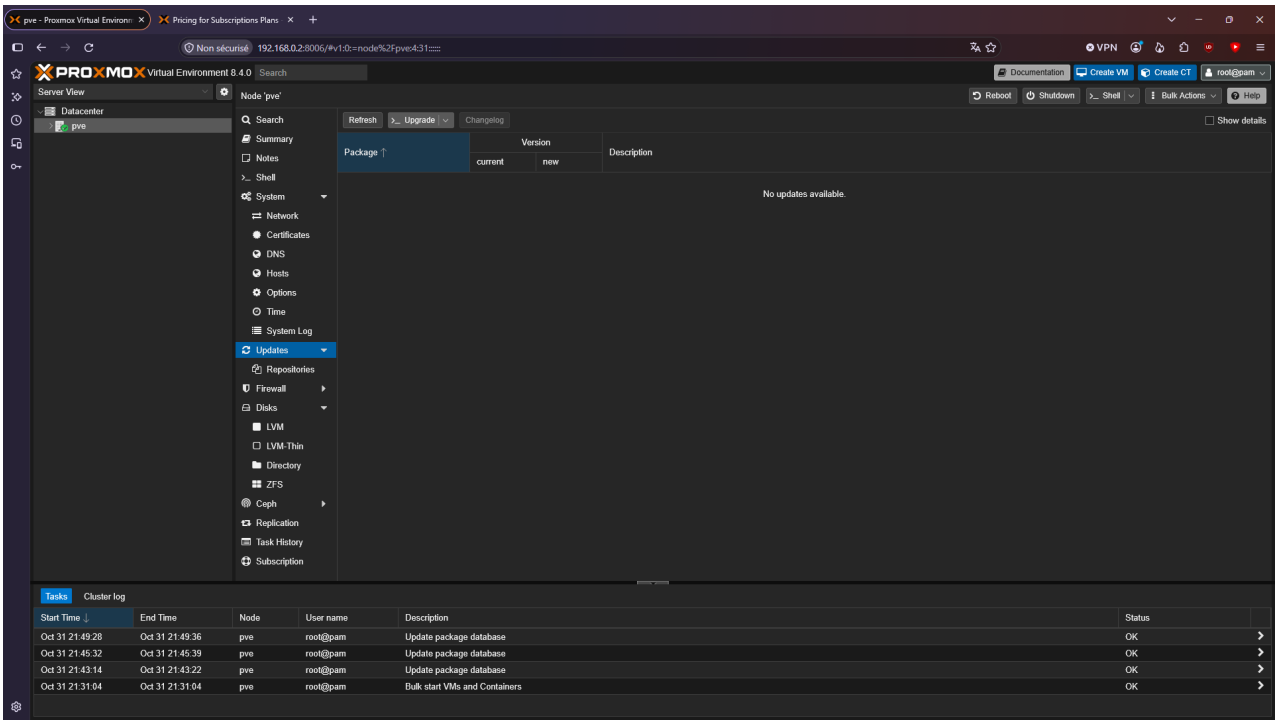
Cliquez sur **Add**, puis choisissez **No-Subscription** :

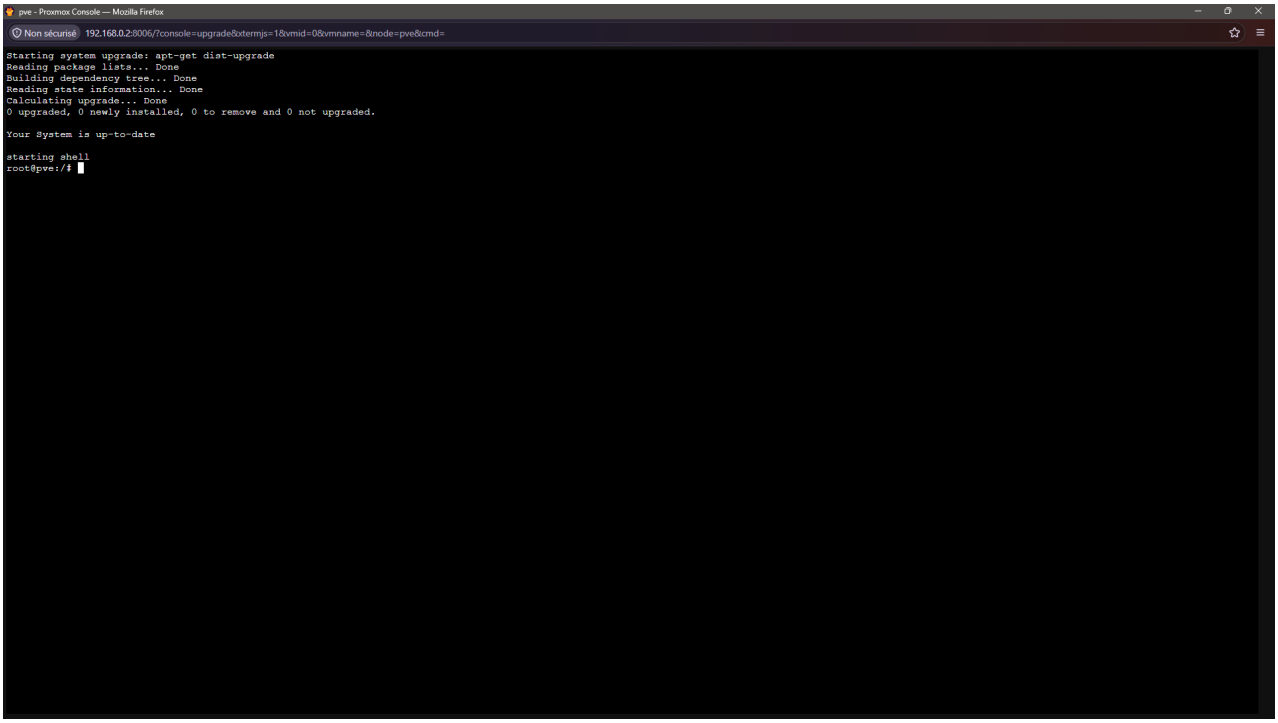


Allez ensuite dans **Updates** puis cliquez sur **Refresh** :



Puis cliquez sur **Update** :

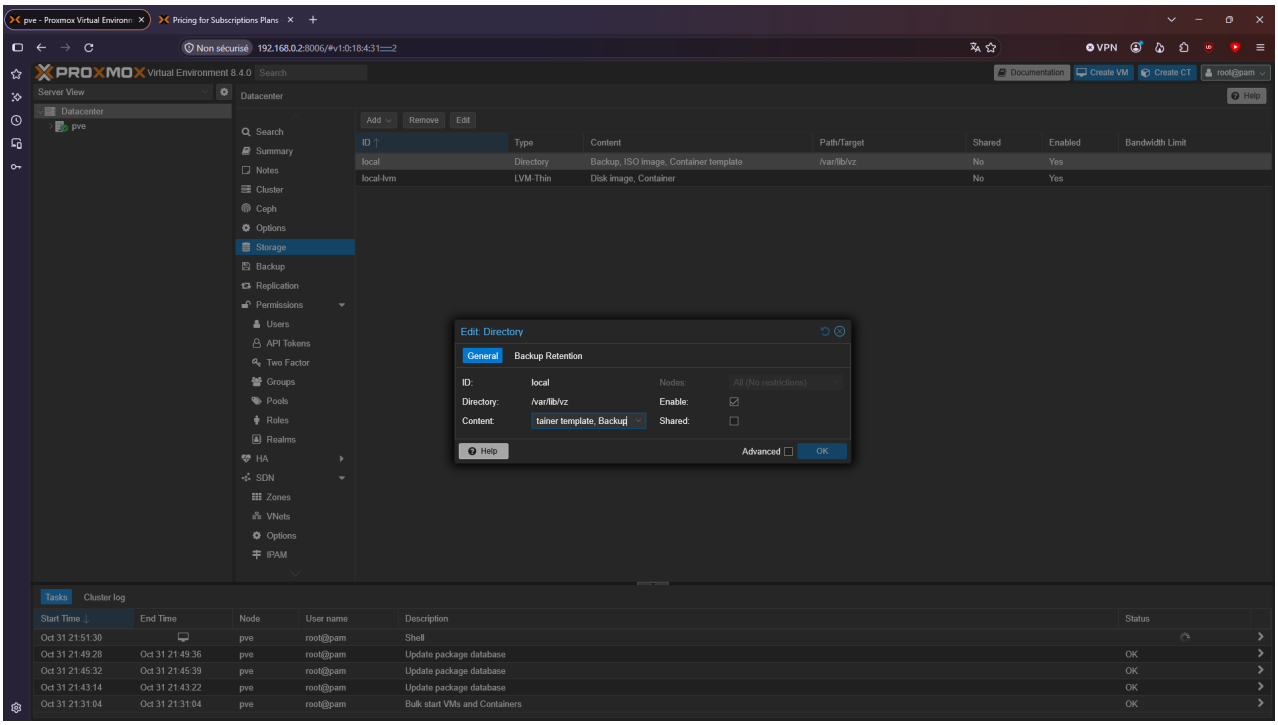




Nous avons ainsi installer et mis à jour Proxmox VE, pensez à exporter la VM comme Template du projet.

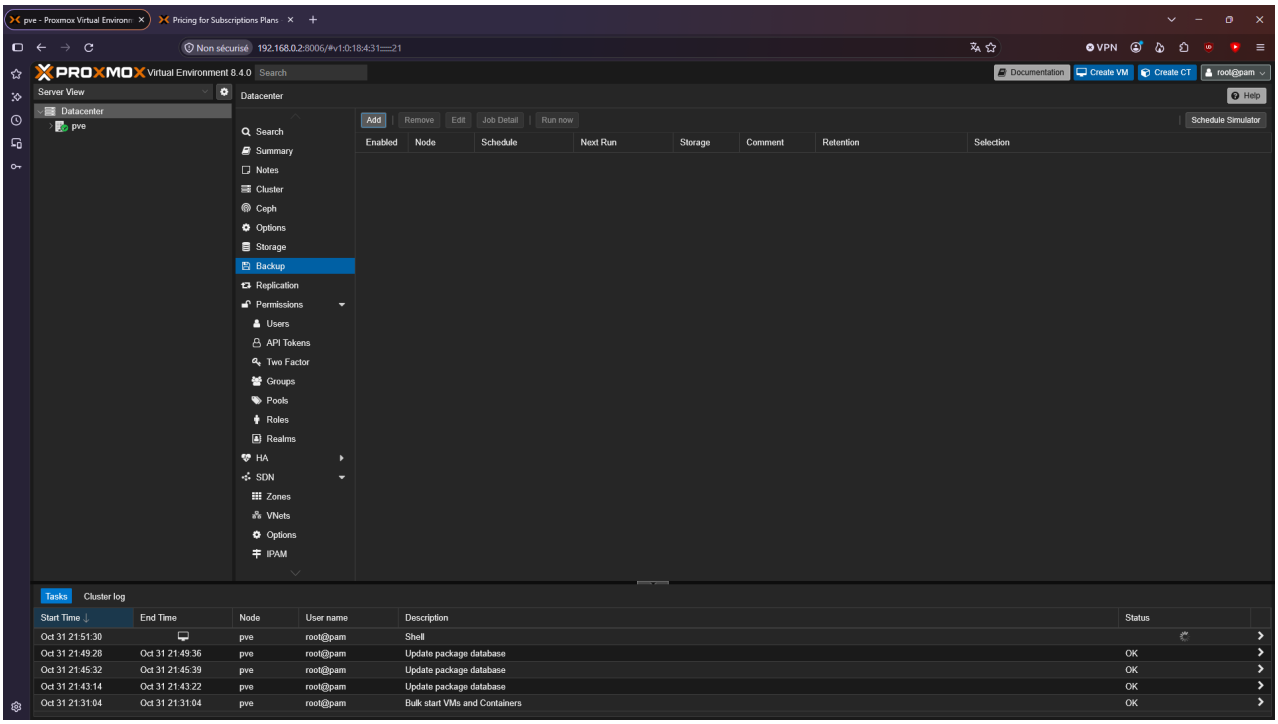
3.2. Configurer les sauvegardes de secours sur Proxmox

Nous allons maintenant configurer les sauvegardes des VM sur le disque virtuel interne de Proxmox. Pour cela, allez dans **Datacenter -> Storage -> local -> Edit** :

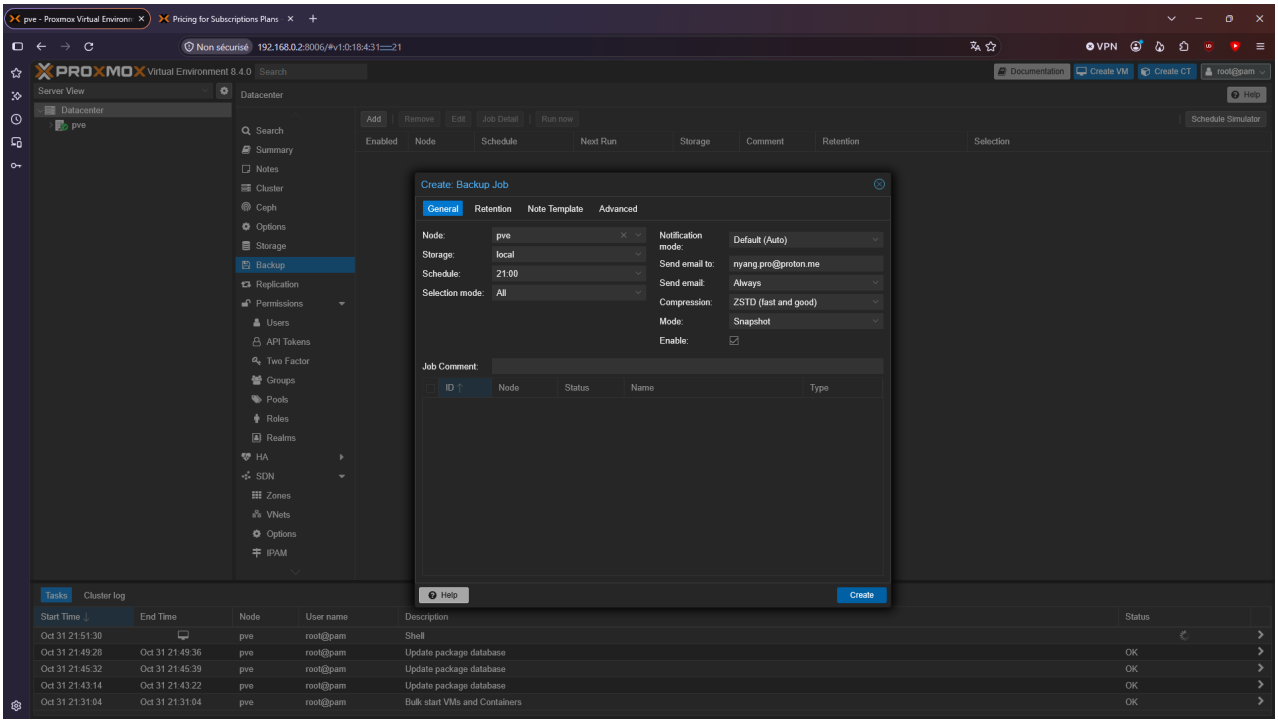


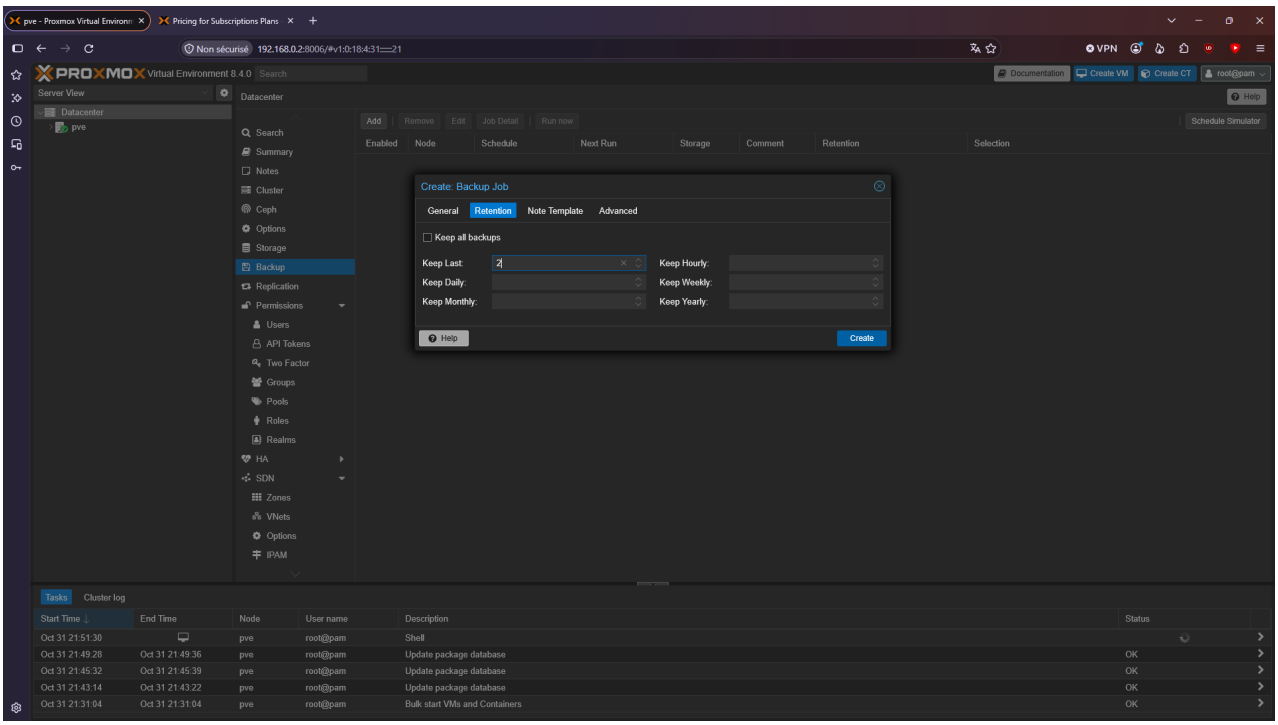
Assurez-vous que **Content** contient **Backup** avant de procéder à l'étape suivante.

Ouvrez **Datacenter -> Backup -> Add** :

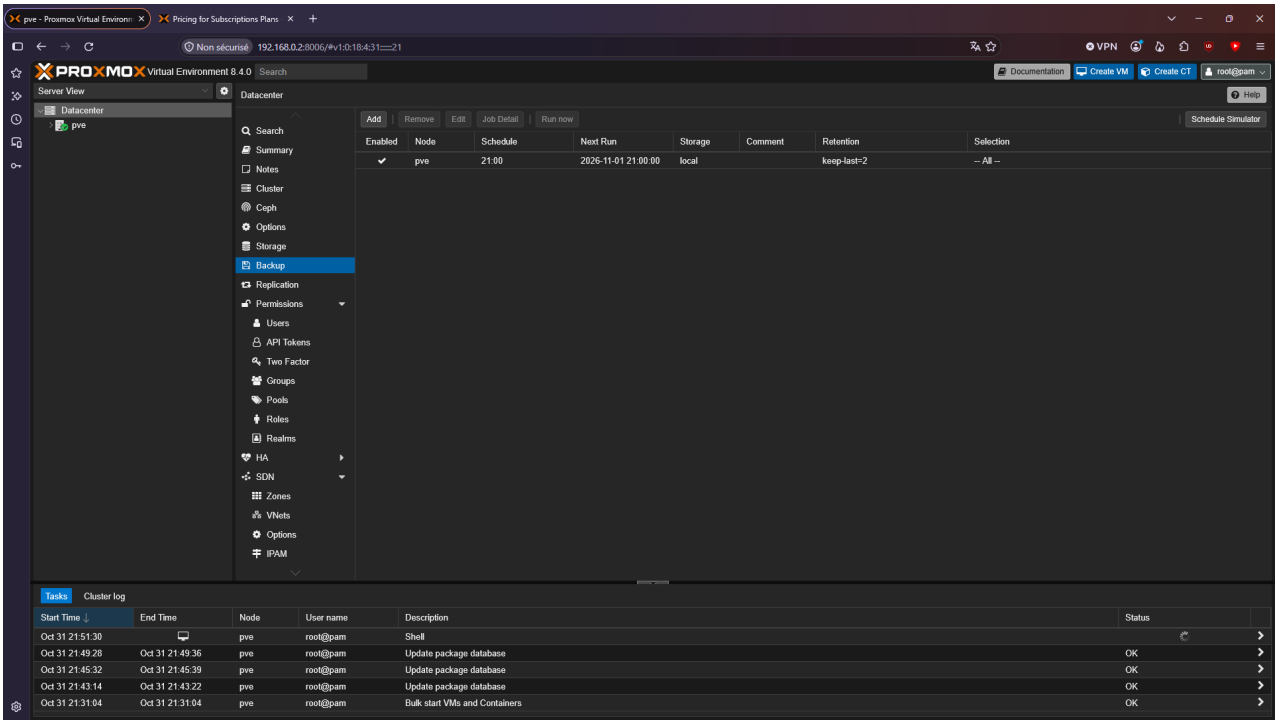


Ajoutez les paramètres suivants :





Puis créez la sauvegarde de secours.



Vous avez donc mis en place une sauvegarde de secours pour vos VM automatiquement.

3.3. Activer la transmission des VLAN (WIP)

Sur la vSwitch de OPNsense, exécutez les commandes suivantes (le script de l'installation PVE fait déjà ça) :

```
# Allow tagged VLANs 20 and 30
Set-VMNetworkAdapterVlan -VMName "pve" `
    -VMNetworkAdapterName "Network Adapter" `
    -Trunk -AllowedVlanIdList "20,30" -NativeVlanId 0
```

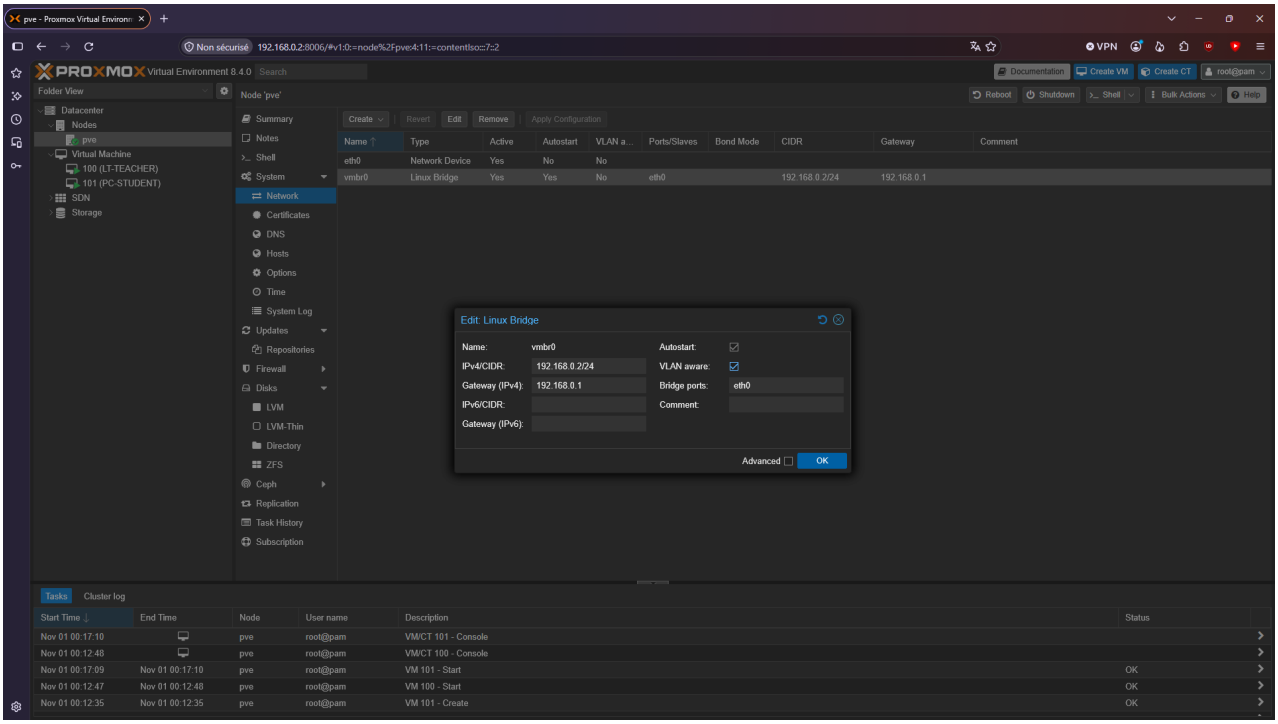
Vérifiez ensuite avec :

```
Get-VMNetworkAdapterVlan -VMName "pve"
```

Vous devriez obtenir :

```
VMName VMNetworkAdapterName Mode  VlanList
------ -------------------- ----  --------
pve    Network Adapter      Trunk 0,20,30
```

Puis sur Proxmox, allez dans le noeud pve -> System -> Network -> vmbr0, et configurez la comme ci-dessous :

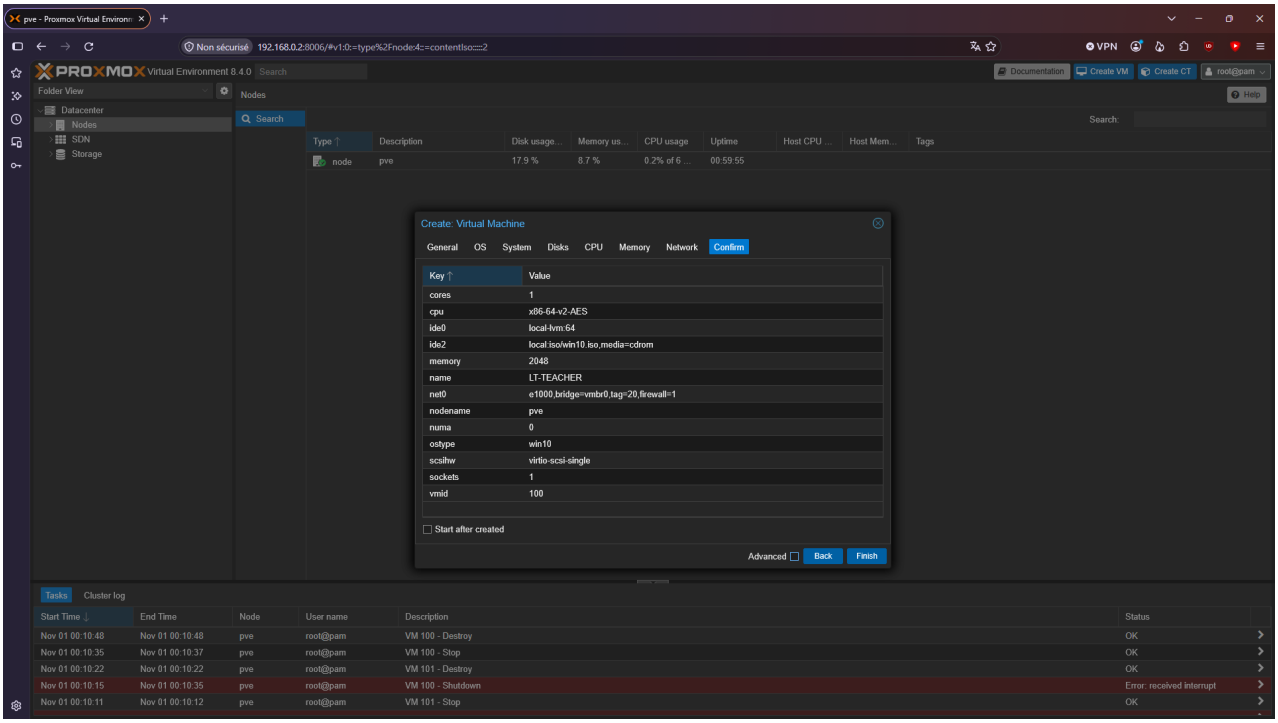


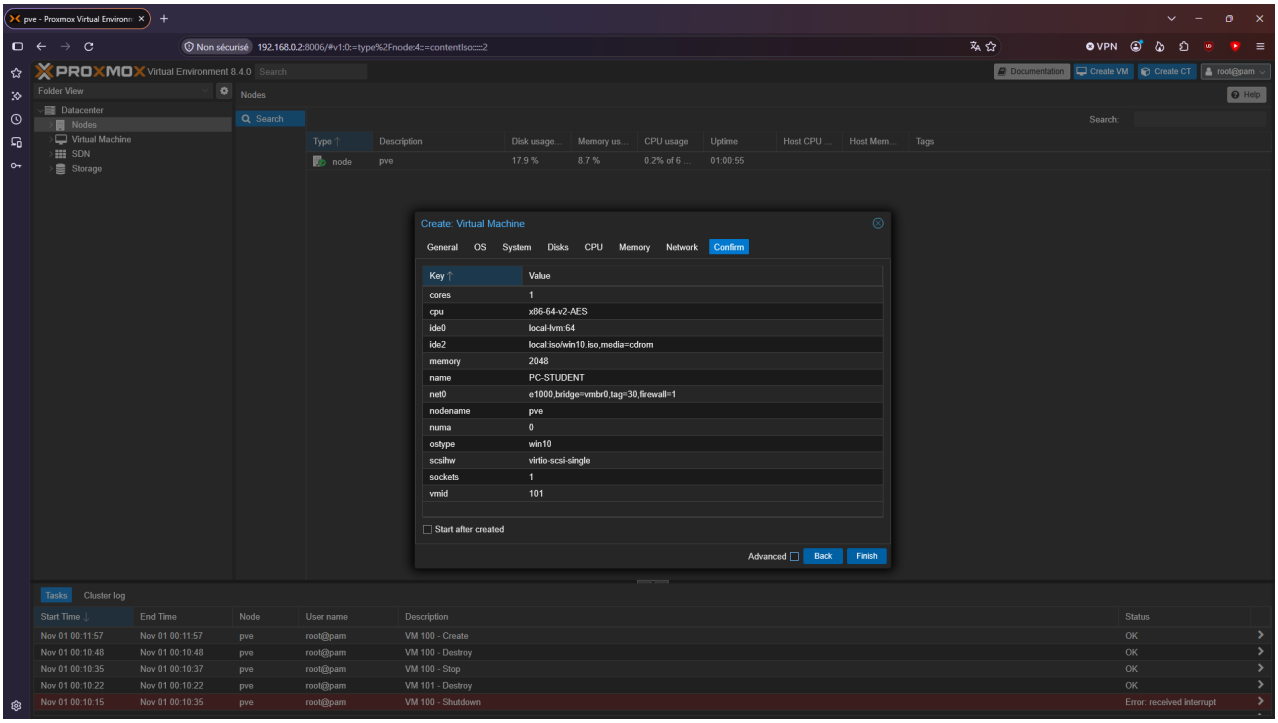
Création des VM Windows 10 sur Proxmox VE (REVOIR)

Nous allons mettre en place les postes de travail de l'infrastructure afin de vérifier le bon fonctionnement du serveur DHCP sur OPNsense et préparer le serveur DC.

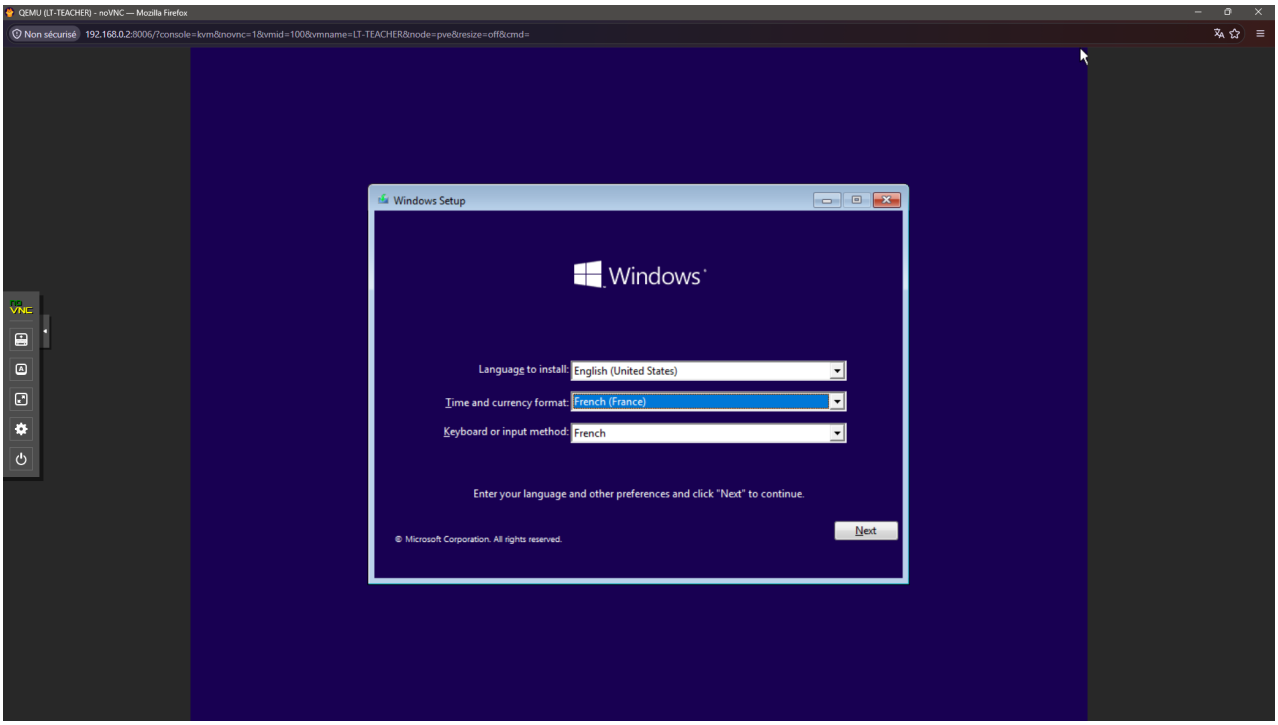
1. Créer les VM Windows 10 pour le professeur et étudiant

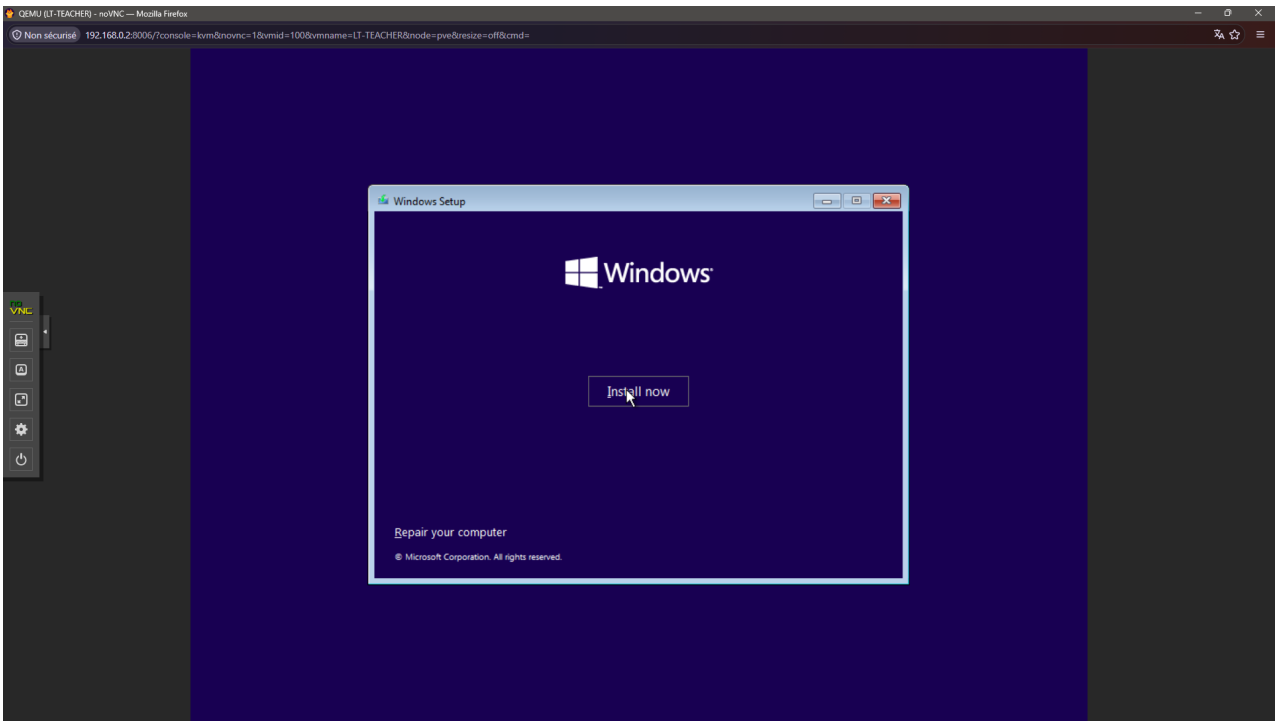
Nous allons installer deux VM Windows 10, une pour le professeur et l'autre pour l'élève. Récupérez d'abord l'ISO de Windows 10 à https://www.microsoft.com/fr-fr/software-download/windows10. Pour créer une VM Windows 10 sous Proxmox, cliquez sur **Create VM** et créez la configuration suivante :

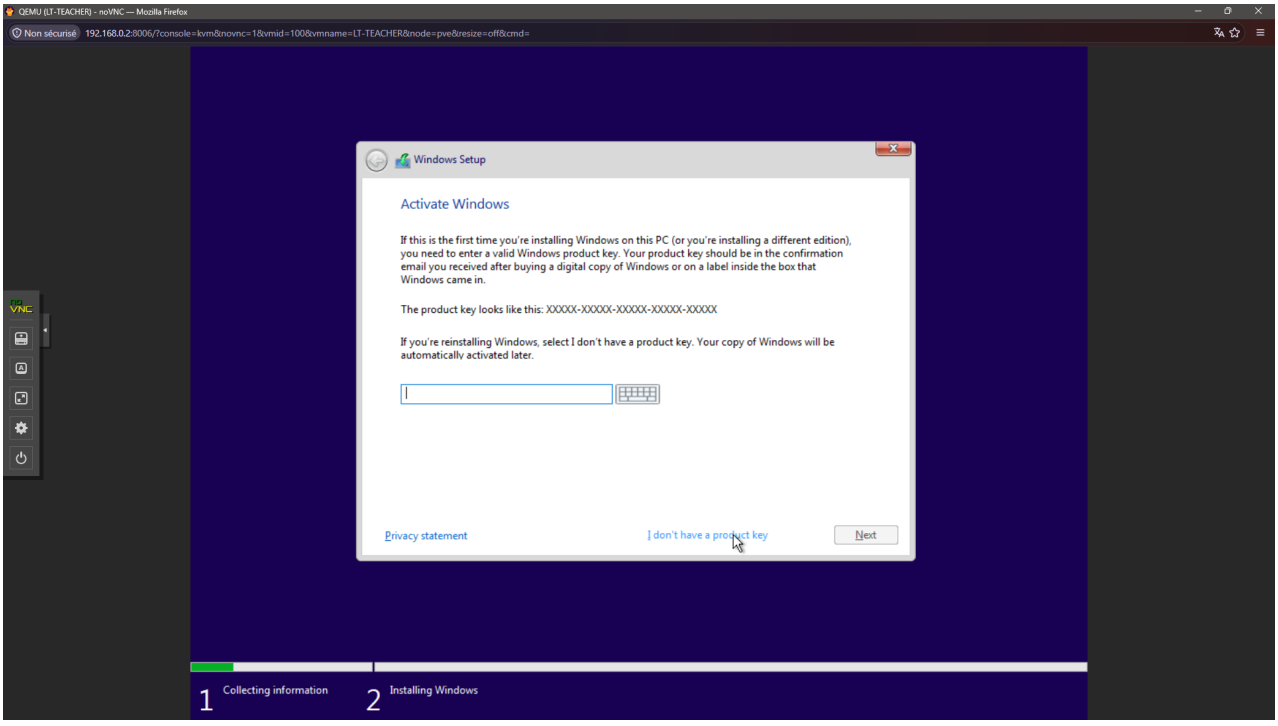




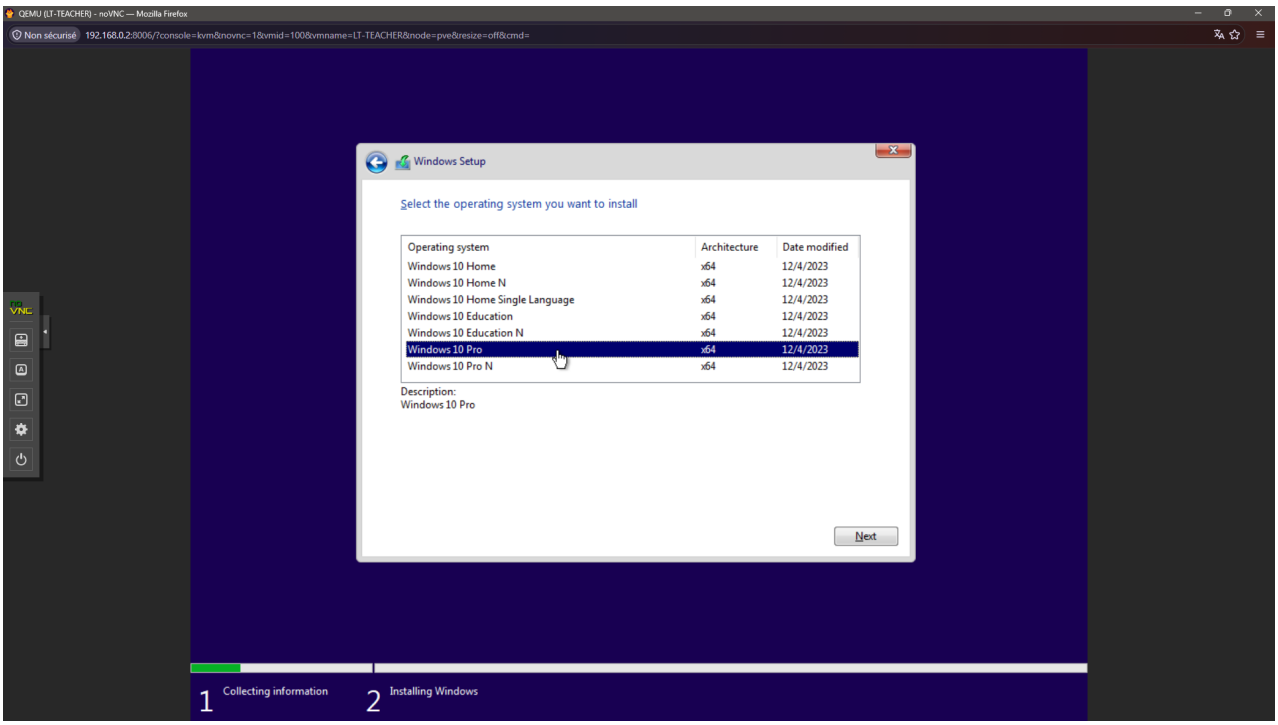
Lancez les VM avec un double clic jusqu'à que le démarrage arrive à **Windows Setup**. Suivez la configuration suivante :

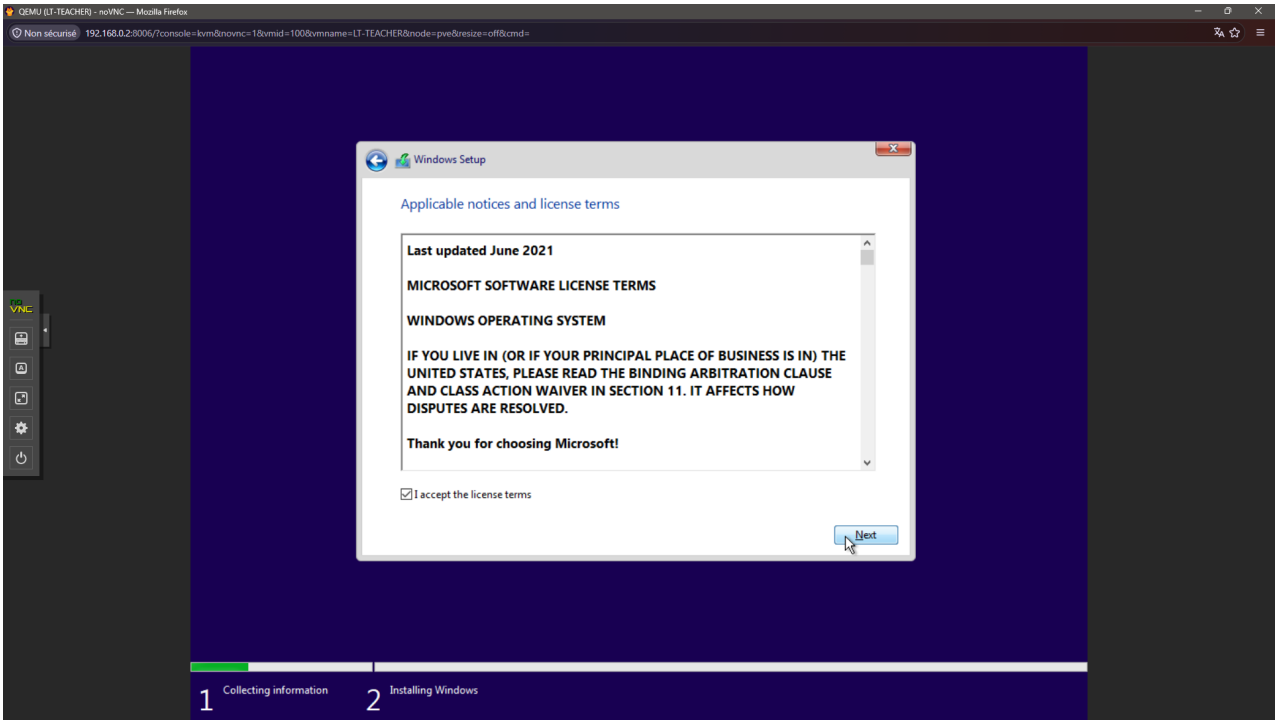


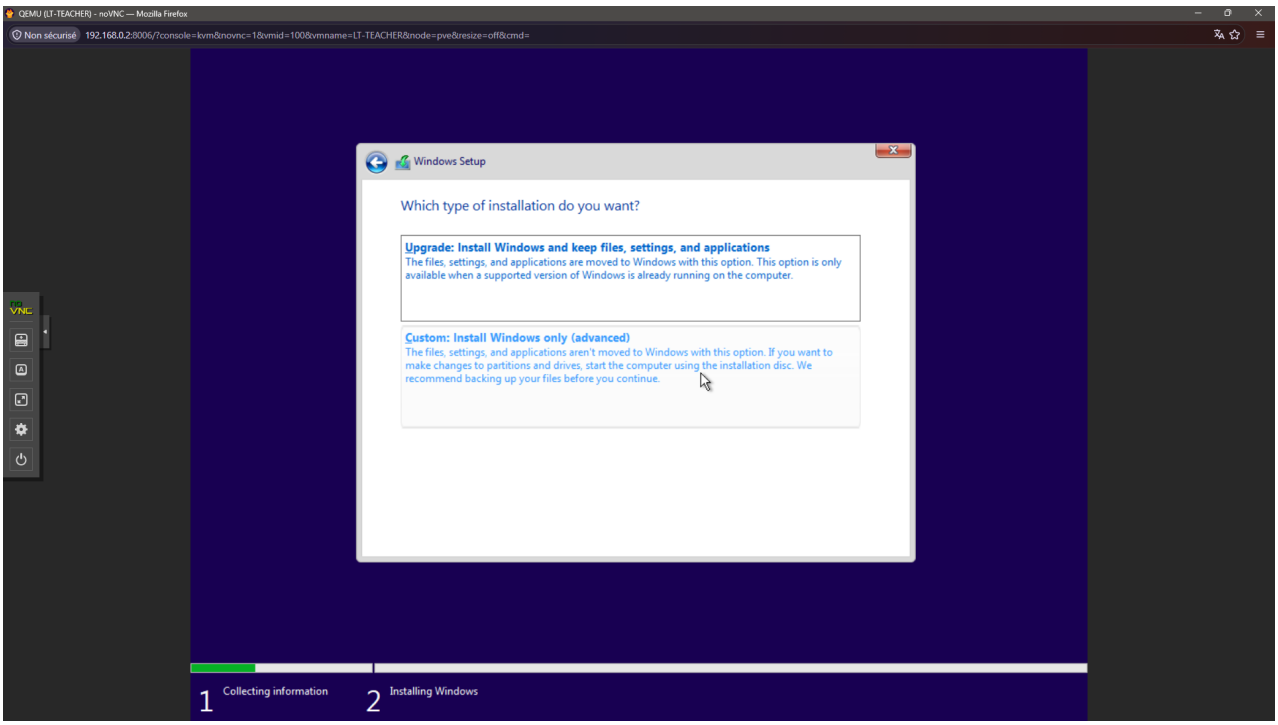




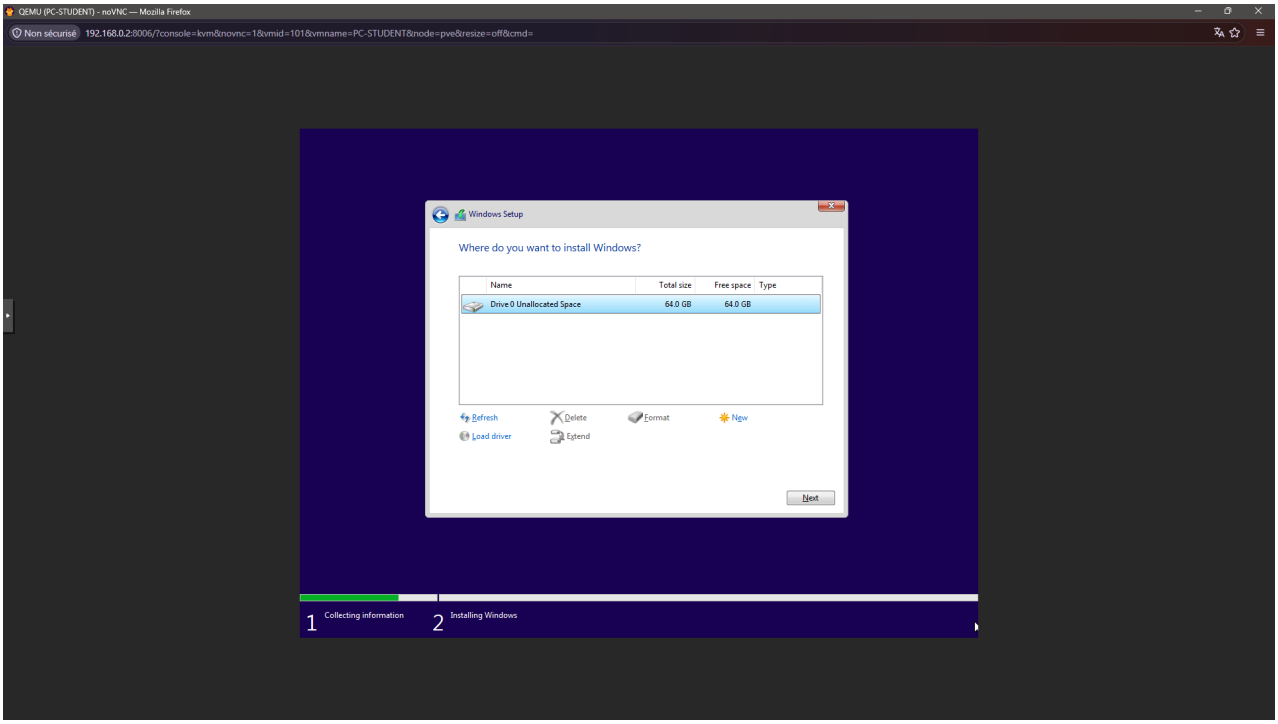
Choisissez Windows 10 Pro :

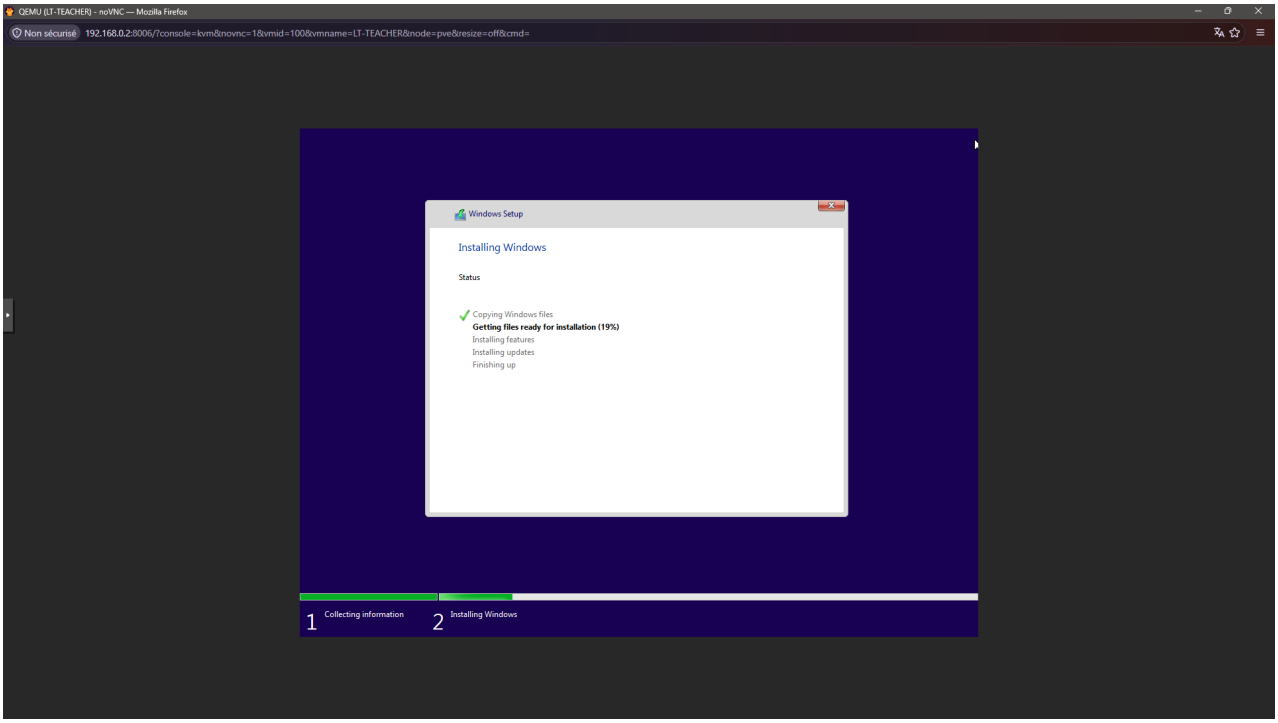

QEMU (LT-TEACHER) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=lvm&novnc=1&vmid=100&vmname=LT-TEACHER&node=pve&resize=off&cmd=
Windows Setup
Select the operating system you want to install
Operating system
Architecture
Date modified
Windows 10 Home x64 12/4/2023
Windows 10 Home N x64 12/4/2023
Windows 10 Home Single Language x64 12/4/2023
Windows 10 Education x64 12/4/2023
Windows 10 Education N x64 12/4/2023
Windows 10 Pro x64 12/4/2023
Windows 10 Pro N x64 12/4/2023
Description:
Windows 10 Pro
Next
1 Collecting information
2 Installing Windows

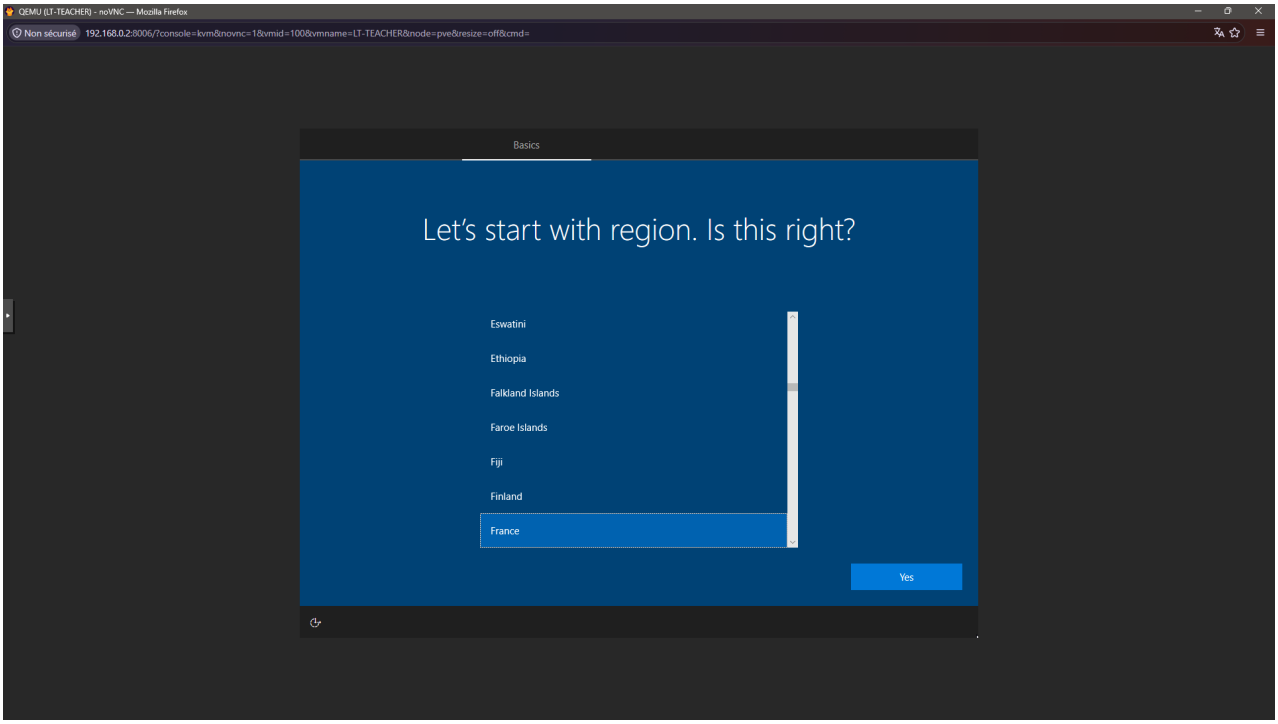

QEMU (LT-TEACHER) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=lvm&novnc=1&vmid=100&vmname=LT-TEACHER&node=pve&resize=off&cmd=
Windows Setup
Applicable notices and license terms
Last updated June 2021
MICROSOFT SOFTWARE LICENSE TERMS
WINDOWS OPERATING SYSTEM
IF YOU LIVE IN (OR IF YOUR PRINCIPAL PLACE OF BUSINESS IS IN) THE UNITED STATES, PLEASE READ THE BINDING ARBITRATION CLAUSE AND CLASS ACTION WAIVER IN SECTION 11. IT AFFECTS HOW DISPUTES ARE RESOLVED.
Thank you for choosing Microsoft!
I accept the license terms
Next
1 Collecting information
2 Installing Windows

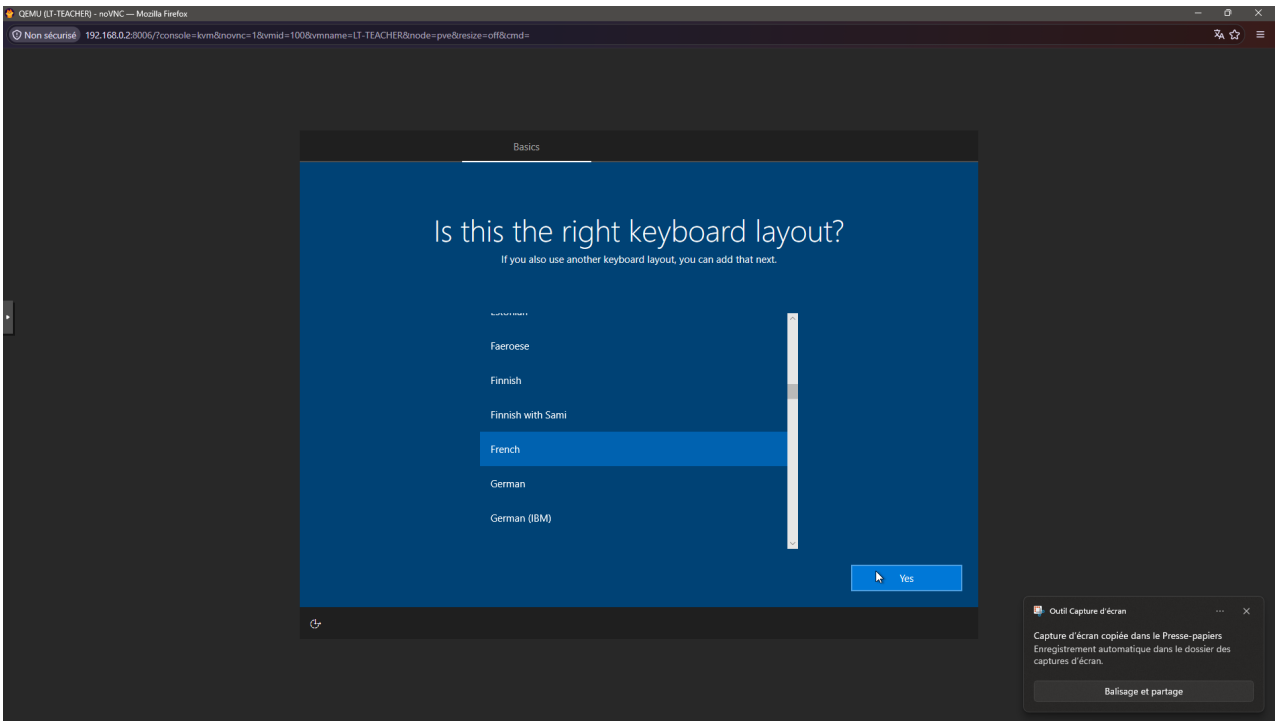


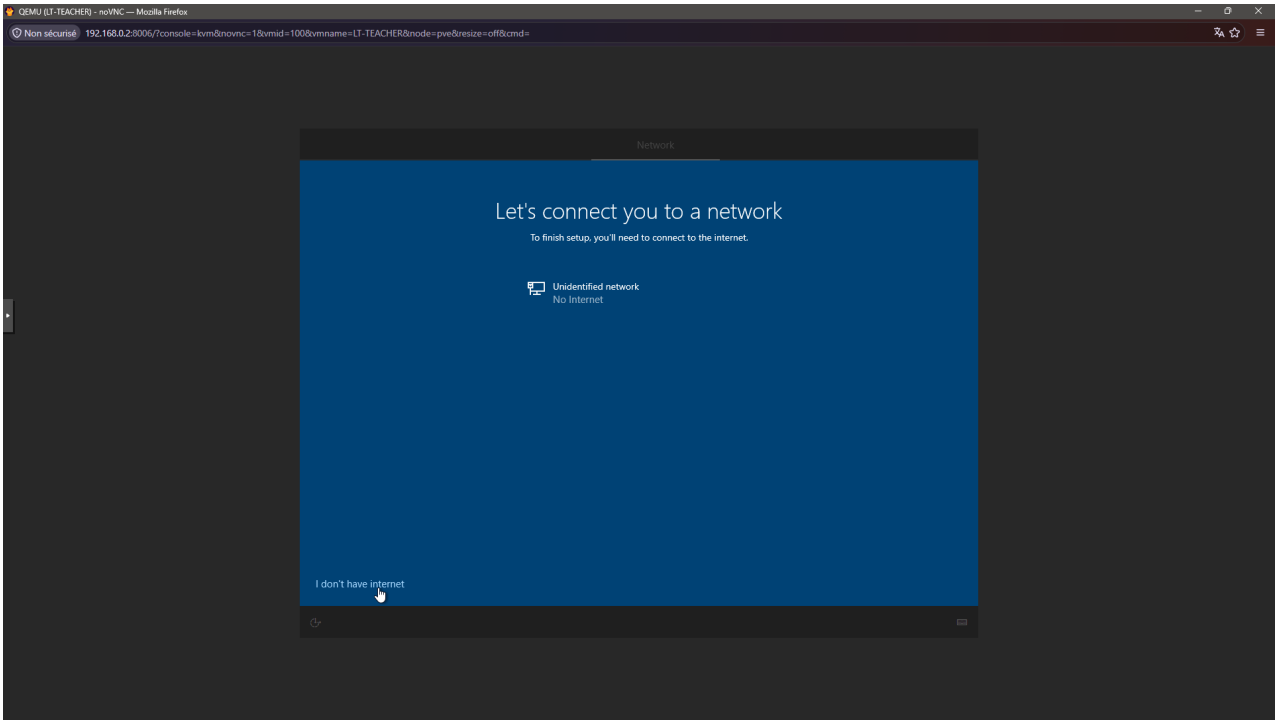
Choisir l'unique vHDD de la VM puis démarrez l'installation :

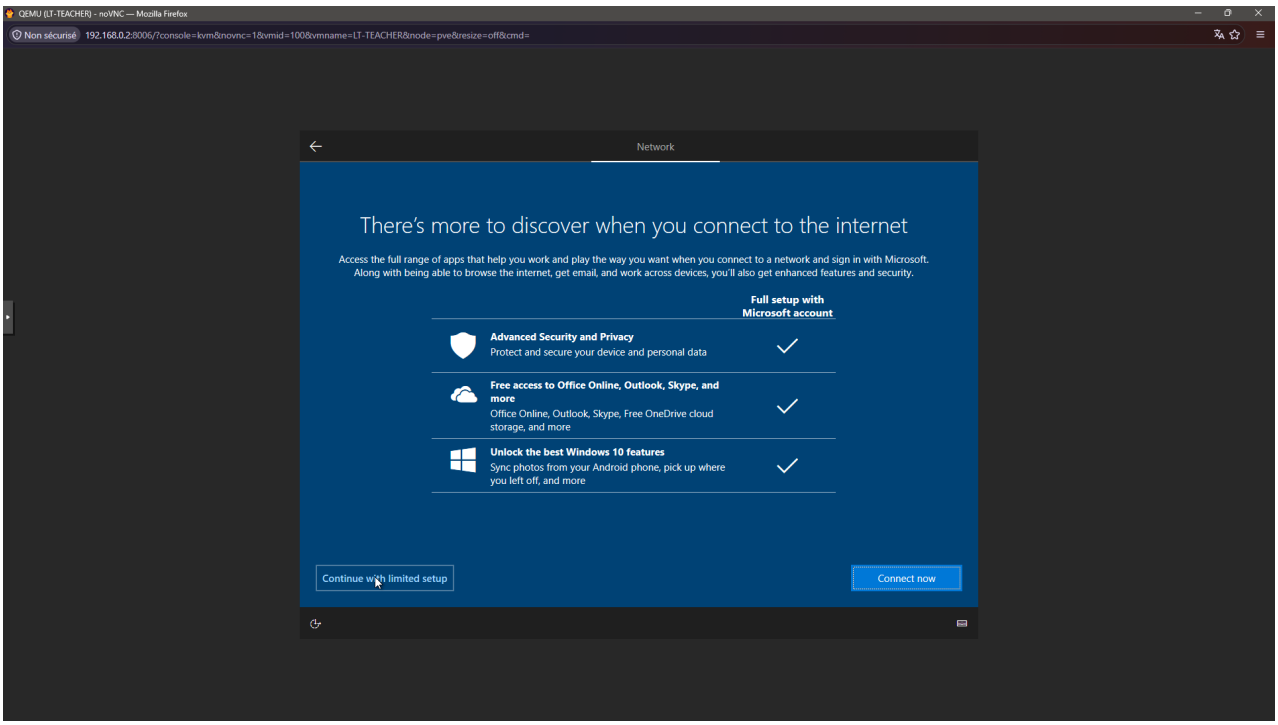



QEMU (LT-TEACHER) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=100&vmname=LT-TEACHER&node=pve&resize=off&cmd=
Windows Setup
Installing Windows
Status
Copying Windows files
Getting files ready for installation (19%)
Installing features
Installing updates
Finishing up
1 Collecting information
2 Installing Windows

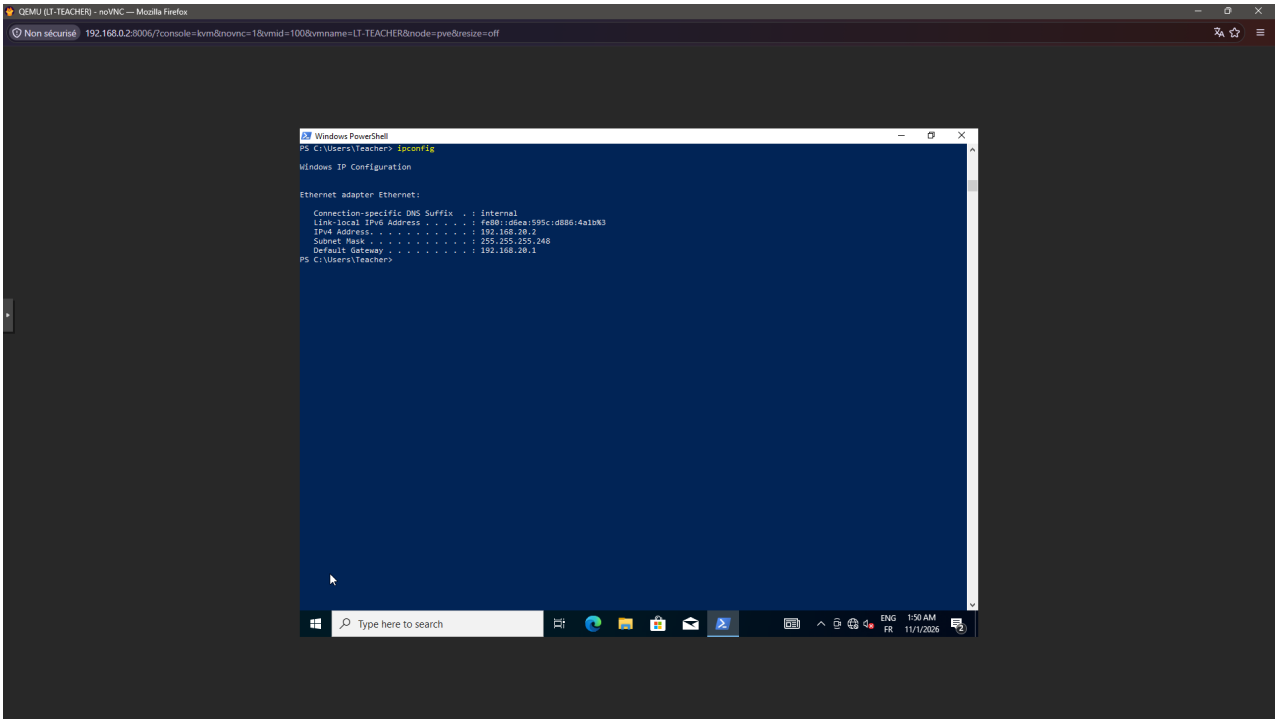

QEMU (LT-TEACHER) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=100&vmname=LT-TEACHER&node=pve&resize=off&cmd=
Basics
Let's start with region. Is this right?
Eswatini
Ethiopia
Falkland Islands
Faroe Islands
Fiji
Finland
France
Yes

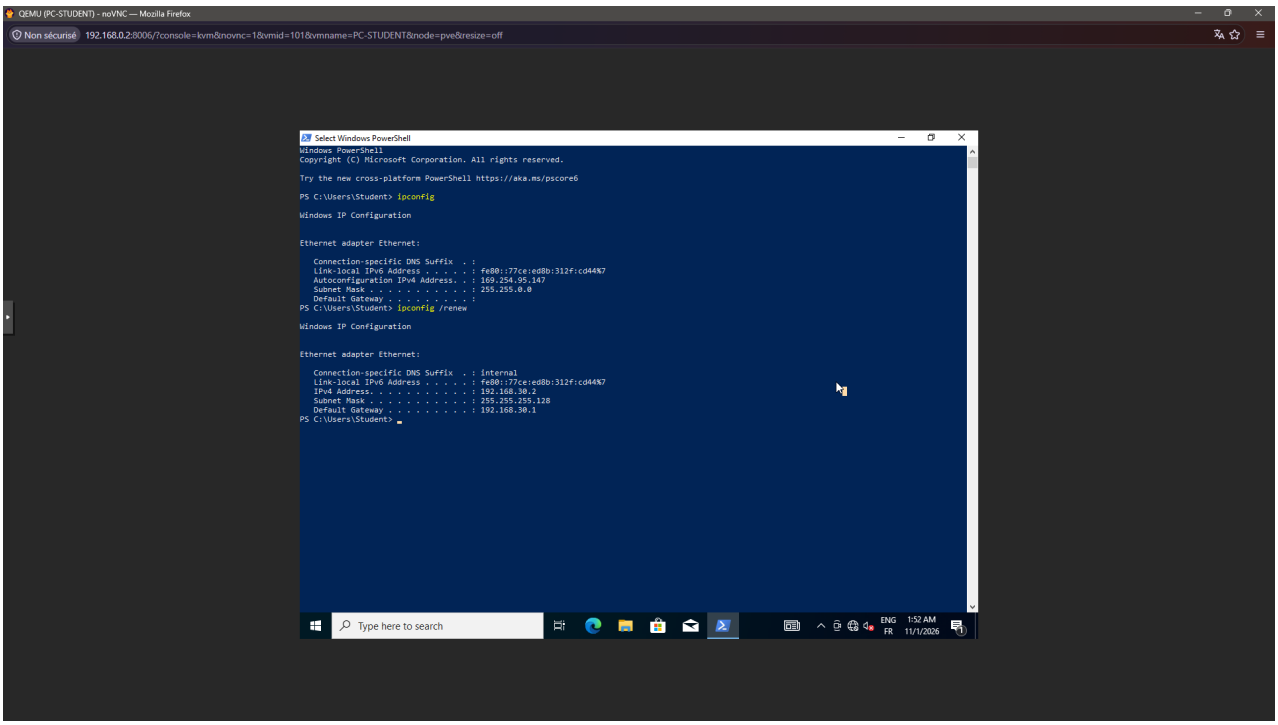

QEMU (LT-TEACHER) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=100&vmname=LT-TEACHER&node=pve&resize=off&cmd=
Basics
Is this the right keyboard layout?
If you also use another keyboard layout, you can add that next.
Faeroese
Finnish
Finnish with Sami
French
German
German (IBM)
Yes
Outil Capture d'écran
Capture d'écran copiée dans le Presse-papiers
Enregistrement automatique dans le dossier des captures d'écran.
Balisage et partage


QEMU (LT-TEACHER) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=100&vmname=LT-TEACHER&node=pve&resize=off&cmd=
Network
Let's connect you to a network
To finish setup, you'll need to connect to the internet.
Unidentified network
No Internet
I don't have internet



Dès que l'installation est terminé et que vous vous retrouvez sur le bureau, vérifiez si vous récupérer votre configuration réseau en DHCP avec **ipconfig**:





Mettez ensuite à jour les deux VM, nous configurons Veyon sur elles plus tard. Nous allons passer à la VM du serveur DC.

2. Configurer les VM (WIP)

Exécutez la commande suivante pour renommer chaque VM :

```
# Teacher's laptop
Rename-Computer -NewName "LT-KIFORMATOU" -Restart

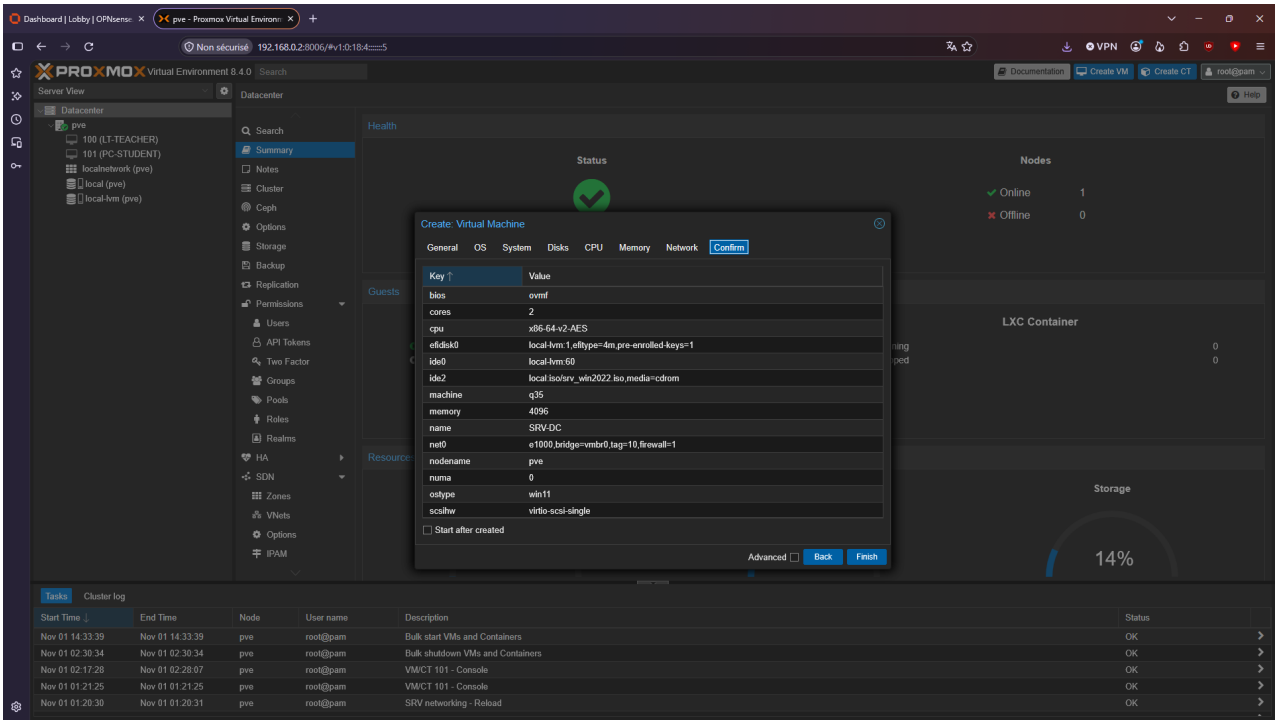

# Student's PC
Rename-Computer -NewName "PC-KIFORMATOU" -Restart
```

Création du serveur Active Directory et DNS (WIP)

Nous allons mettre en place le serveur DC. Il servira à centraliser les comptes et les règles du LAN Kiformatou grâce à l'AD.

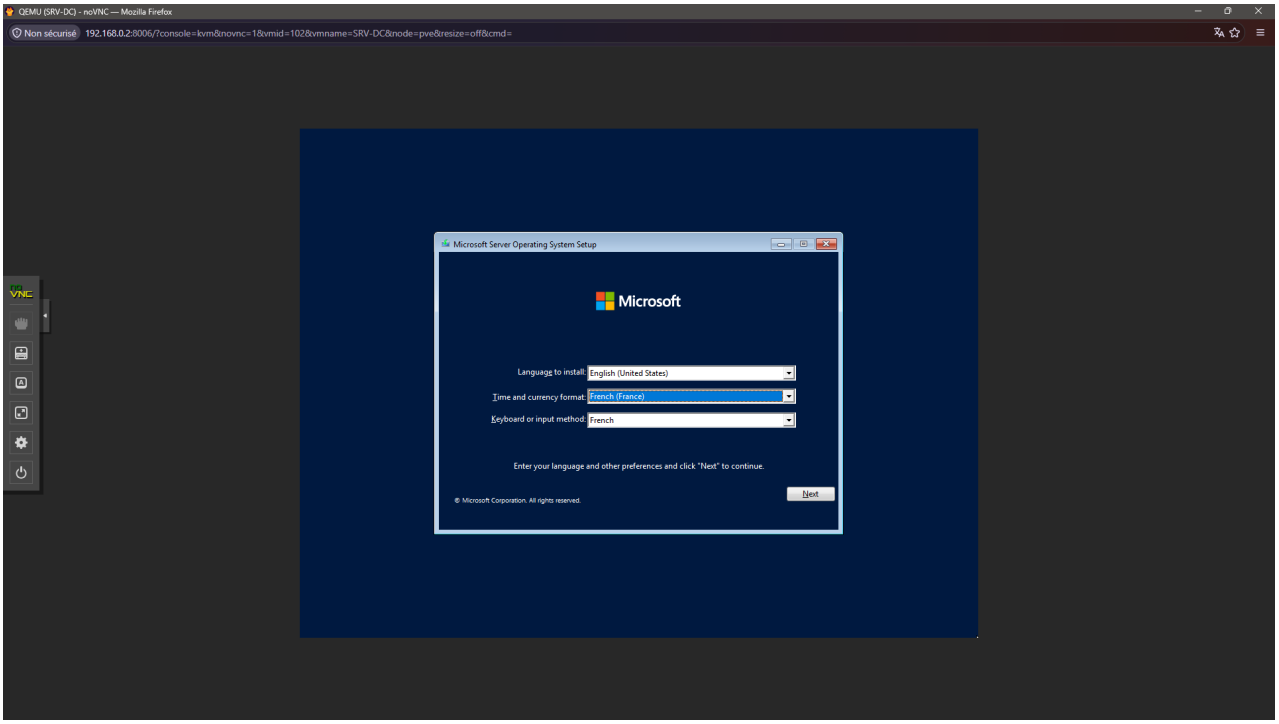
1. Prérequis

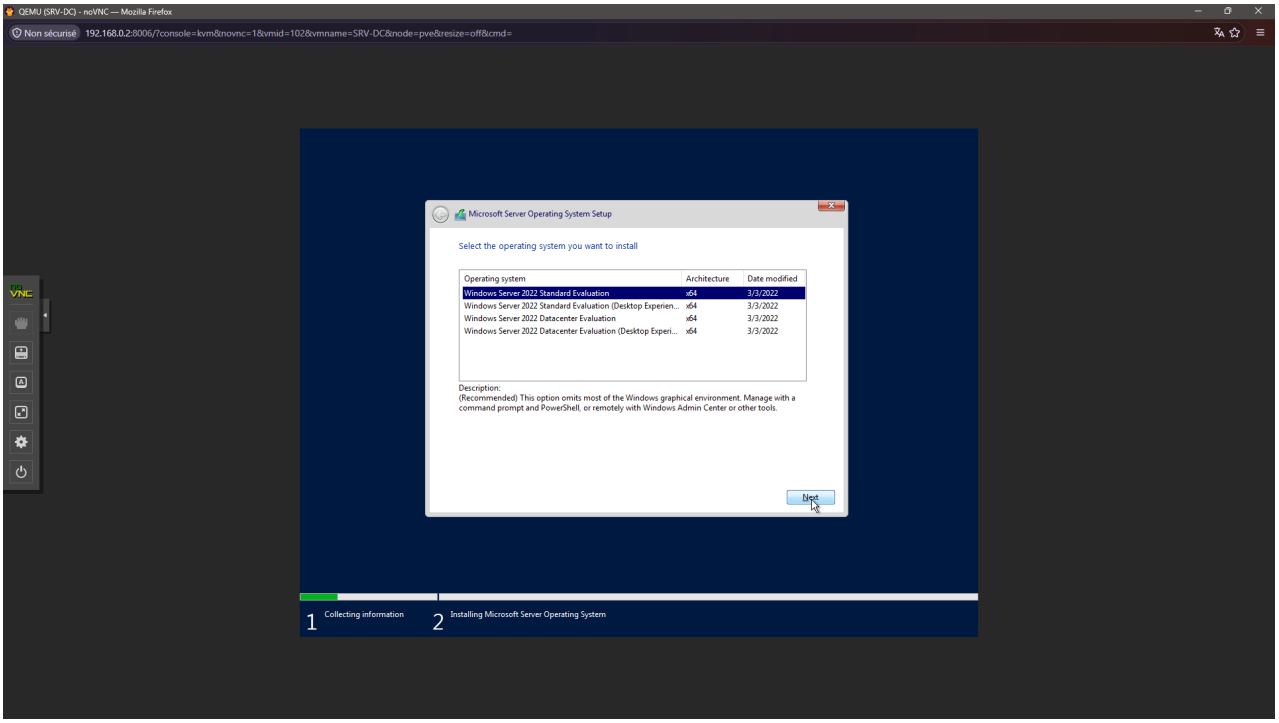
Récupérez l'ISO Windows Server 2022 à https://www.microsoft.com/fr-fr/evalcenter/download-windows-server-2022. Puis créez la VM avec la configuration suivante :



2. Installez la VM

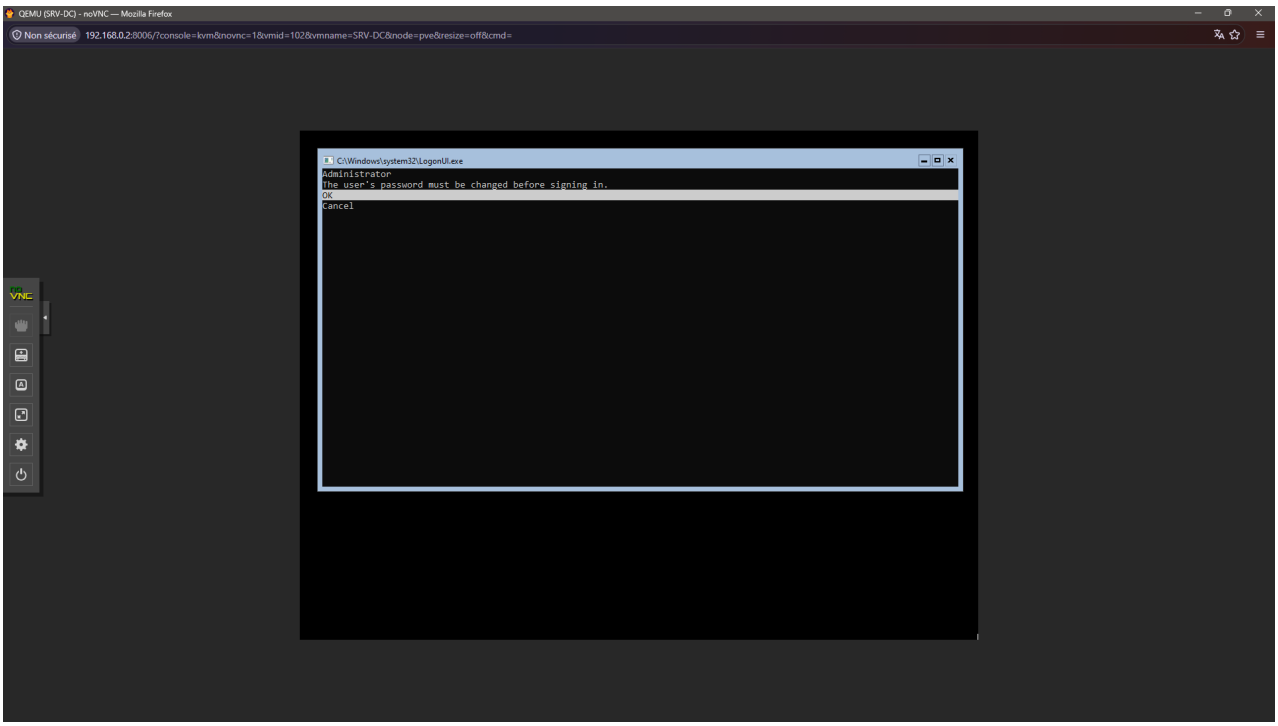
Démarrez la VM SRV-DC. Vous pouvez normalement vous débrouillez pour l'installation, assurez-vous juste de choisir la disposition de clavier AZERTY et la version standard pour avoir le serveur en mode core :



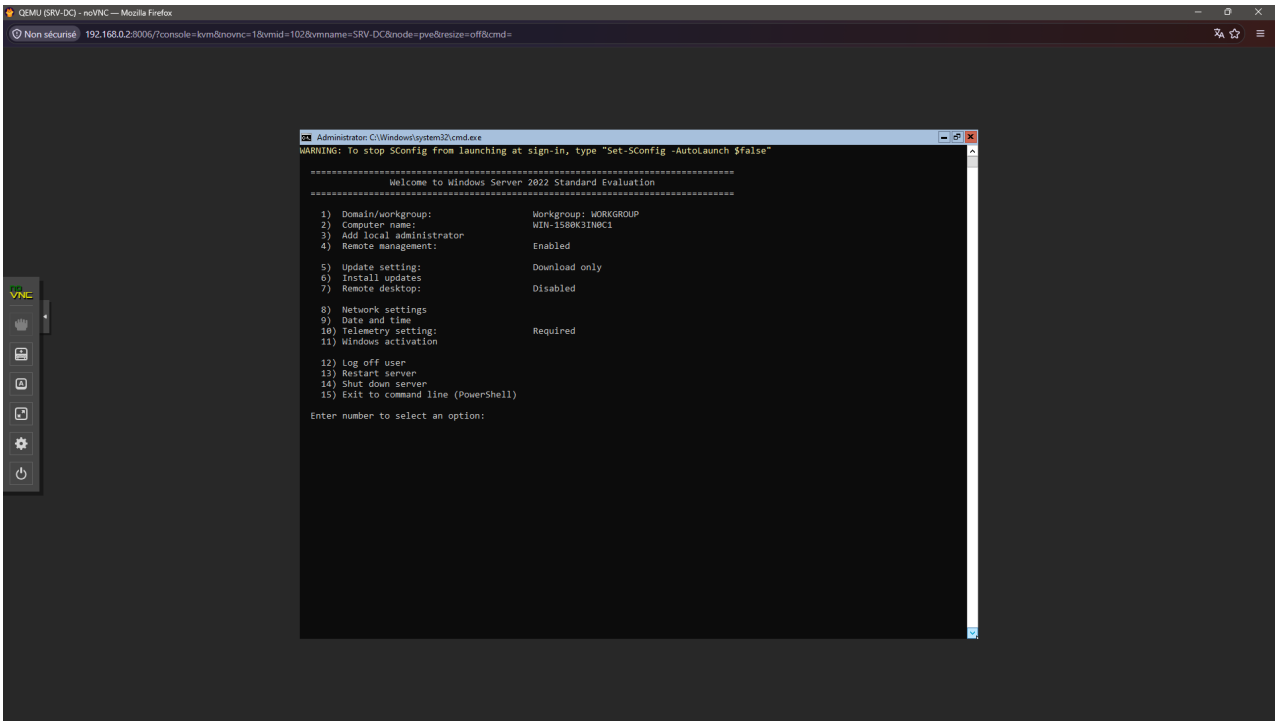


3. Configurer la VM

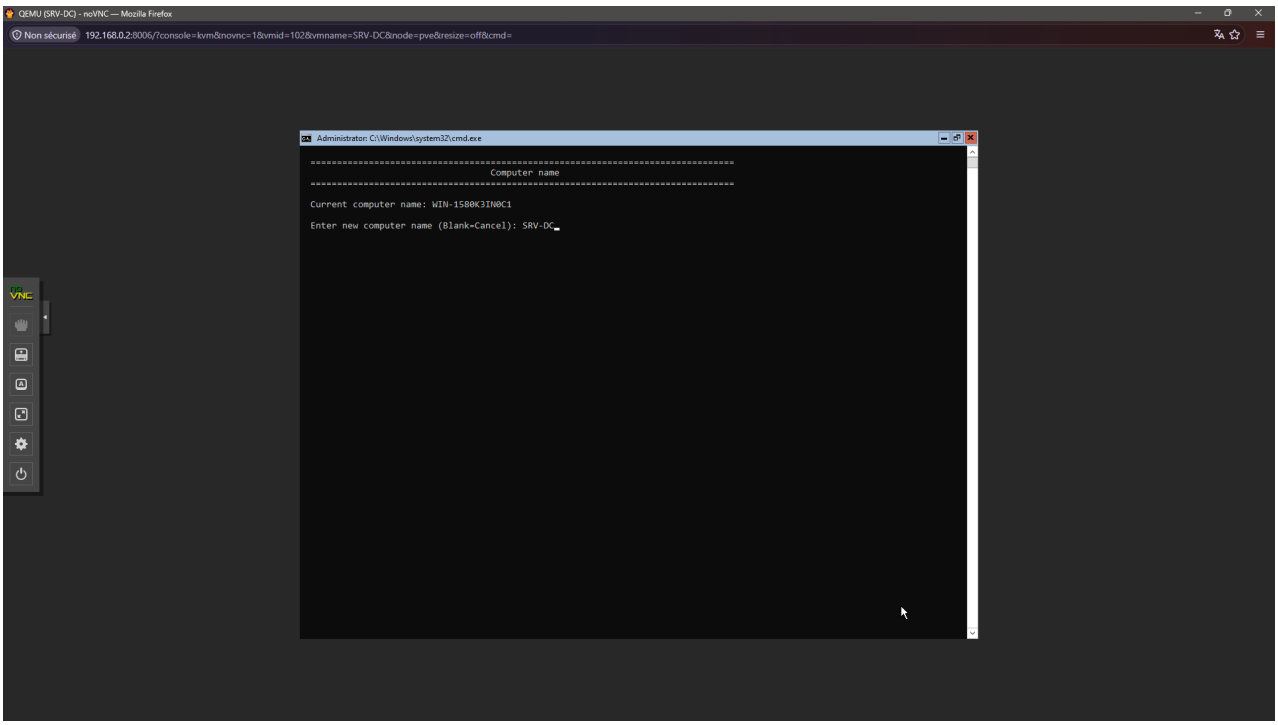
3.1. Changer le nom d'hôte

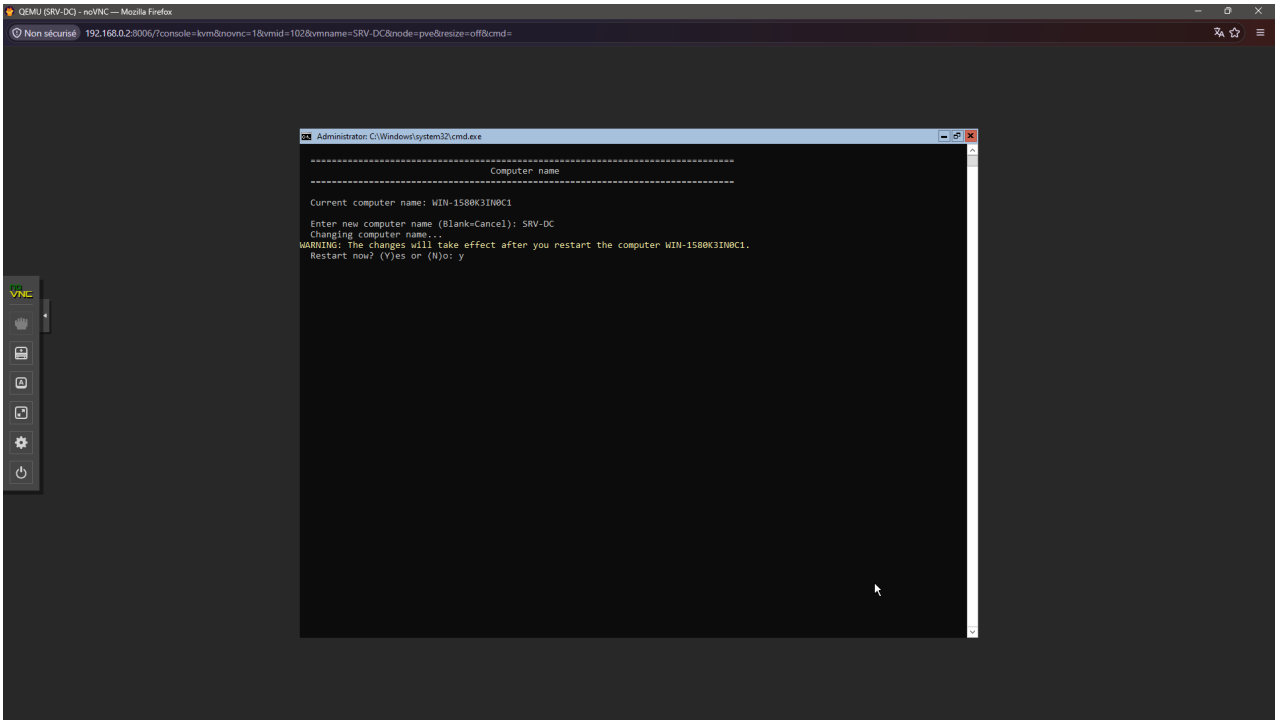


Créez votre mot de passe pour votre compte adminstrateur (le supprimer plus tard)



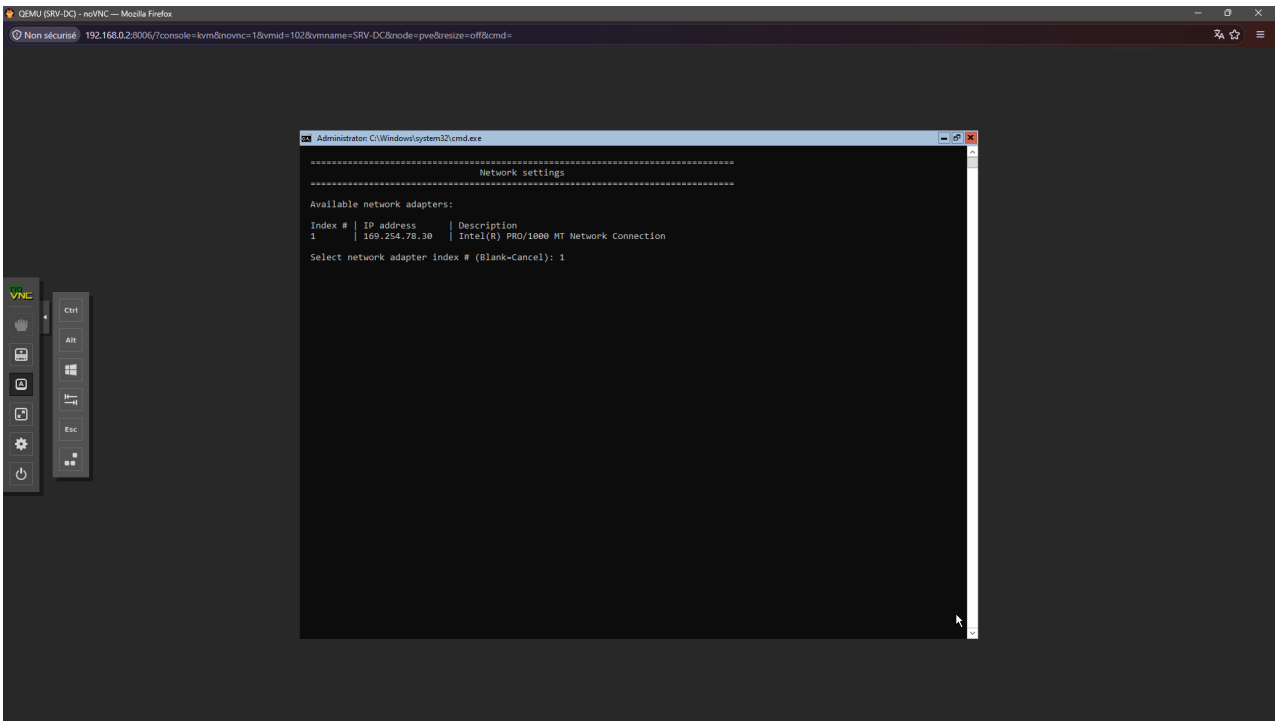
Sur le tableau de bord, modifiez le nom du serveur pour SRV-DC en tapant 2 :

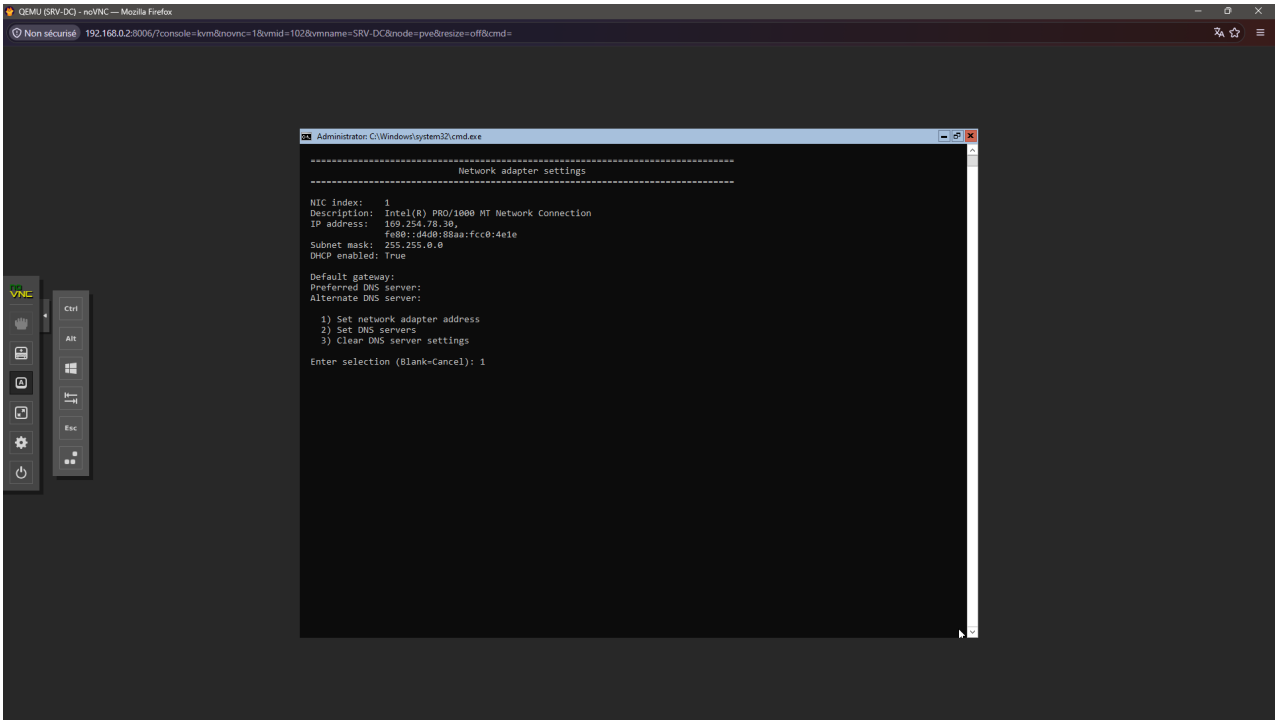


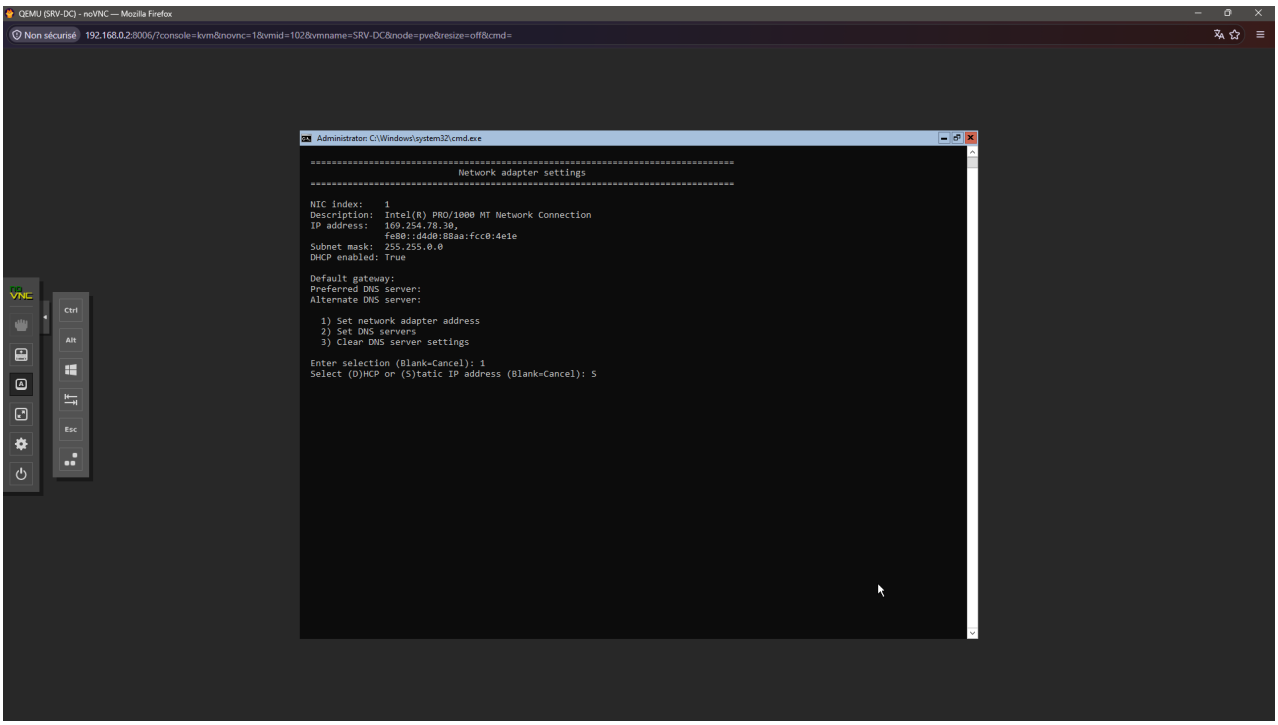


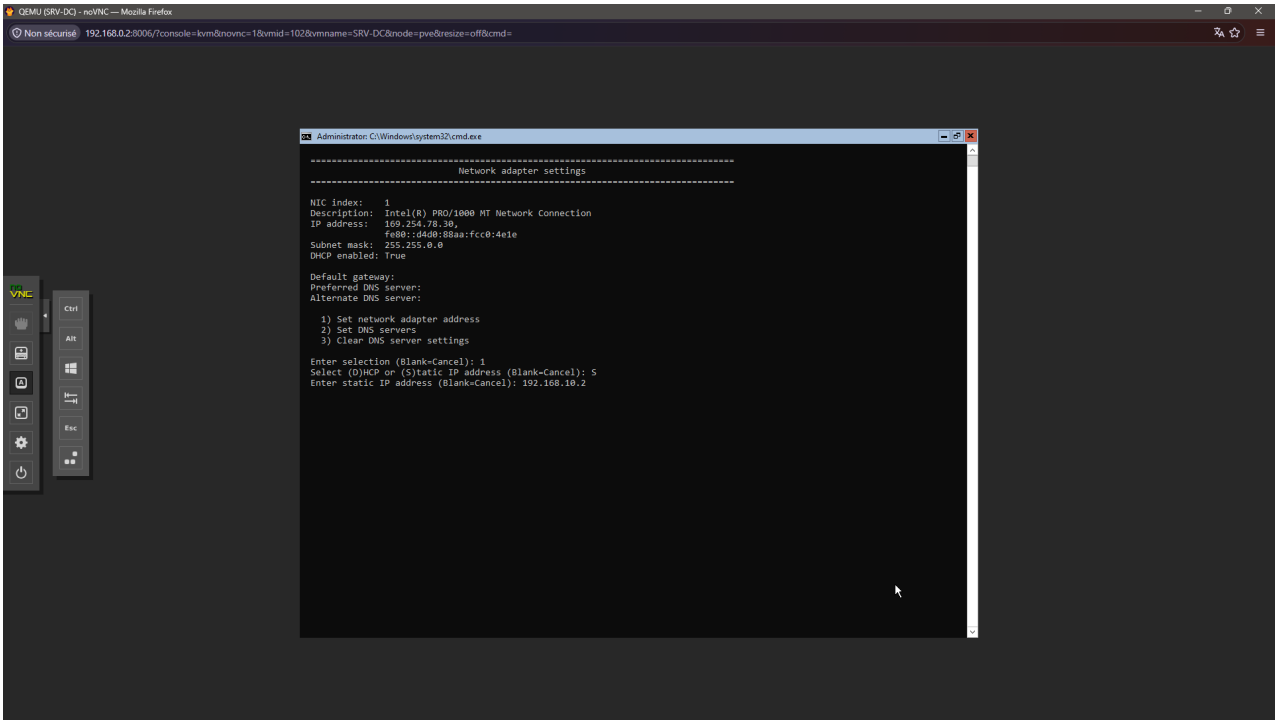
3.2. Mofifier la configuration réseau

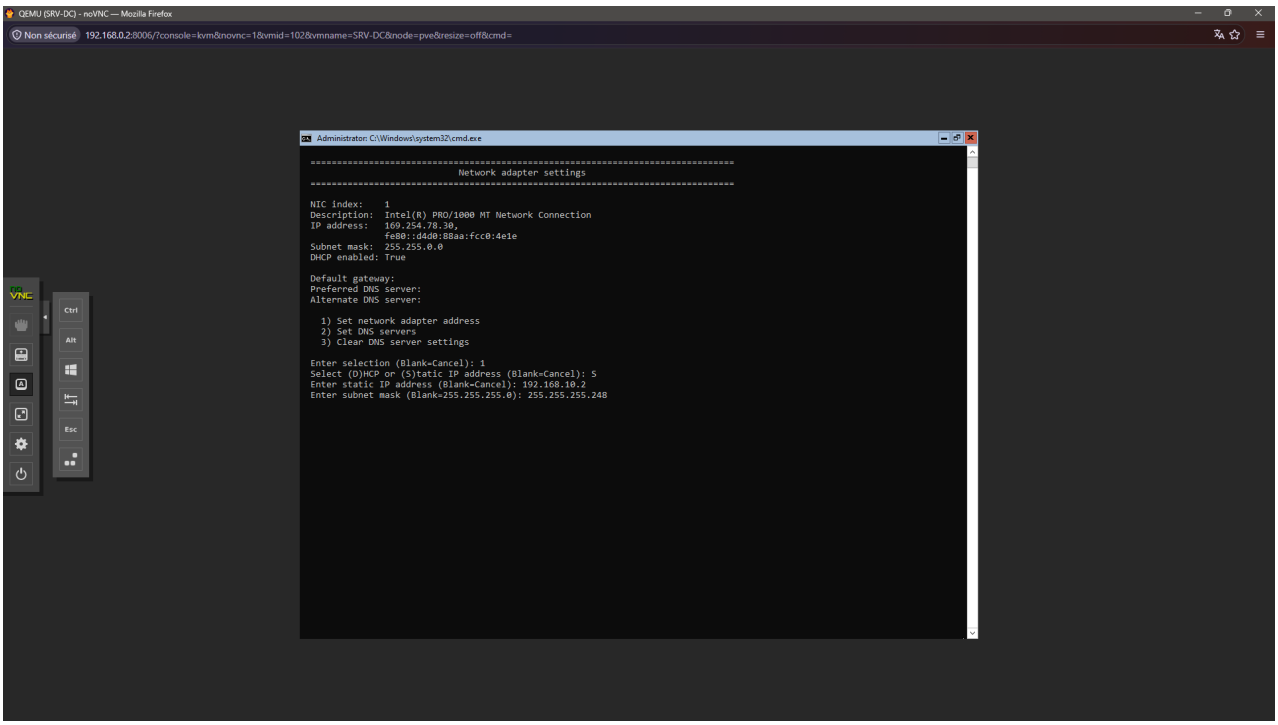
Nous mettrons une adresse statique pour notre serveur DC. Tapez 8 :

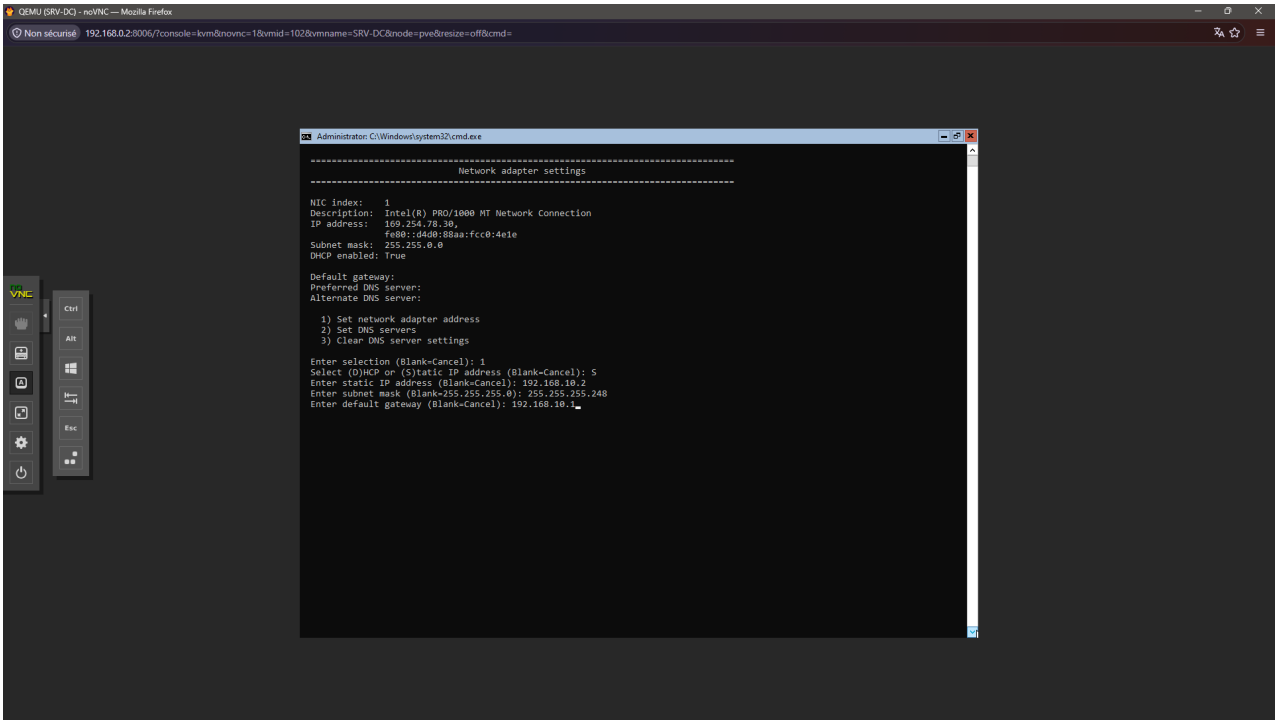

QEMU (SRV-DC) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=102&vmname=SRV-DC&node=pve&resize=off&cmd=
Administrator: C:\Windows\system32\cmd.exe
Network settings
Available network adapters:
Index # | IP address | Description
1 | 169.254.78.30 | Intel(R) PRO/1000 MT Network Connection
Select network adapter index # (Blank=Cancel): 1
Ctrl
Alt
Esc

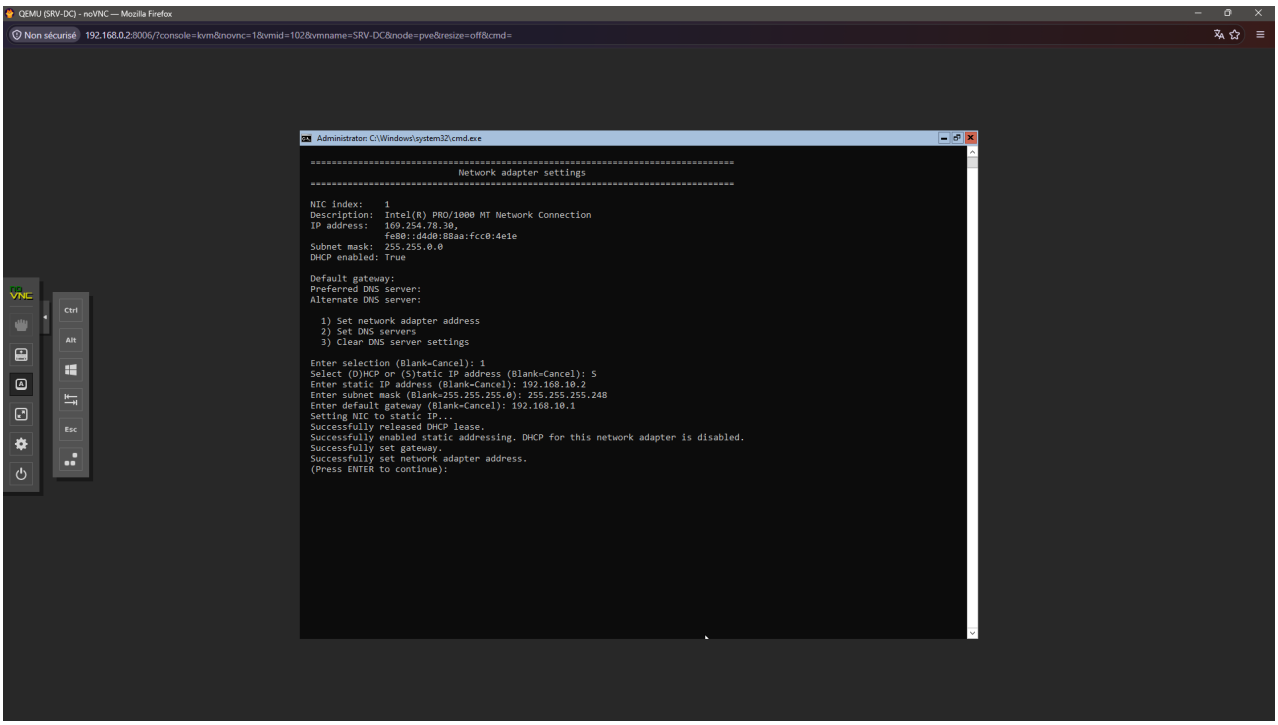

QEMU (SRV-DC) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=102&vmname=SRV-DC&node=pve&resize=off&cmd=
Administrator: C:\Windows\system32\cmd.exe
Network adapter settings
NIC index: 1
Description: Intel(R) PRO/1000 MT Network Connection
IP address: 169.254.78.30,
fe80::d4d0:88aa:fcc0:4e1e
Subnet mask: 255.255.0.0
DHCP enabled: True
Default gateway:
Preferred DNS server:
Alternate DNS server:
1) Set network adapter address
2) Set DNS servers
3) Clear DNS server settings
Enter selection (Blank=Cancel): 1
Ctrl
Alt
Esc


QEMU (SRV-DC) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=102&vmname=SRV-DC&node=pve&resize=off&cmd=
Administrator: C:\Windows\system32\cmd.exe
Network adapter settings
NIC index: 1
Description: Intel(R) PRO/1000 MT Network Connection
IP address: 169.254.78.30,
fe80::d4d0:88aa:fcc0:4e1e
Subnet mask: 255.255.0.0
DHCP enabled: True
Default gateway:
Preferred DNS server:
Alternate DNS server:
1) Set network adapter address
2) Set DNS servers
3) Clear DNS server settings
Enter selection (Blank=Cancel): 1
Select (D)HCP or (S)tatic IP address (Blank=Cancel): S
Ctrl
Alt
Esc

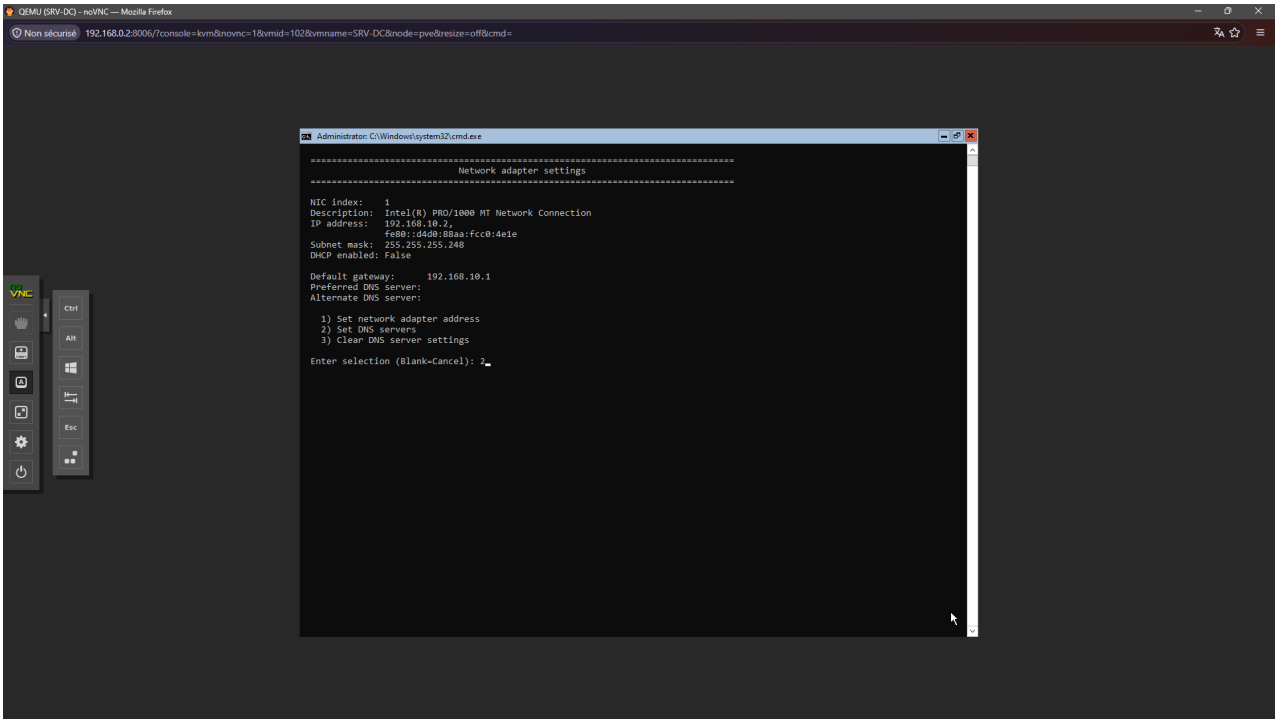

QEMU (SRV-DC) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=102&vmname=SRV-DC&node=pve&resize=off&cmd=
Administrator: C:\Windows\system32\cmd.exe
Network adapter settings
NIC index: 1
Description: Intel(R) PRO/1000 MT Network Connection
IP address: 169.254.78.30,
fe80::d4d0:88aa:fcc0:4e1e
Subnet mask: 255.255.0.0
DHCP enabled: True
Default gateway:
Preferred DNS server:
Alternate DNS server:
1) Set network adapter address
2) Set DNS servers
3) Clear DNS server settings
Enter selection (Blank=Cancel): 1
Select (D)HCP or (S)tatic IP address (Blank=Cancel): S
Enter static IP address (Blank=Cancel): 192.168.10.2
Ctrl
Alt
Esc

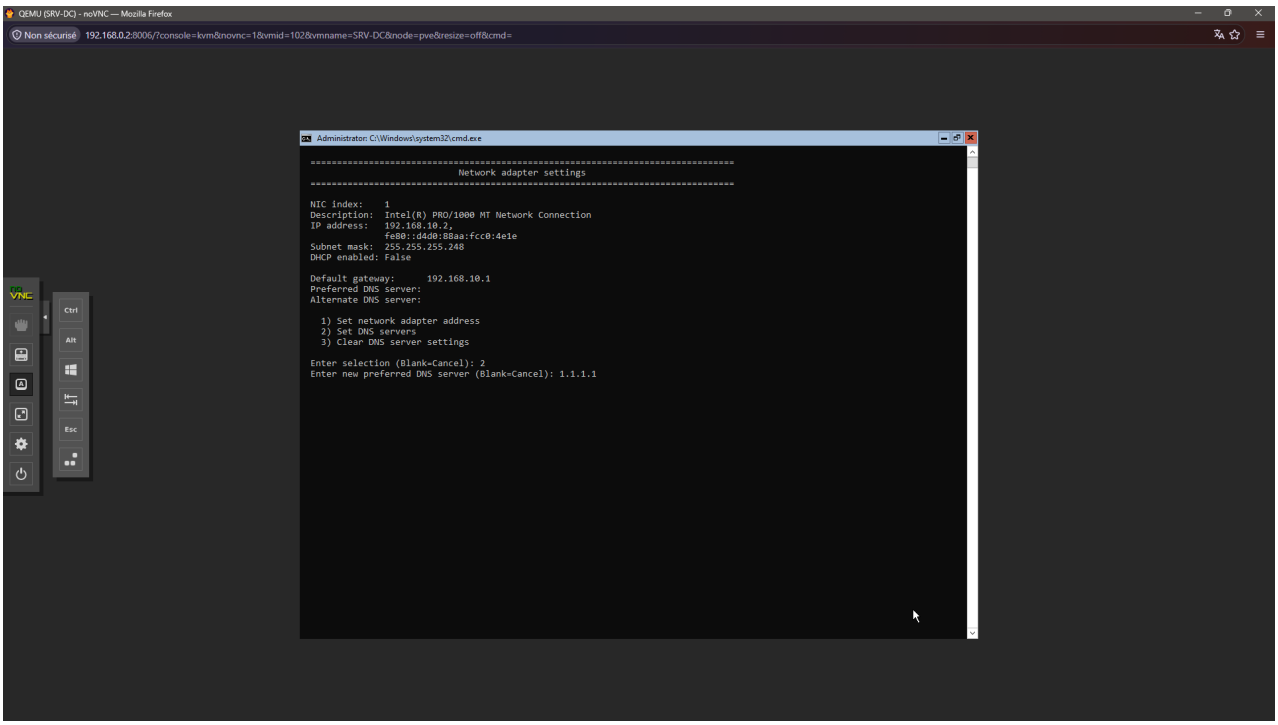

QEMU (SRV-DC) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=102&vmname=SRV-DC&node=pve&resize=off&cmd=
Administrator: C:\Windows\system32\cmd.exe
Network adapter settings
NIC index: 1
Description: Intel(R) PRO/1000 MT Network Connection
IP address: 169.254.78.30,
fe80::d4d0:88aa:fcc0:4e1e
Subnet mask: 255.255.0.0
DHCP enabled: True
Default gateway:
Preferred DNS server:
Alternate DNS server:
1) Set network adapter address
2) Set DNS servers
3) Clear DNS server settings
Enter selection (Blank=Cancel): 1
Select (D)HCP or (S)tatic IP address (Blank=Cancel): S
Enter static IP address (Blank=Cancel): 192.168.10.2
Enter subnet mask (Blank=255.255.255.0): 255.255.255.248
Ctrl
Alt
Esc

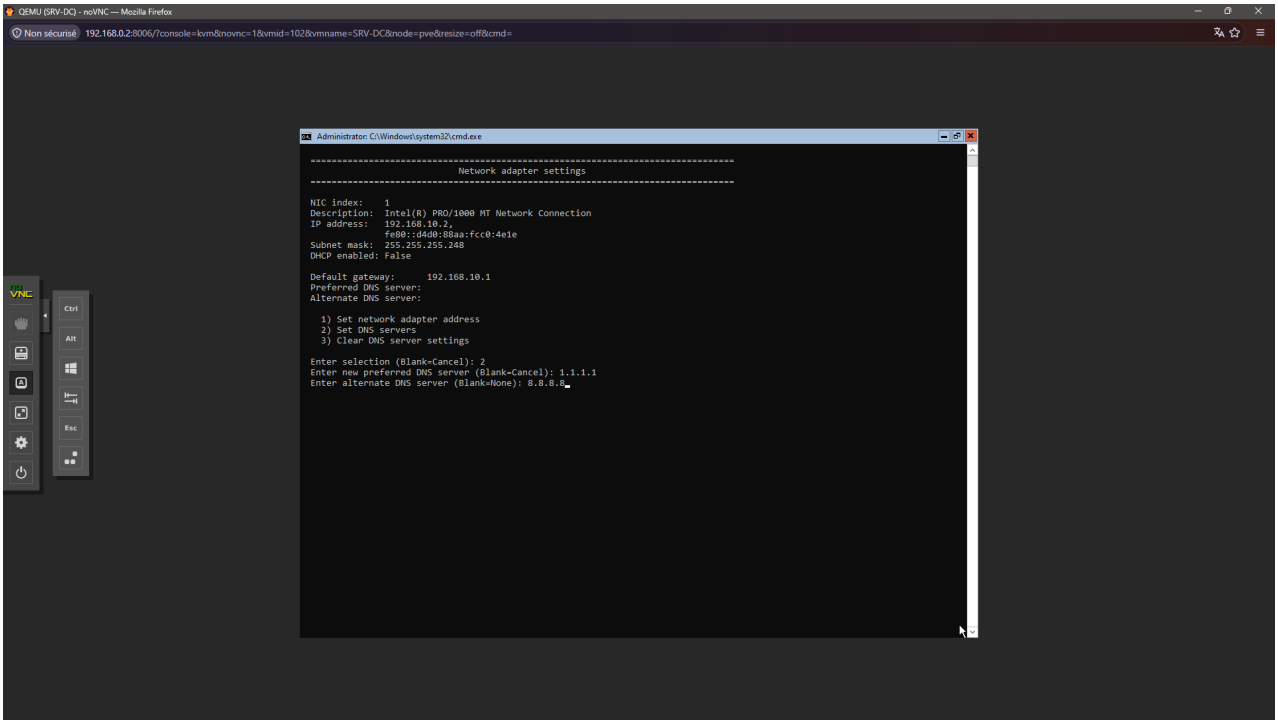

QEMU (SRV-DC) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=102&vmname=SRV-DC&node=pve&resize=off&cmd=
Administrator: C:\Windows\system32\cmd.exe
Network adapter settings
NIC index: 1
Description: Intel(R) PRO/1000 MT Network Connection
IP address: 169.254.78.30,
fe80::d4d0:88aa:fcc0:4e1e
Subnet mask: 255.255.0.0
DHCP enabled: True
Default gateway:
Preferred DNS server:
Alternate DNS server:
1) Set network adapter address
2) Set DNS servers
3) Clear DNS server settings
Enter selection (Blank=Cancel): 1
Select (D)HCP or (S)tatic IP address (Blank=Cancel): S
Enter static IP address (Blank=Cancel): 192.168.10.2
Enter subnet mask (Blank=255.255.255.0): 255.255.255.248
Enter default gateway (Blank=Cancel): 192.168.10.1
Ctrl
Alt
Esc

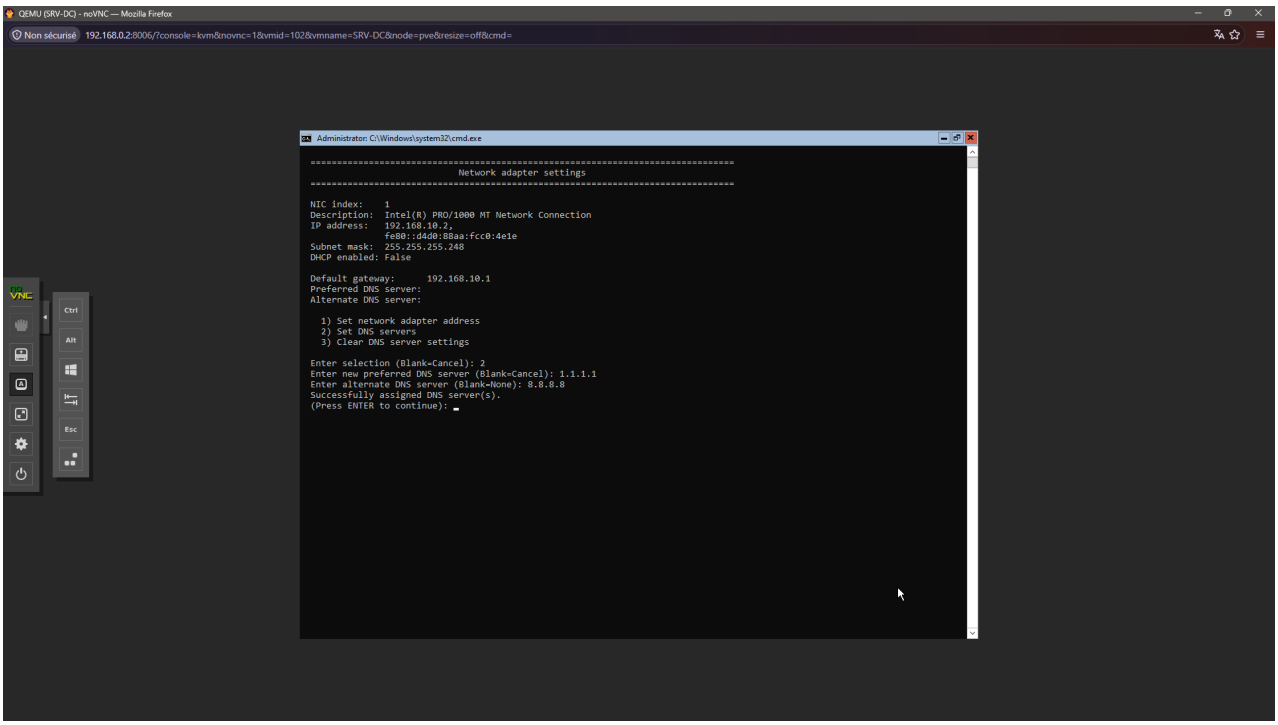


3.3. Configurer les serveurs DNS



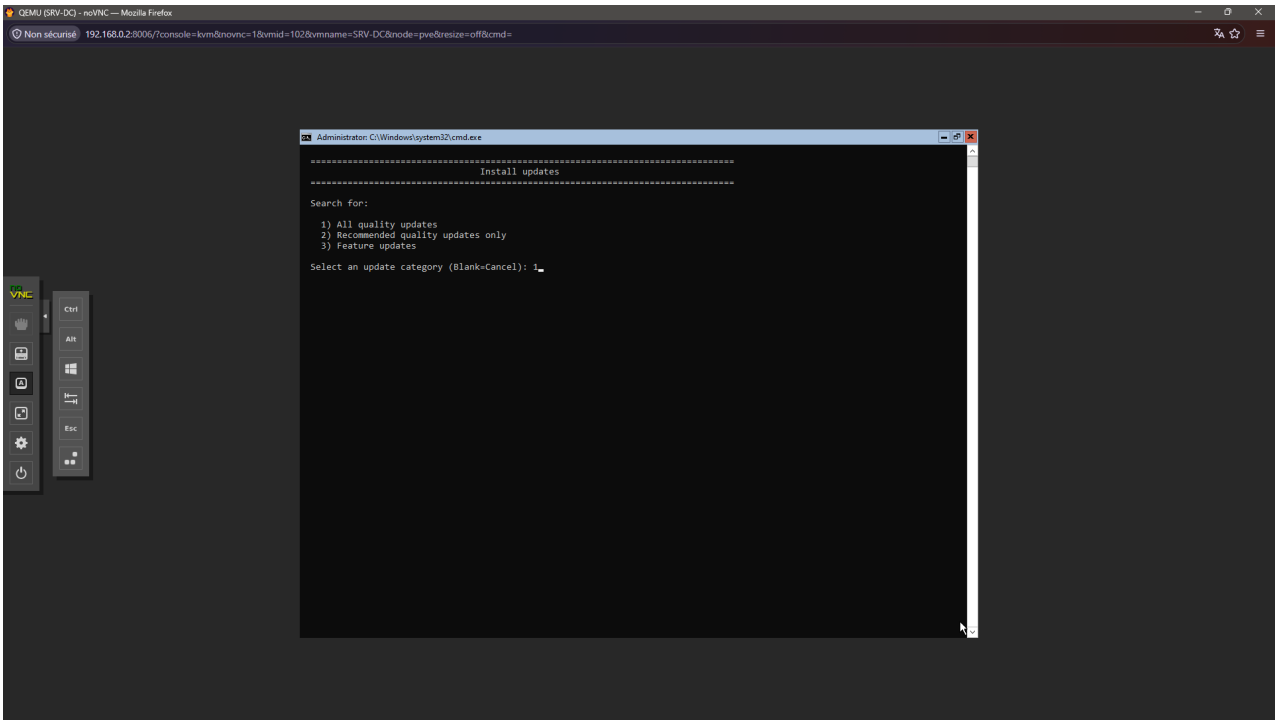

QEMU (SRV-DC) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=102&vmname=SRV-DC&node=pve&resize=off&cmd=
Administrator: C:\Windows\system32\cmd.exe
Network adapter settings
NIC index: 1
Description: Intel(R) PRO/1000 MT Network Connection
IP address: 192.168.10.2,
fe80::d4d0:88aa:fcc0:4e1e
Subnet mask: 255.255.255.248
DHCP enabled: False
Default gateway: 192.168.10.1
Preferred DNS server:
Alternate DNS server:
1) Set network adapter address
2) Set DNS servers
3) Clear DNS server settings
Enter selection (Blank=Cancel): 2
Enter new preferred DNS server (Blank=Cancel): 1.1.1.1
Ctrl
Alt
Esc

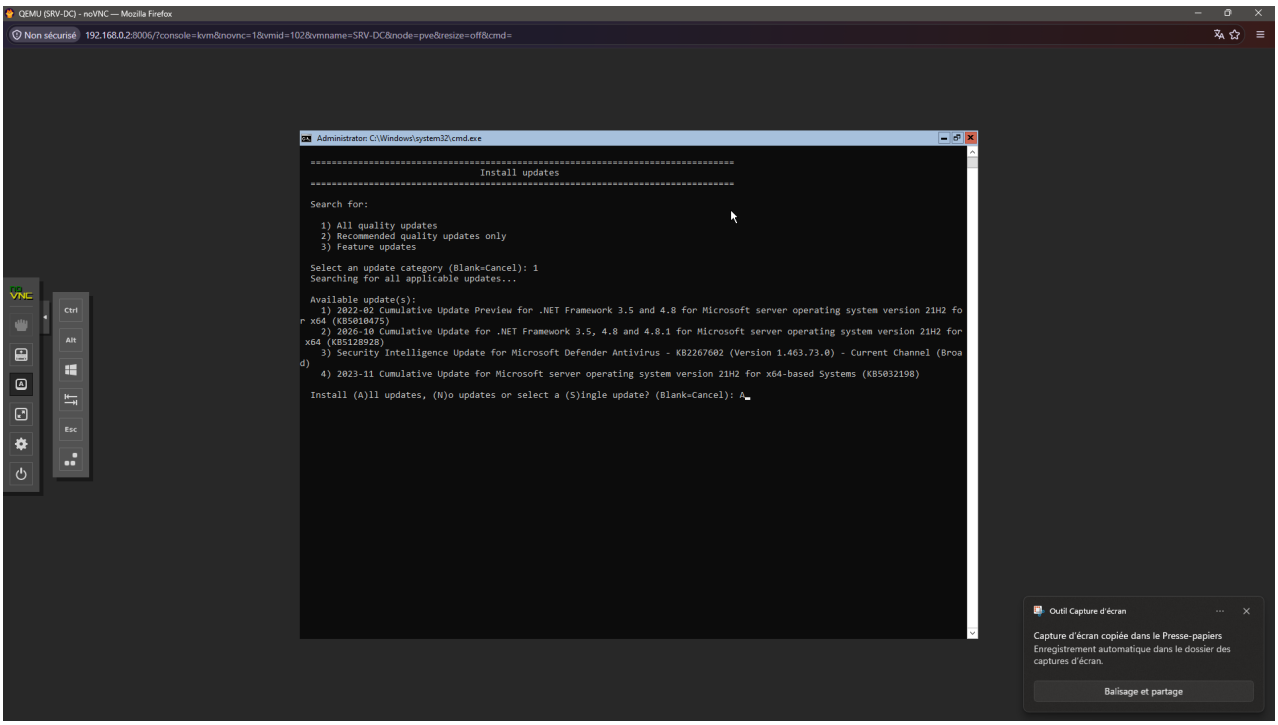

QEMU (SRV-DC) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=102&vmname=SRV-DC&node=pve&resize=off&cmd=
Administrator: C:\Windows\system32\cmd.exe
Network adapter settings
NIC index: 1
Description: Intel(R) PRO/1000 MT Network Connection
IP address: 192.168.10.2,
fe80::d4d0:88aa:fcc0:4e1e
Subnet mask: 255.255.255.248
DHCP enabled: False
Default gateway: 192.168.10.1
Preferred DNS server:
Alternate DNS server:
1) Set network adapter address
2) Set DNS servers
3) Clear DNS server settings
Enter selection (Blank=Cancel): 2
Enter new preferred DNS server (Blank=Cancel): 1.1.1.1
Enter alternate DNS server (Blank=None): 8.8.8.8
Ctrl
Alt
Esc

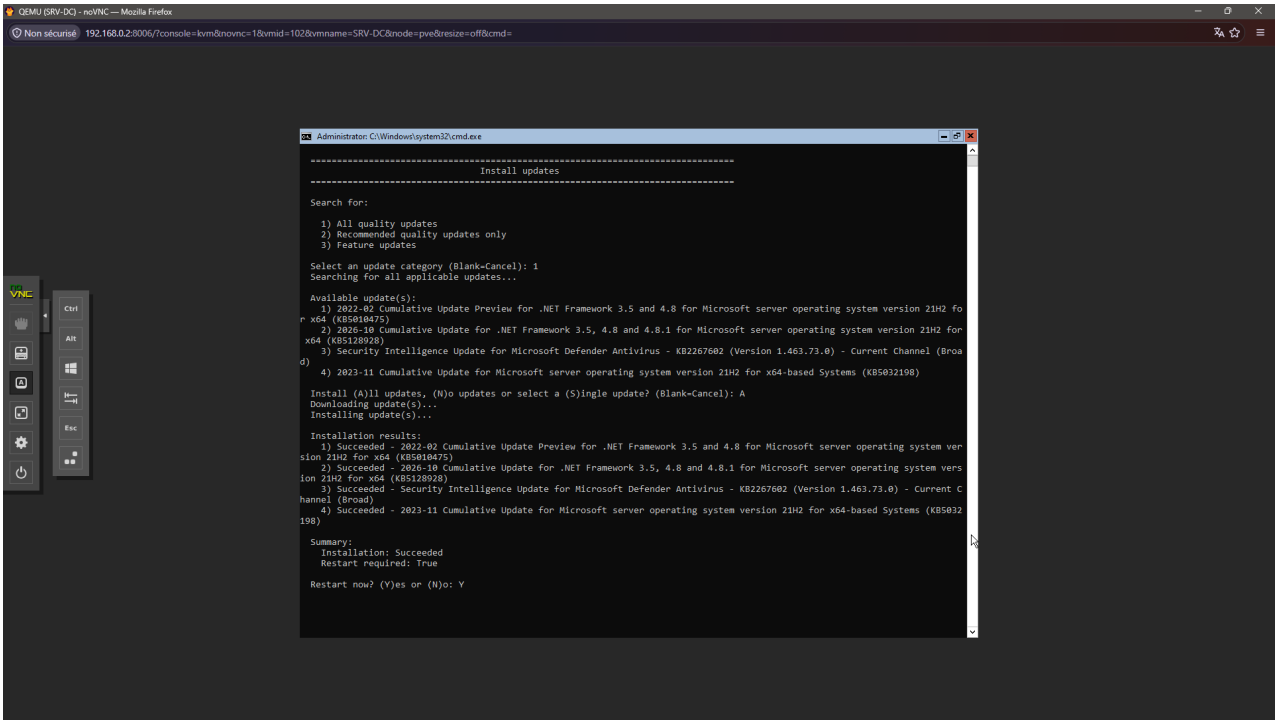


3.4. Mettre à jour le serveur Windows

Nous allons mettre à jour le serveur Windows, tapez 6 puis sélectionnez l'option 1 :







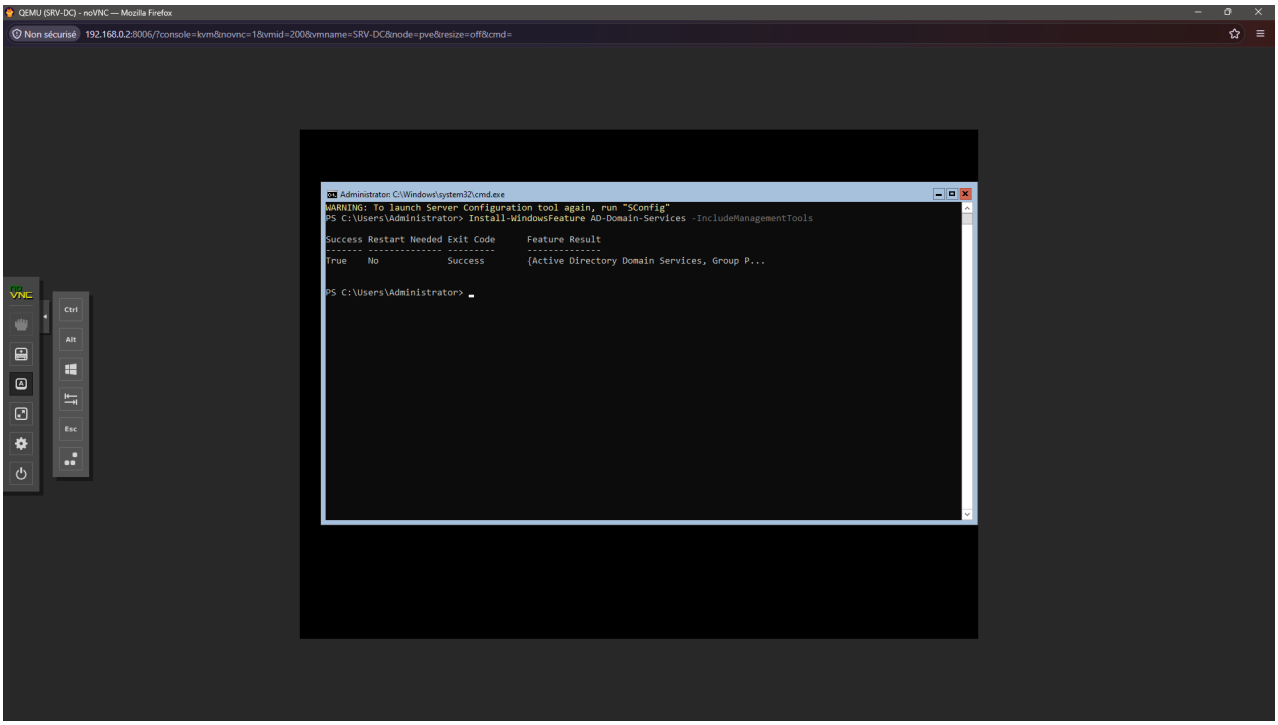
3.5. Préparer le serveur DC

Après le redémarrage, tapez 15 dans le tableau de bord pour accéder à PowerShell. Exécutez ensuite :

```
# Install AD DS
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools


# Check installation
Get-WindowsFeature AD-Domain-Services
```

```
Get-Command INstall-ADDSForest
```

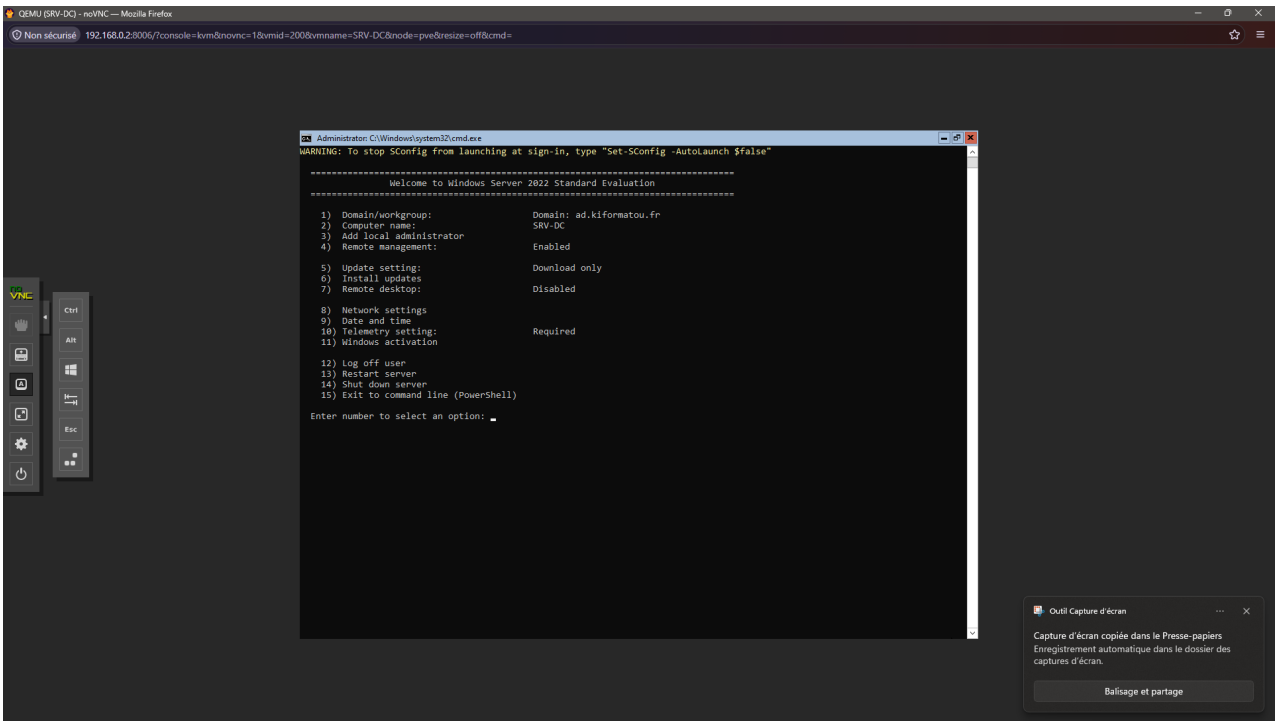


3.6. Créer un nouveau domaine

Nous allons crée une forêt et le domaine ad.kiformatou.fr. Toujours dans PowerShell, exécutez les commandes suivantes :

```
Install-ADDSForest -DomainName "ad.kiformatou.fr" -
DomainNetbiosName "KIFORMATOU" -InstallDns
```

Redémarrez

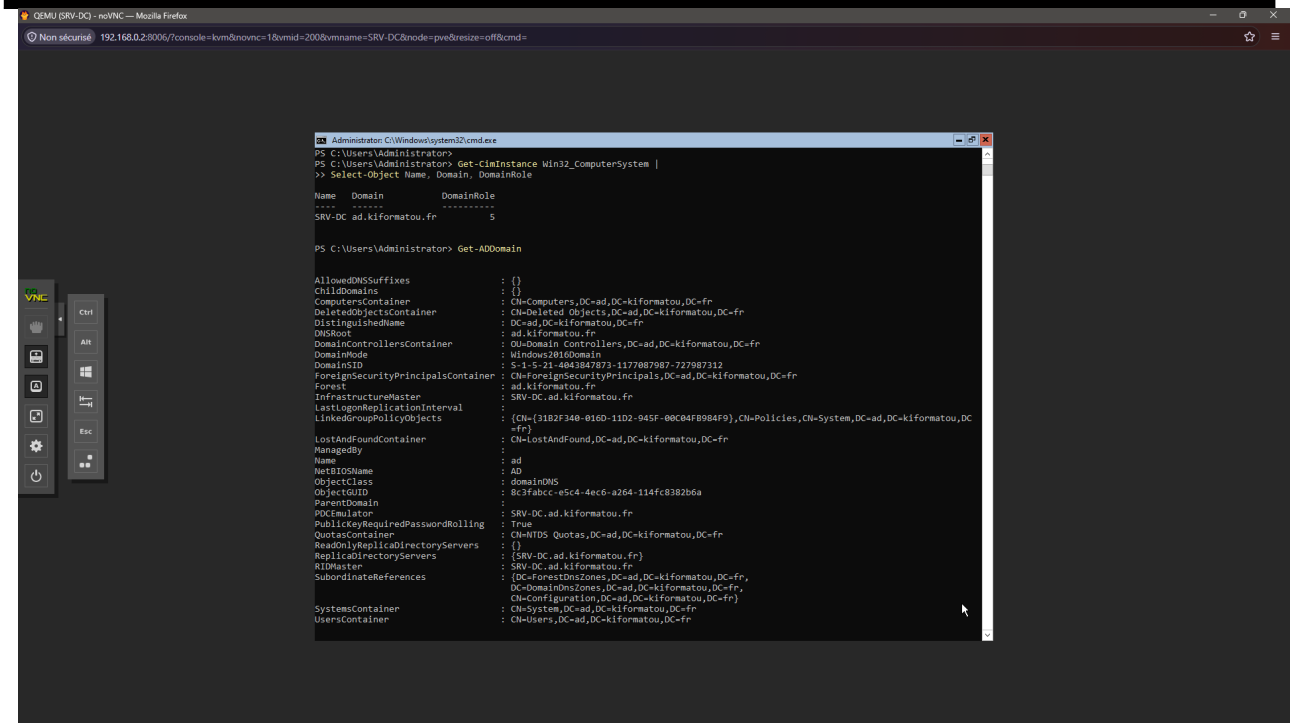


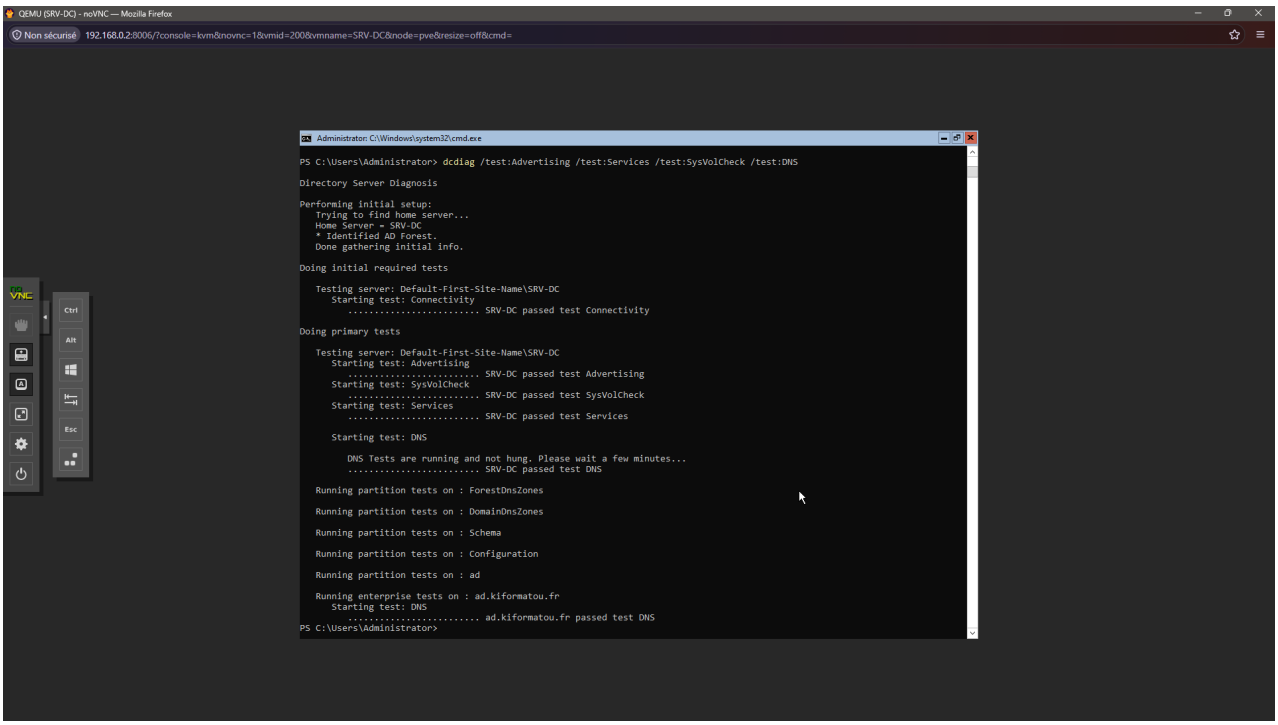
Vérifiez avec :

```
Get-ADDomain

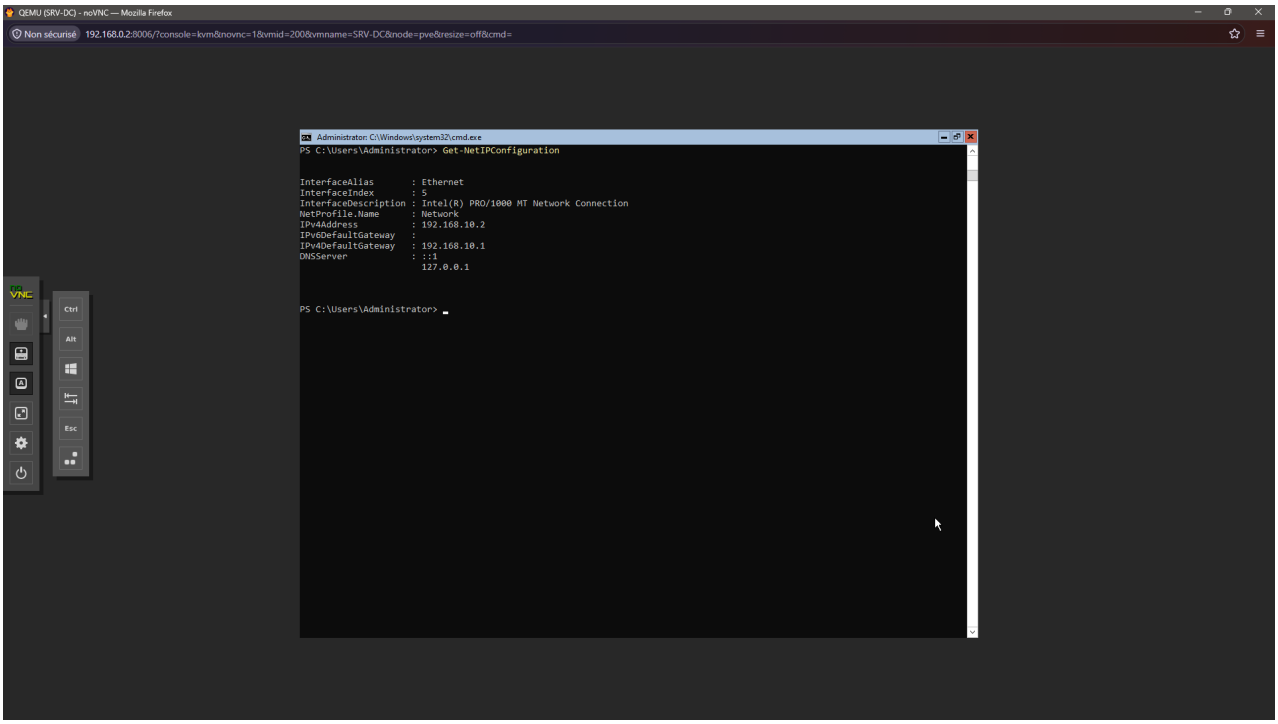
Get-ADDomainController -Filter *

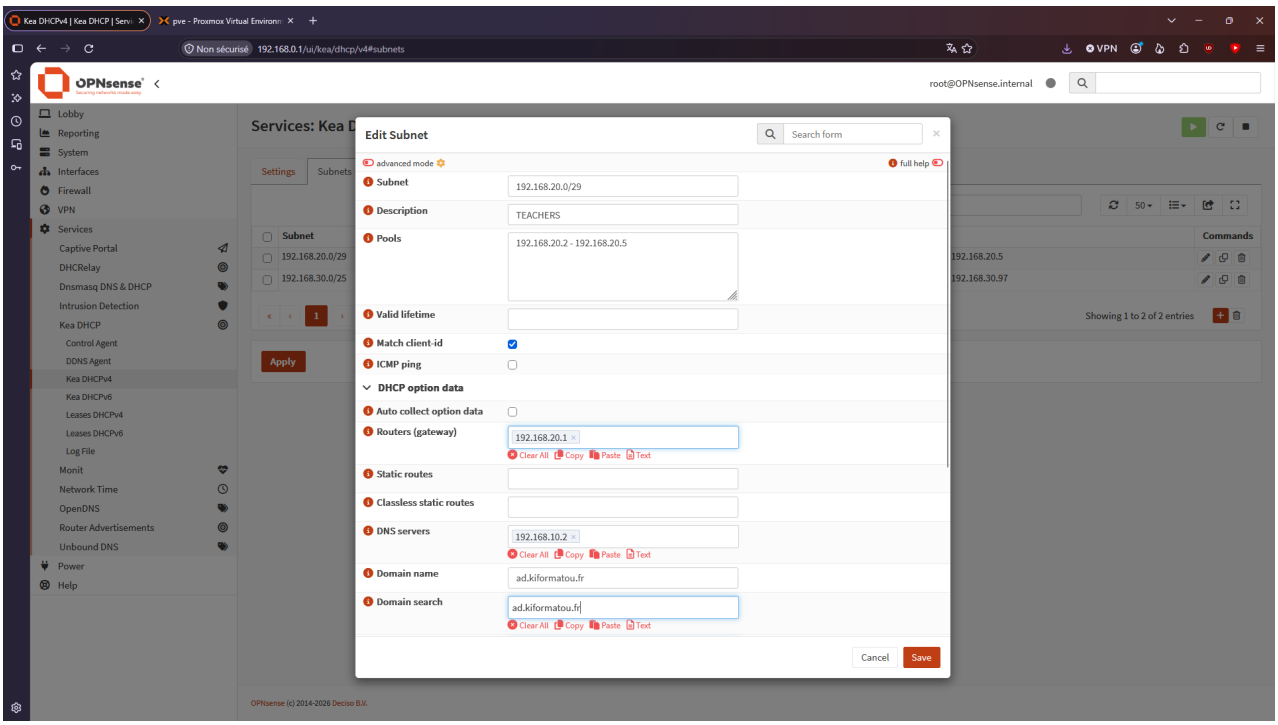
dcdiag /test:DNS
```

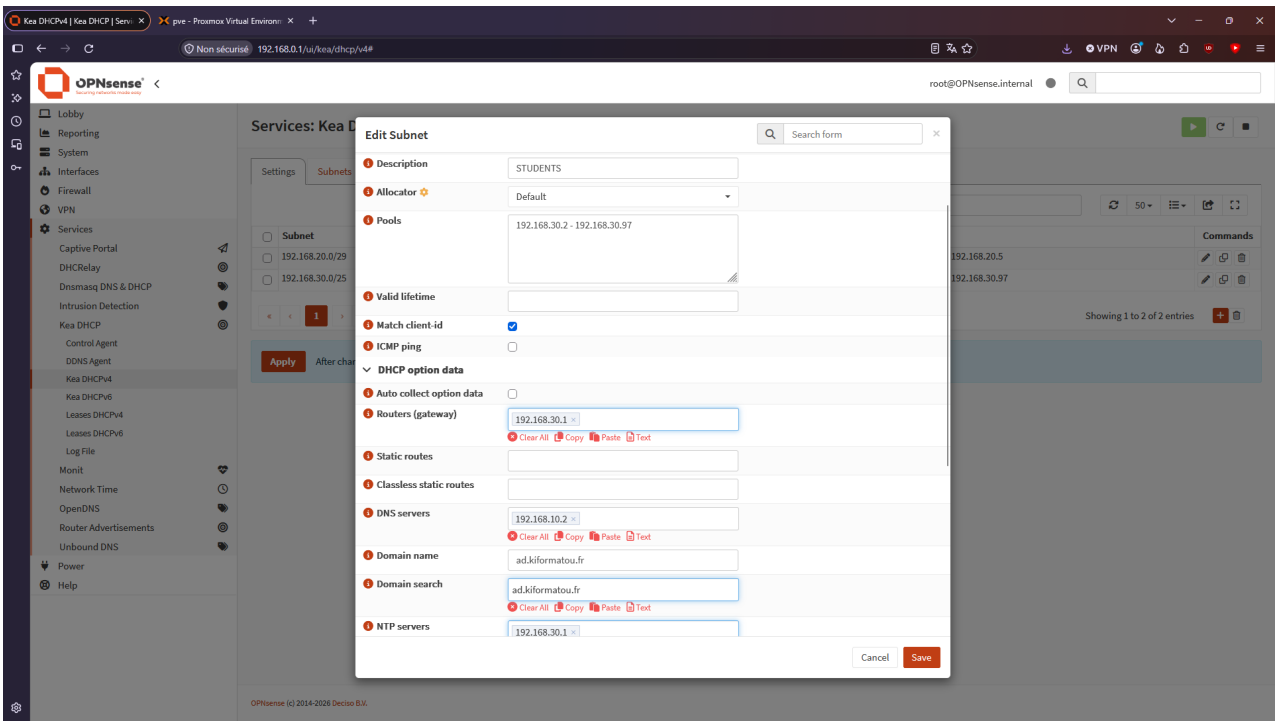


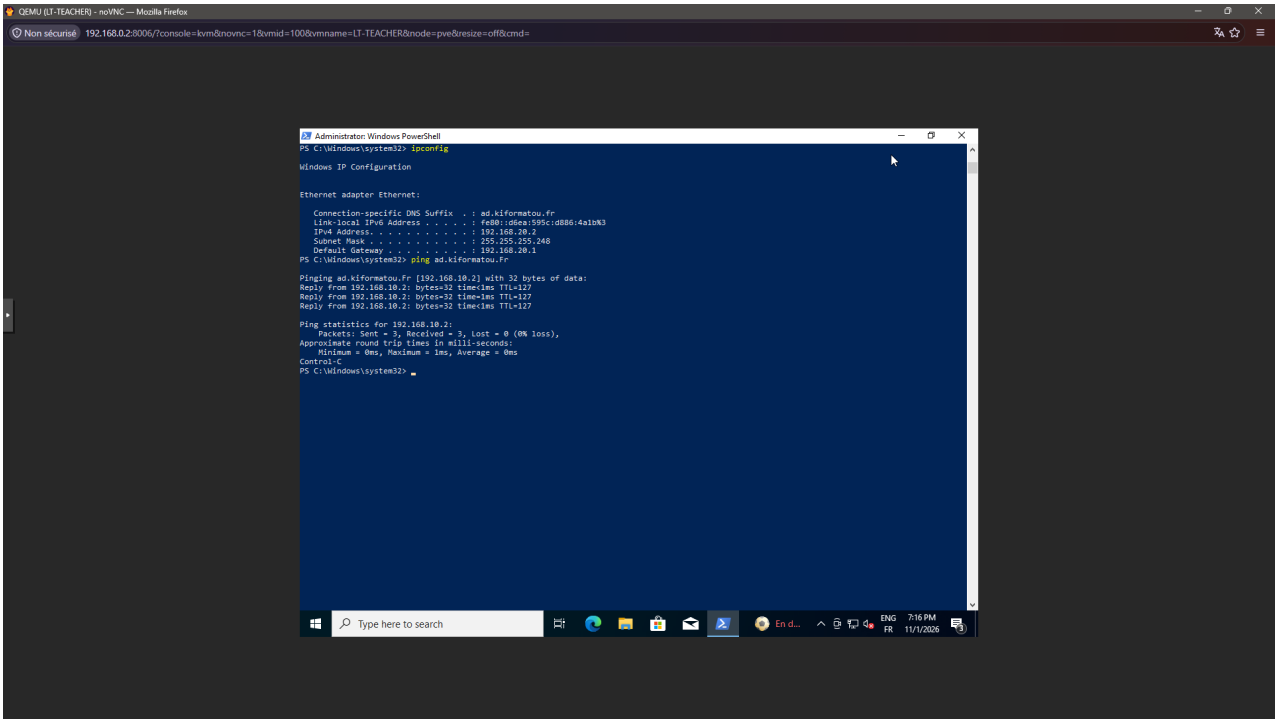


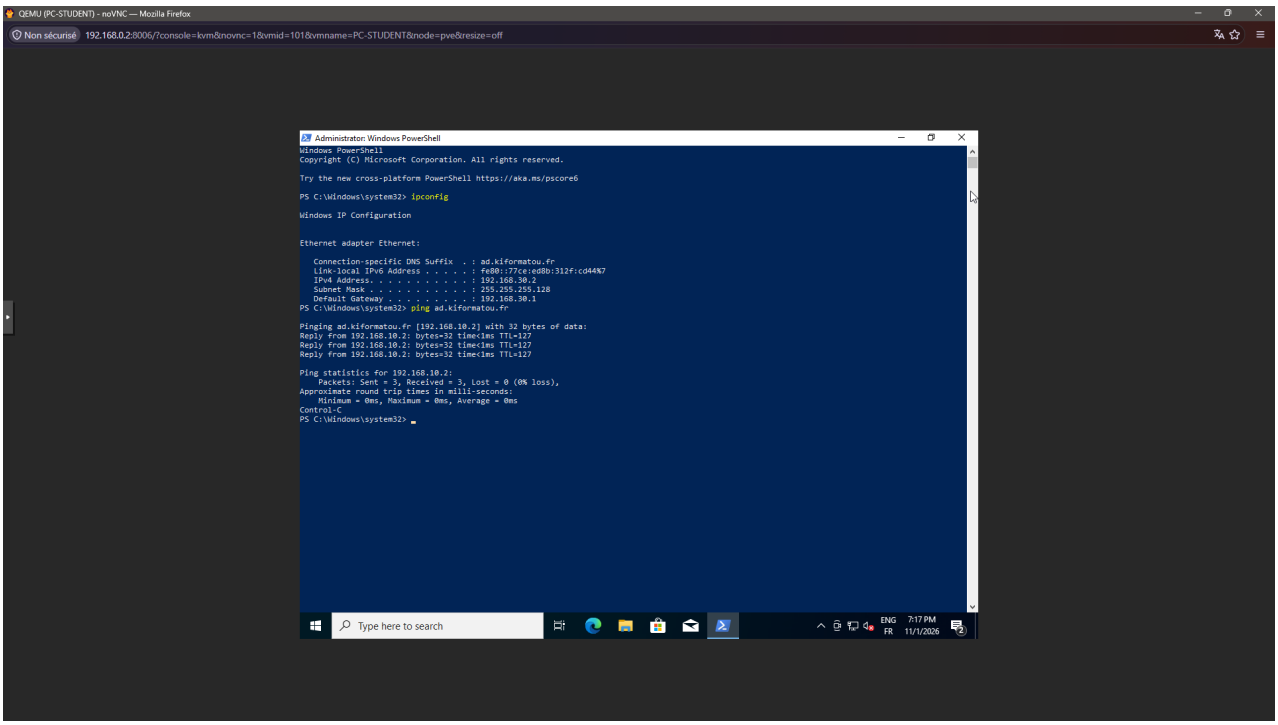
3.7. Joindre un poste de travail au domain




Edit Subnet
Search form
advanced mode
full help
Subnet
192.168.20.0/29
Description
TEACHERS
Pools
192.168.20.2 - 192.168.20.5
Valid lifetime
Match client-id
ICMP ping
DHCP option data
Auto collect option data
Routers (gateway)
192.168.20.1
Clear All Copy Paste Text
Static routes
Classless static routes
DNS servers
192.168.10.2
Domain name
ad.kiformatou.fr
Domain search
ad.kiformatou.fr
Cancel
Save

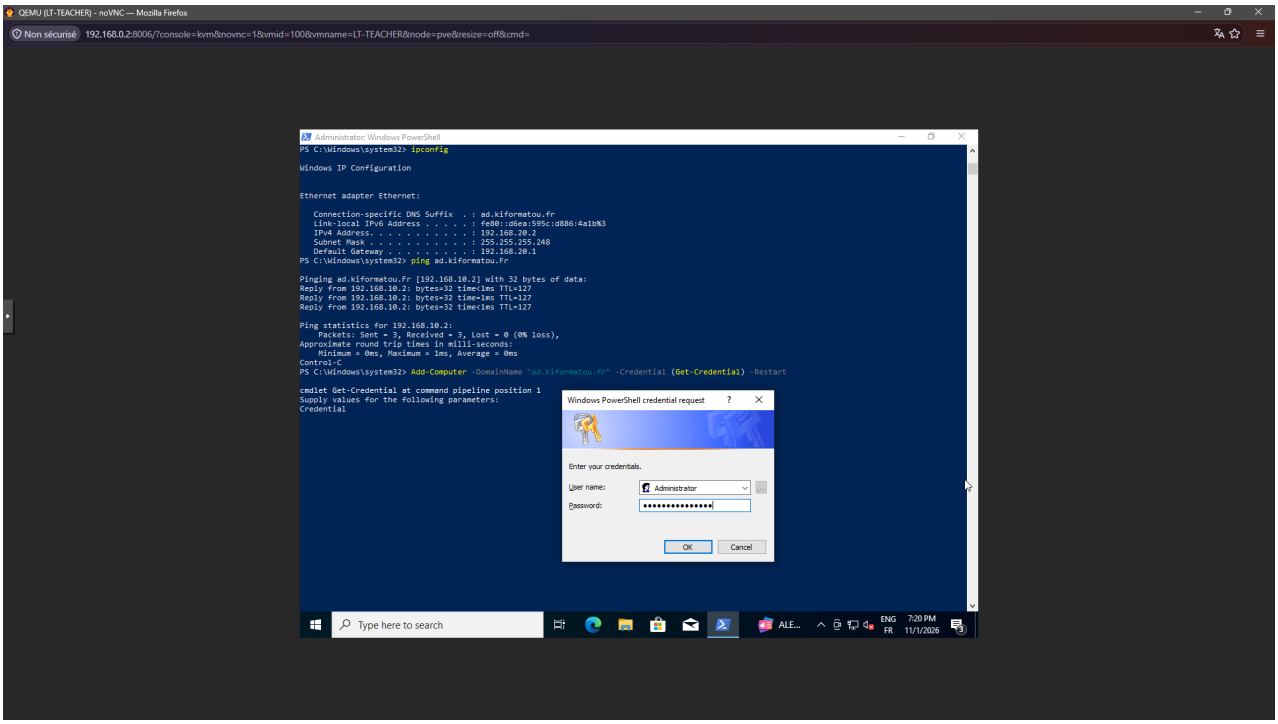

Edit Subnet
Search form
Description
STUDENTS
Allocator
Default
Pools
192.168.30.2 - 192.168.30.97
Valid lifetime
Match client-id
ICMP ping
DHCP option data
Auto collect option data
Routers (gateway)
192.168.30.1
Static routes
Classless static routes
DNS servers
192.168.10.2
Domain name
ad.kiformatou.fr
Domain search
ad.kiformatou.fr
NTP servers
192.168.30.1
Cancel
Save


Administrator: Windows PowerShell
PS C:\Windows\system32> ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : ad.kiformatou.fr
Link-local IPv6 Address . . . . . : fe80::d6ea:595c:d886:4a1b%3
IPv4 Address. . . . . : 192.168.20.2
Subnet Mask . . . . . : 255.255.255.248
Default Gateway . . . . . : 192.168.20.1
PS C:\Windows\system32> ping ad.kiformatou.fr
Pinging ad.kiformatou.fr [192.168.10.2] with 32 bytes of data:
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time=1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.10.2:
Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
Control-C
PS C:\Windows\system32>
Type here to search

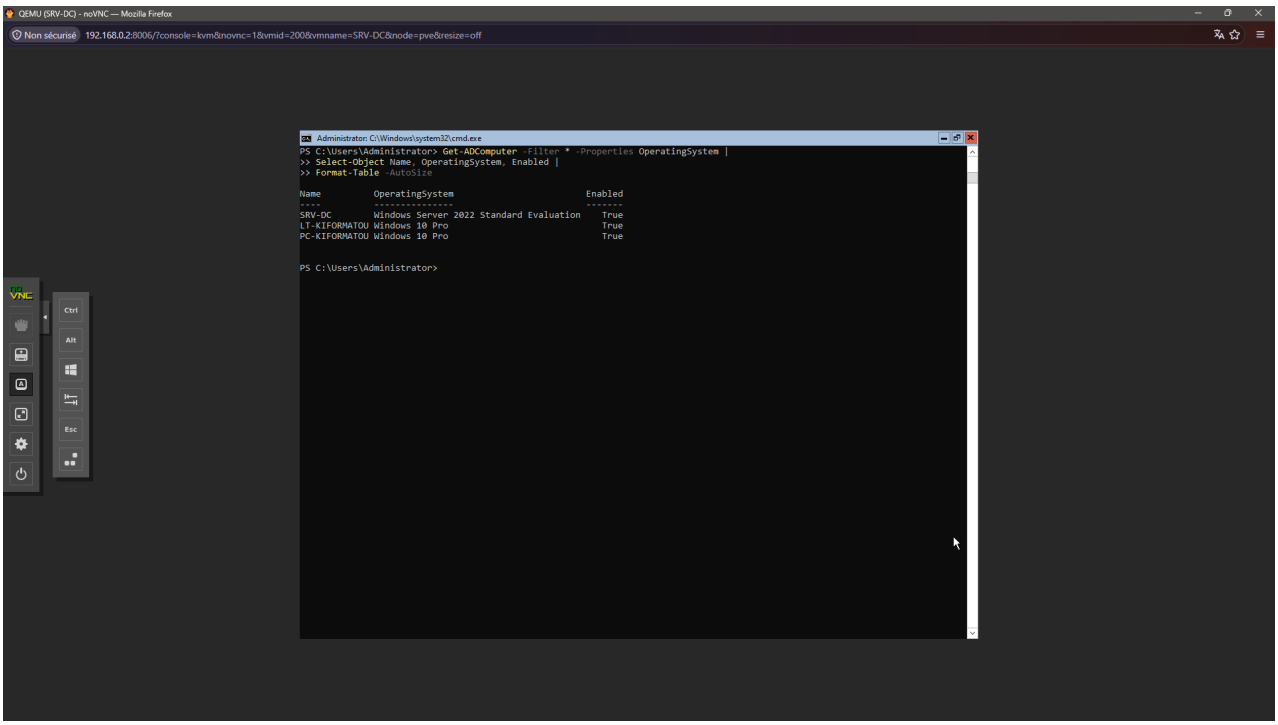


Ajoutez chaque PC au serveur DC en exécutant :

```
Add-Computer -DomainName "ad.kiformatou.fr" -Credential (Get-
Credential) -Restart
```



Vérifiez ensuite sur le serveur DC :



Vous avez avez terminé avec les PC. Il faudra également ajouter les serveurs à l'AD.

3.8. Créer les OU, utilisateurs, groupes et GPO (WIP)

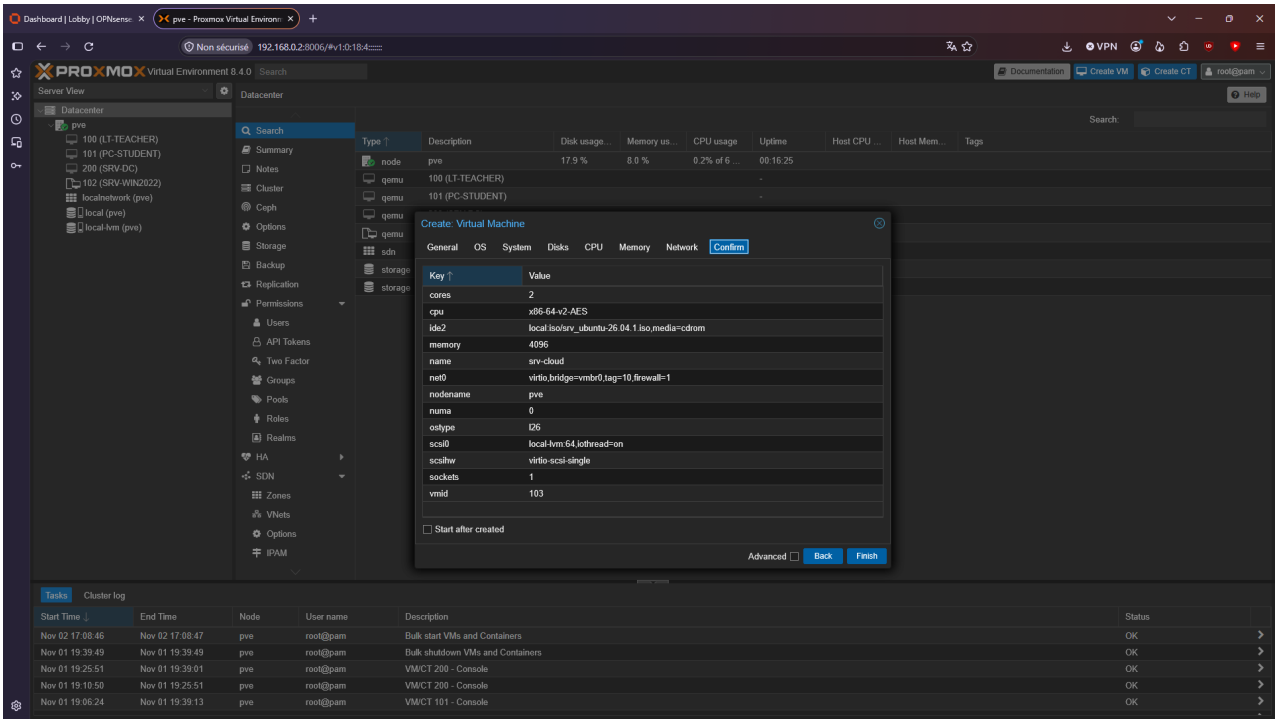
Création du serveur Nextcloud (WIP)

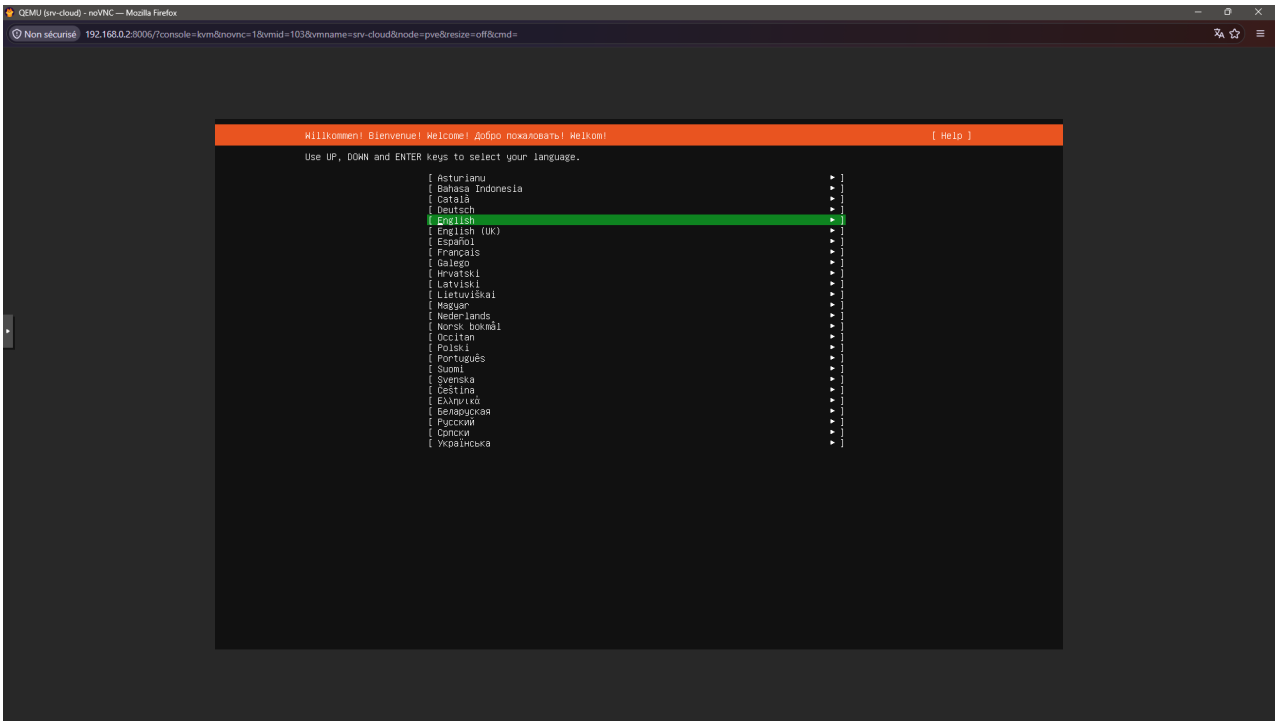
Le serveur Nextcloud est un espace de stockage et de collaboration en ligne, comme Google Drive ou Dropbox. Il permet notamment de :

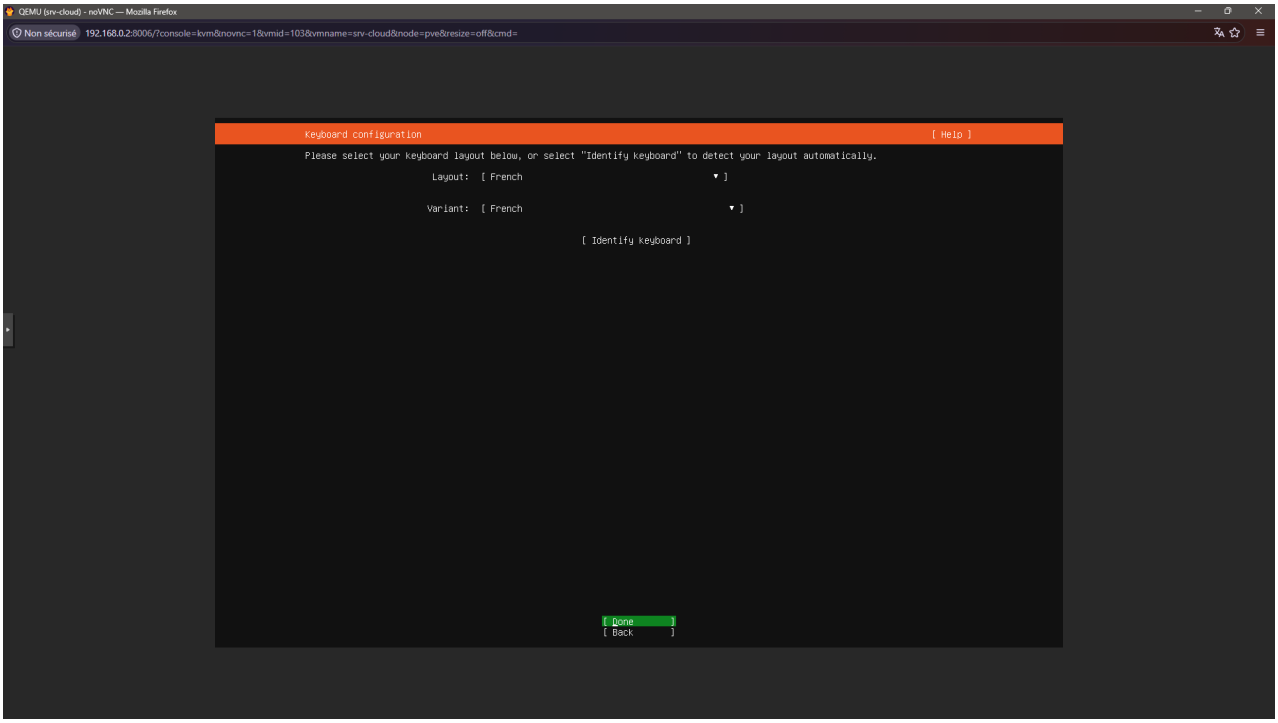
- Stocker et synchroniser des fichiers entre un PC et un téléphone.
- Partager des fichiers avec un lien, un mot de passe et une date d'expiration.
- Synchroniser les contaxts et calendriers.
- Ajouter des fonctions comme l'édition collaborative de documents ou les appels vidéo.

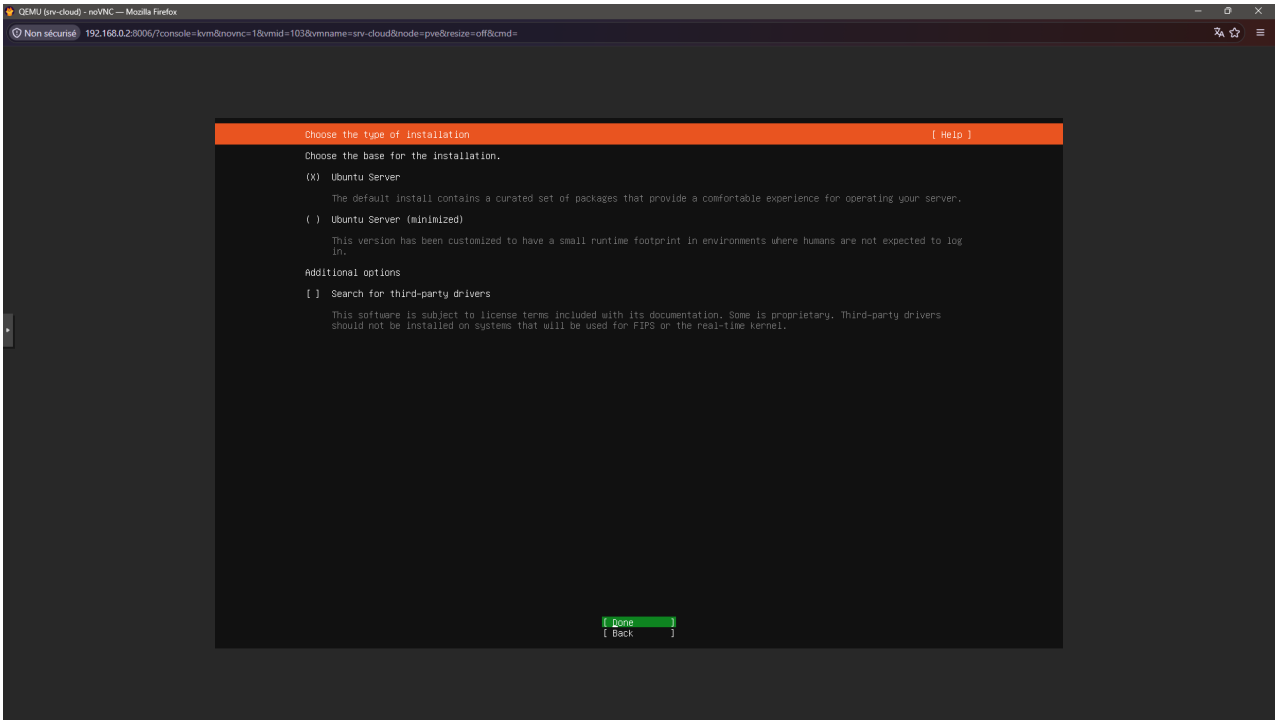
VMName	CPU	RAM (Go)	HDD (Go)
srv-cloud	2	4 / 4096	64

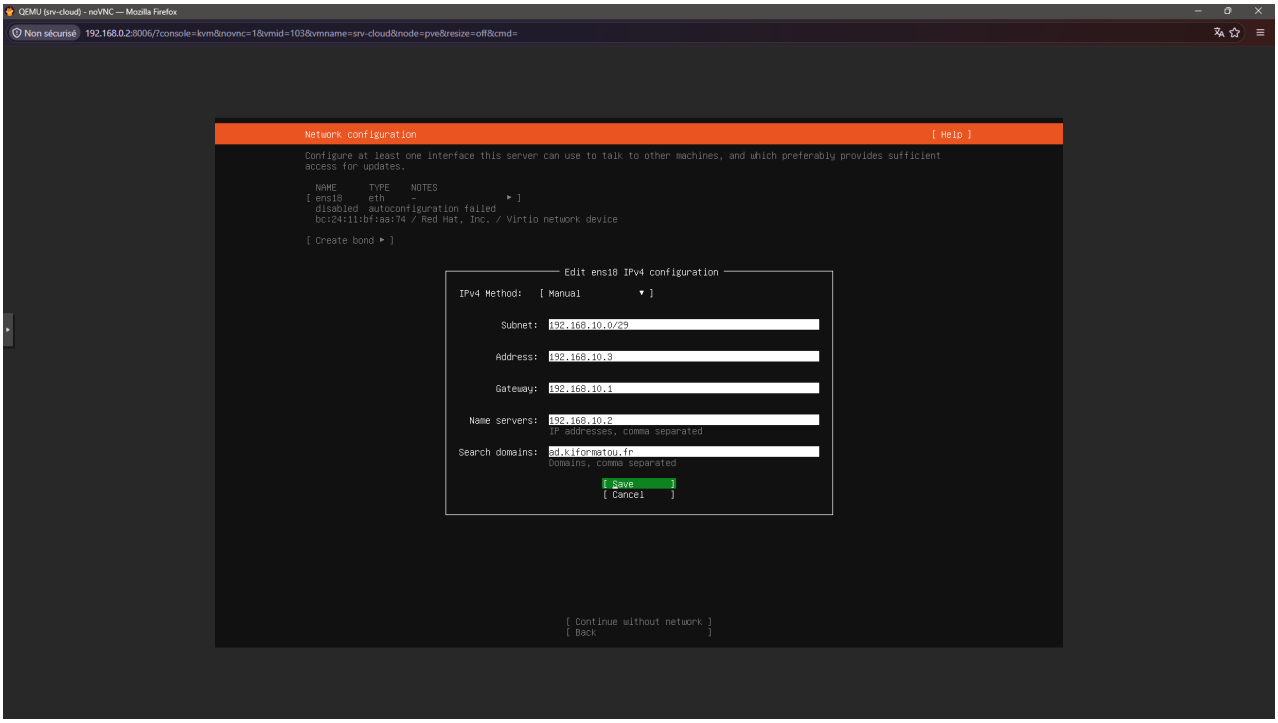
1. Créer la VM Nextcloud

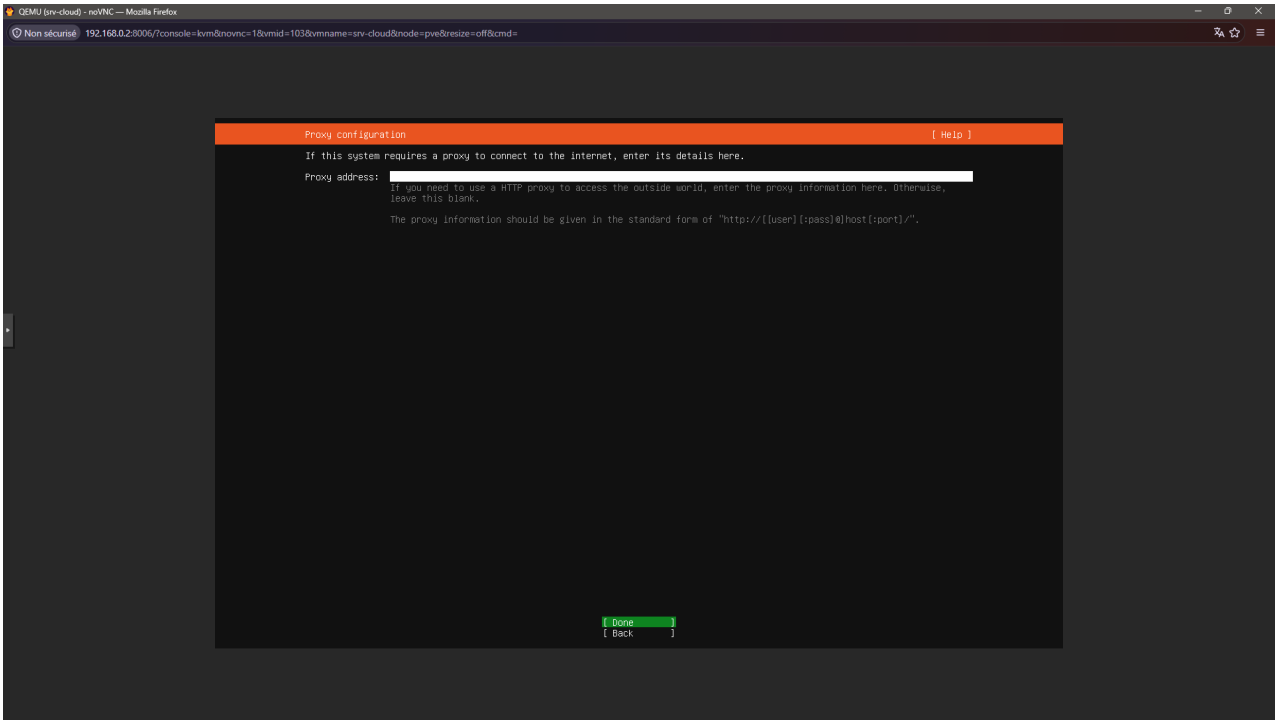


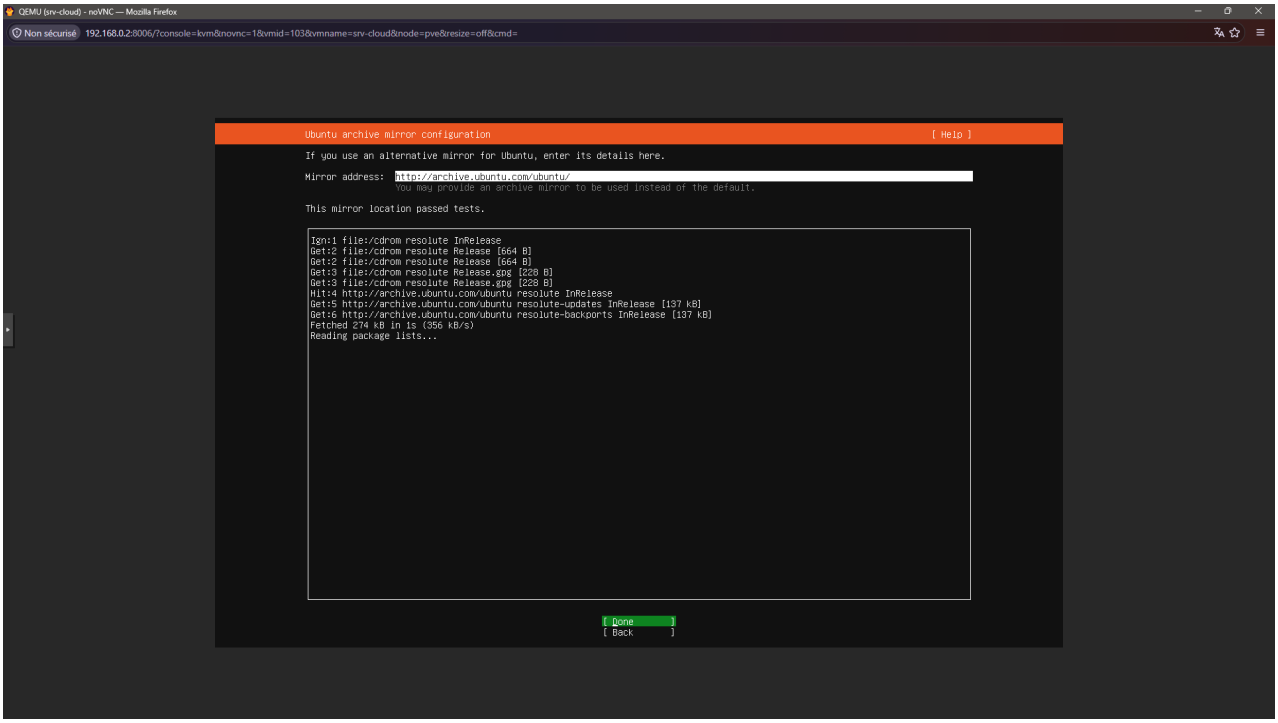


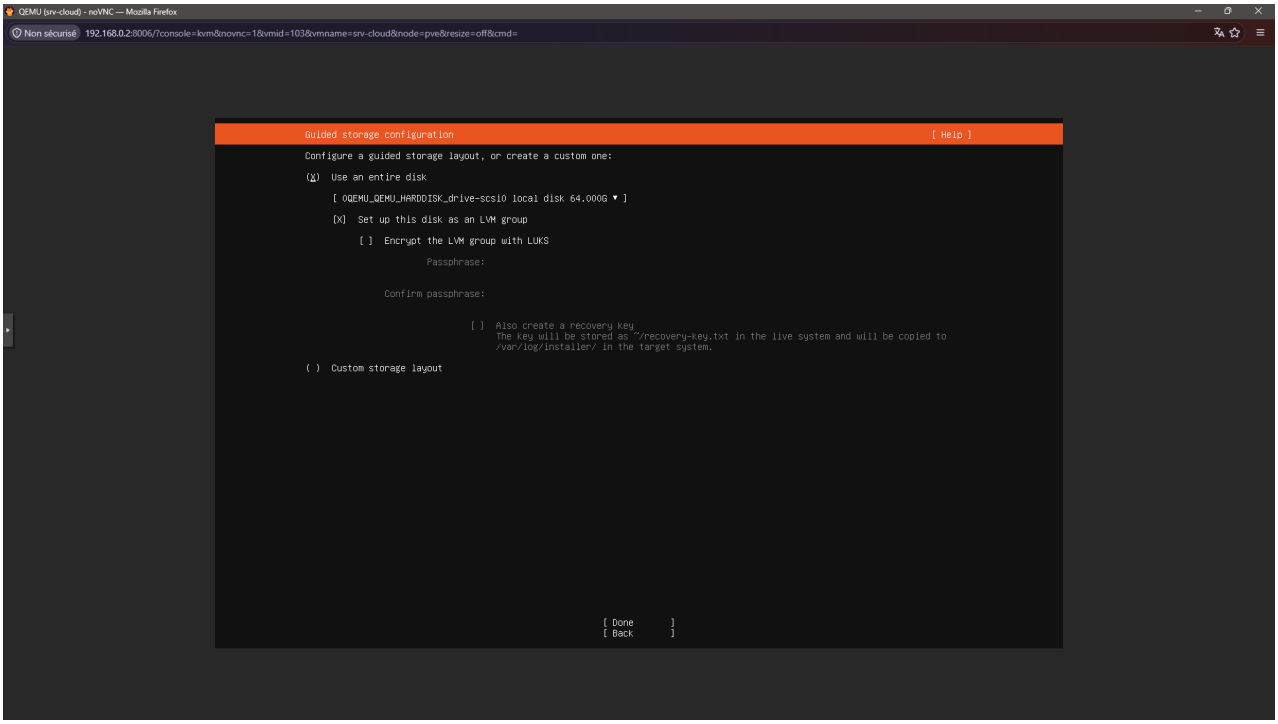

QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
Keyboard configuration
[Help]
Please select your keyboard layout below, or select "Identify keyboard" to detect your layout automatically.
Layout: [French ▾]
Variant: [French ▾]
[Identify keyboard]
[Done]
[Back]

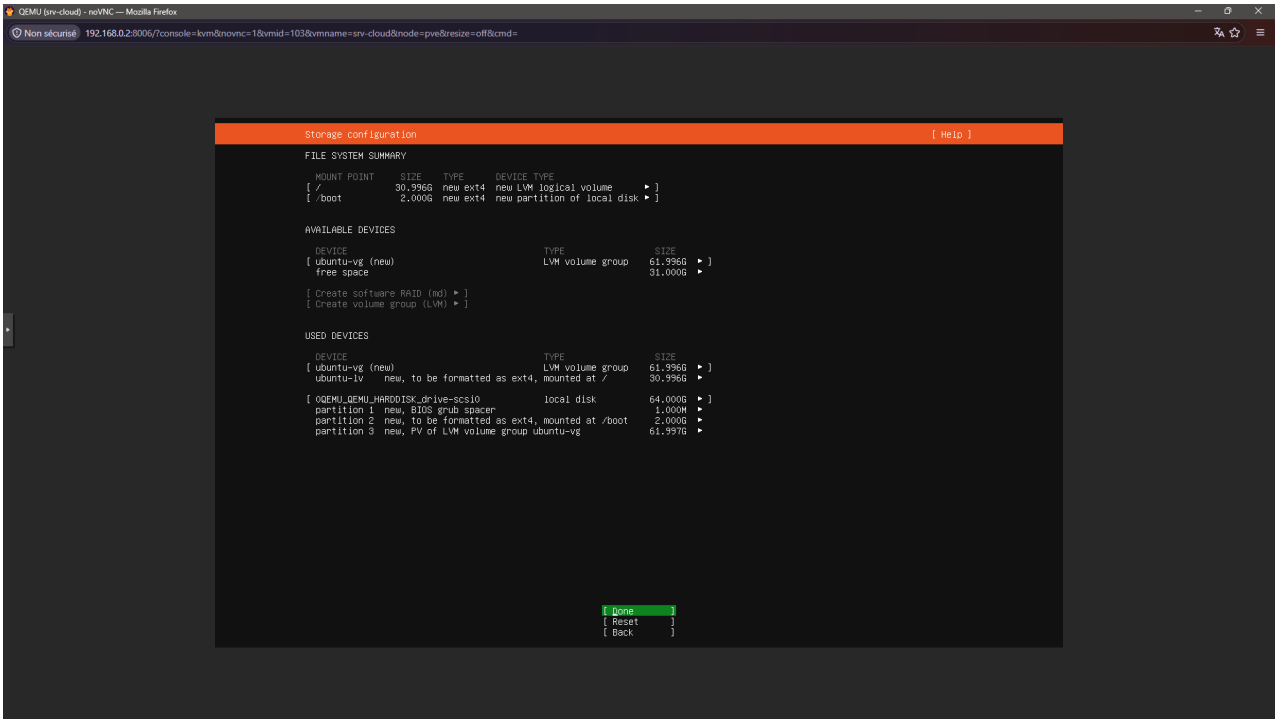

QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
Choose the type of installation
[Help]
Choose the base for the installation.
(X) Ubuntu Server
The default install contains a curated set of packages that provide a comfortable experience for operating your server.
( ) Ubuntu Server (minimized)
This version has been customized to have a small runtime footprint in environments where humans are not expected to log in.
Additional options
[ ] Search for third-party drivers
This software is subject to license terms included with its documentation. Some is proprietary. Third-party drivers should not be installed on systems that will be used for FIPS or the real-time kernel.
[Done]
[Back]

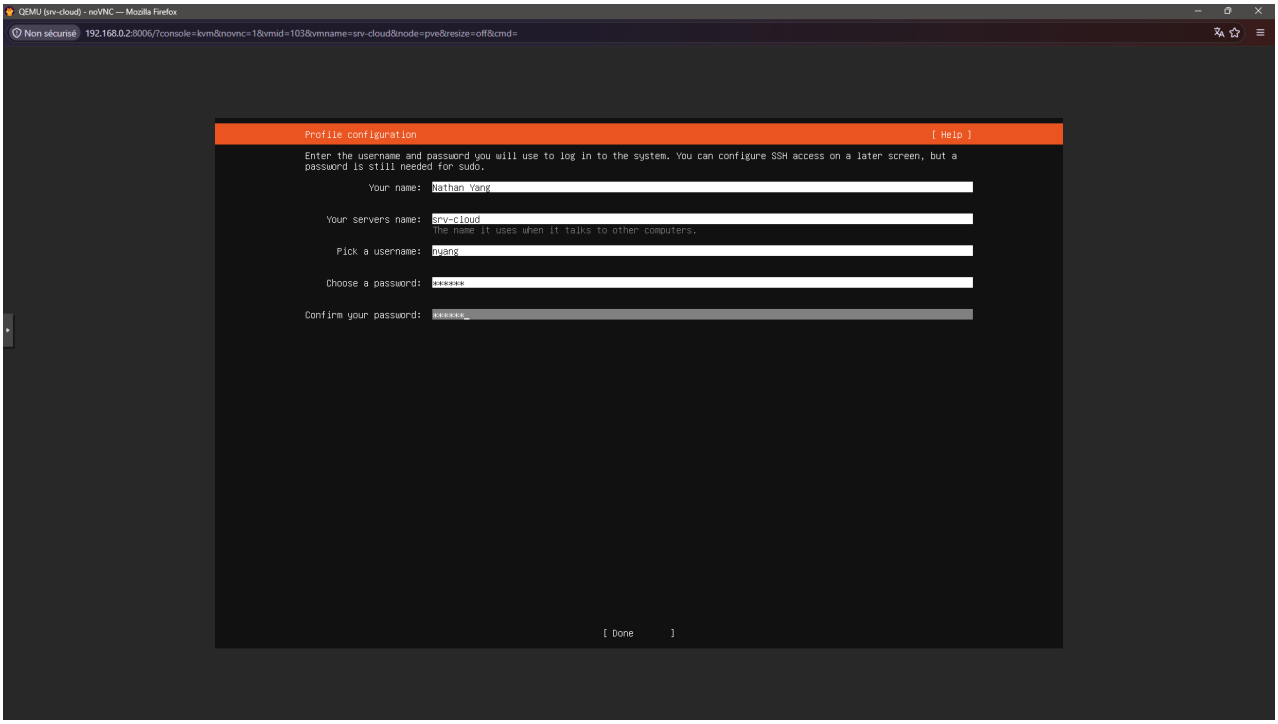

QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
Network configuration
[Help]
Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient access for updates.
NAME TYPE NOTES
[ens18 eth - ▸]
disabled autoconfiguration failed
bc:24:11:bf:aa:74 / Red Hat, Inc. / Virtio network device
[Create bond ▸]
Edit ens18 IPv4 configuration
IPv4 Method: [Manual ▾]
Subnet: 192.168.10.0/29
Address: 192.168.10.3
Gateway: 192.168.10.1
Name servers: 192.168.10.2
IP addresses, comma separated
Search domains: ad.kiformatou.fr
Domains, comma separated
[Save]
[Cancel]
[Continue without network]
[Back]

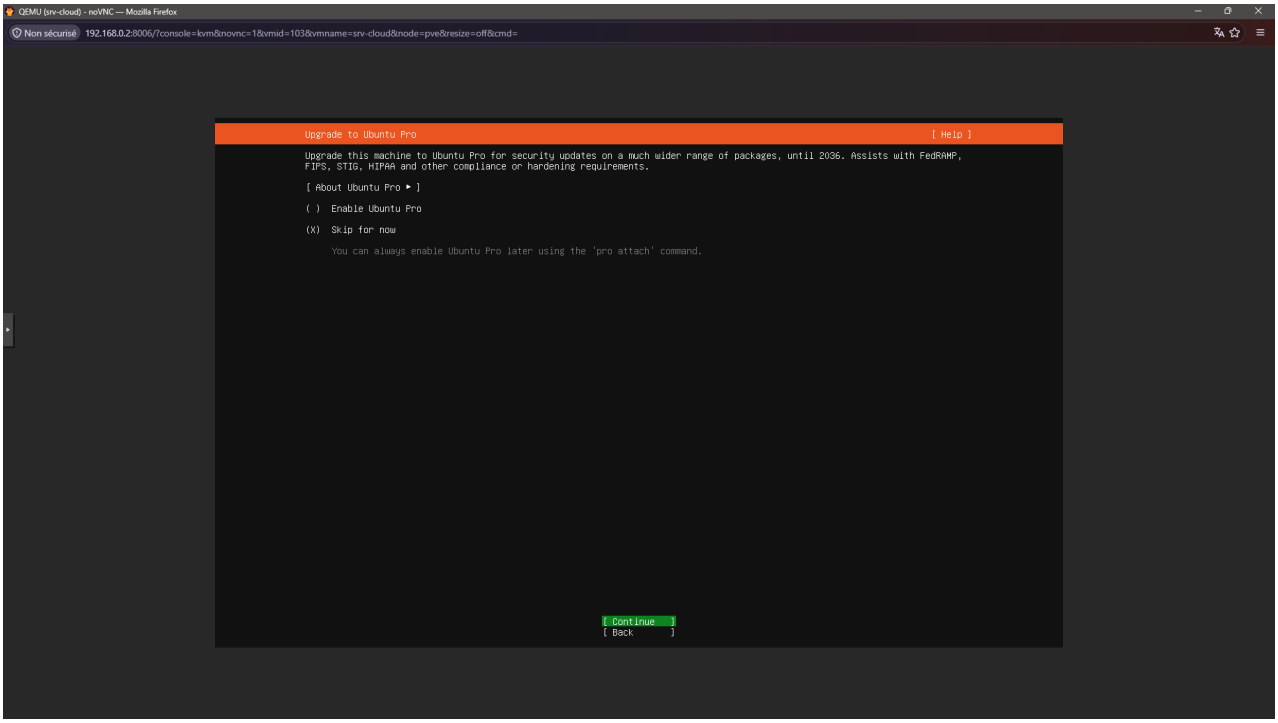

QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
Proxy configuration
[Help]
If this system requires a proxy to connect to the internet, enter its details here.
Proxy address:
If you need to use a HTTP proxy to access the outside world, enter the proxy information here. Otherwise, leave this blank.
The proxy information should be given in the standard form of "http://[[user][:pass]@]host[:port]/".
[Done]
[Back]

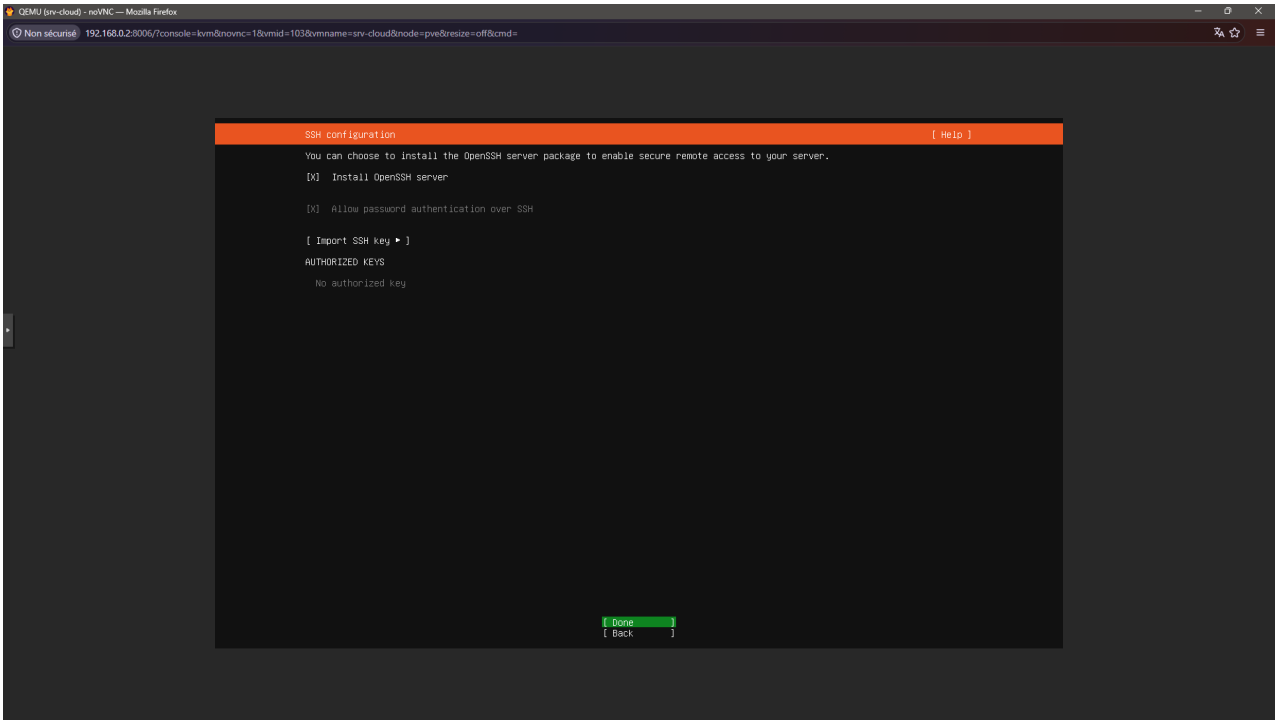

QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
Ubuntu archive mirror configuration
[Help]
If you use an alternative mirror for Ubuntu, enter its details here.
Mirror address: http://archive.ubuntu.com/ubuntu/
You may provide an archive mirror to be used instead of the default.
This mirror location passed tests.
Ign:1 file:/cdrom resolute InRelease
Get:2 file:/cdrom resolute Release [664 B]
Get:2 file:/cdrom resolute Release [664 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Hit:4 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Fetched 274 kB in 1s (356 kB/s)
Reading package lists...
[Done]
[Back]

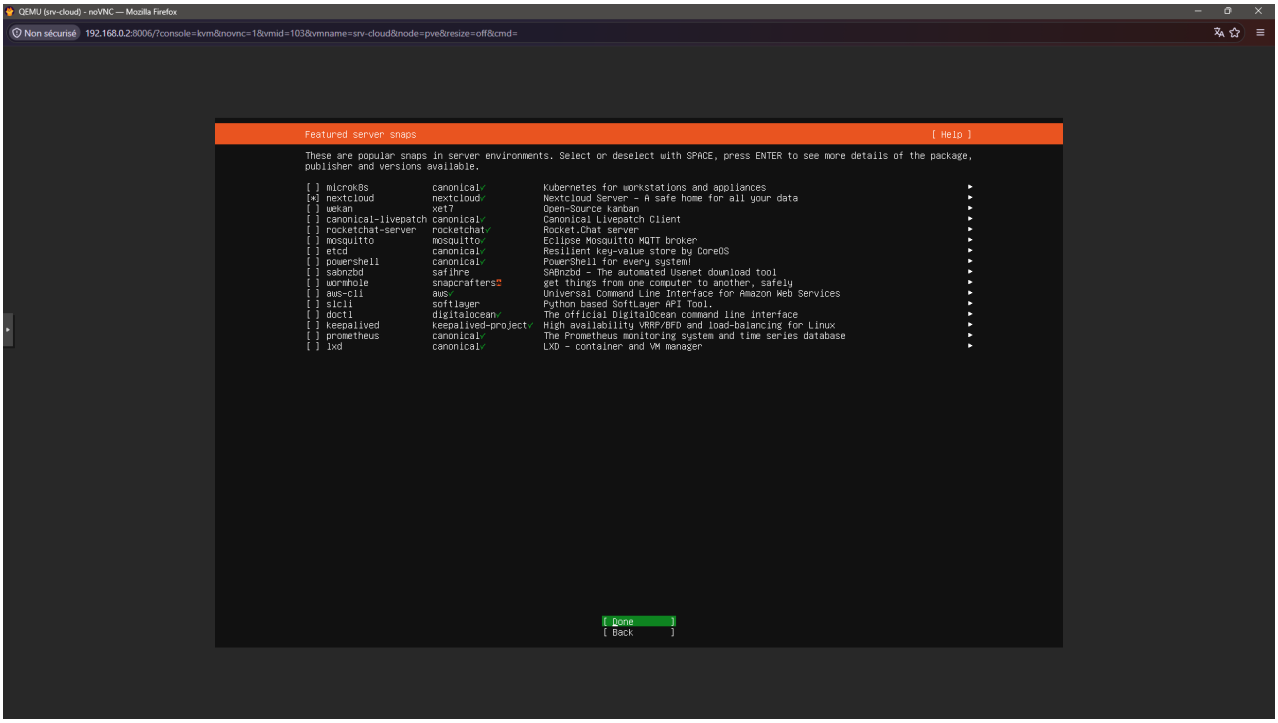

QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
Guided storage configuration
[Help]
Configure a guided storage layout, or create a custom one:
(X) Use an entire disk
[QEMU_QEMU_HARDDISK_drive-scsi0 local disk 64.000G ▾]
[X] Set up this disk as an LVM group
[ ] Encrypt the LVM group with LUKS
Passphrase:
Confirm passphrase:
[ ] Also create a recovery key
The key will be stored as ~/recovery-key.txt in the live system and will be copied to /var/log/installer/ in the target system.
( ) Custom storage layout
[Done]
[Back]

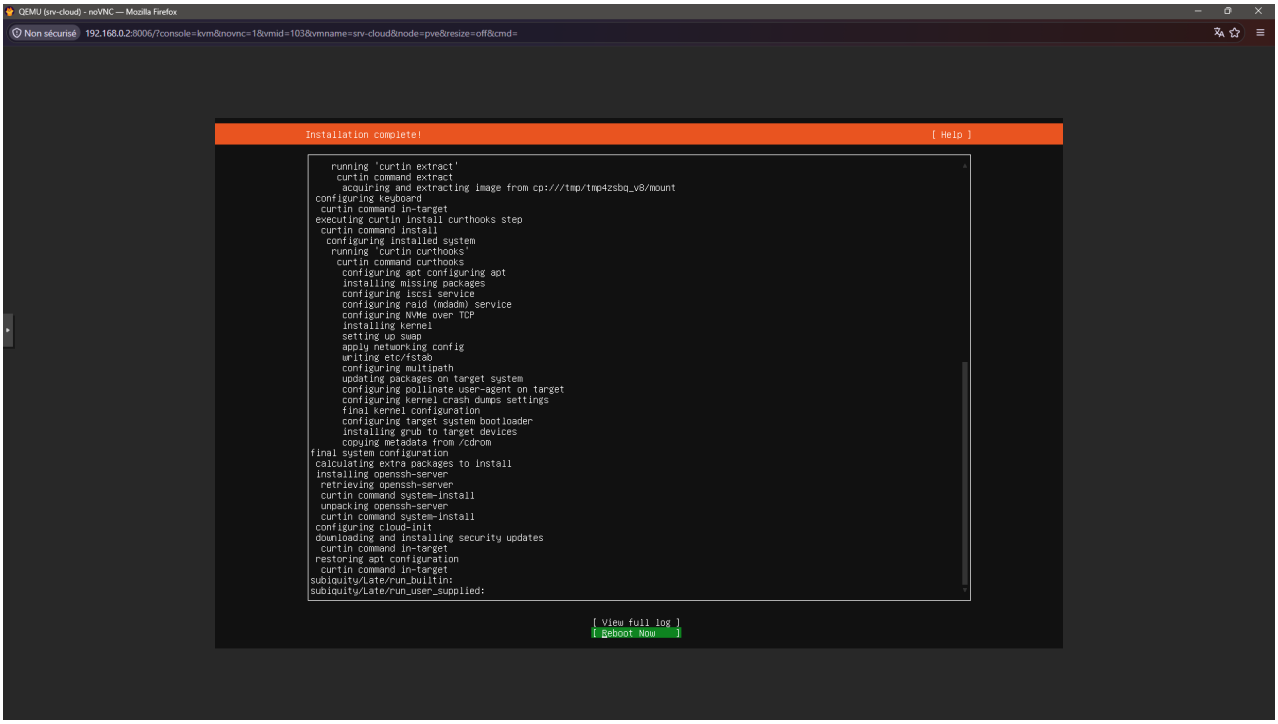

QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
Storage configuration
[Help]
FILE SYSTEM SUMMARY
MOUNT POINT SIZE TYPE DEVICE TYPE
[/ 30.996G new ext4 new LVM logical volume ▸]
[/boot 2.000G new ext4 new partition of local disk ▸]
AVAILABLE DEVICES
DEVICE TYPE SIZE
[ubuntu-vg (new) LVM volume group 61.996G ▸]
free space 31.000G ▸
[Create software RAID (md) ▸]
[Create volume group (LVM) ▸]
USED DEVICES
DEVICE TYPE SIZE
[ubuntu-vg (new) LVM volume group 61.996G ▸]
ubuntu-lv new, to be formatted as ext4, mounted at / 30.996G ▸
[QEMU_QEMU_HARDDISK_drive-scsi0 local disk 64.000G ▸]
partition 1 new, BIOS grub spacer 1.000M ▸
partition 2 new, to be formatted as ext4, mounted at /boot 2.000G ▸
partition 3 new, PV of LVM volume group ubuntu-vg 61.997G ▸
[Done]
[Reset]
[Back]


QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
Profile configuration
[Help]
Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.
Your name: Nathan Yang
Your servers name: srv-cloud
The name it uses when it talks to other computers.
Pick a username: nyang
Choose a password: ******
Confirm your password: ******_
[Done]

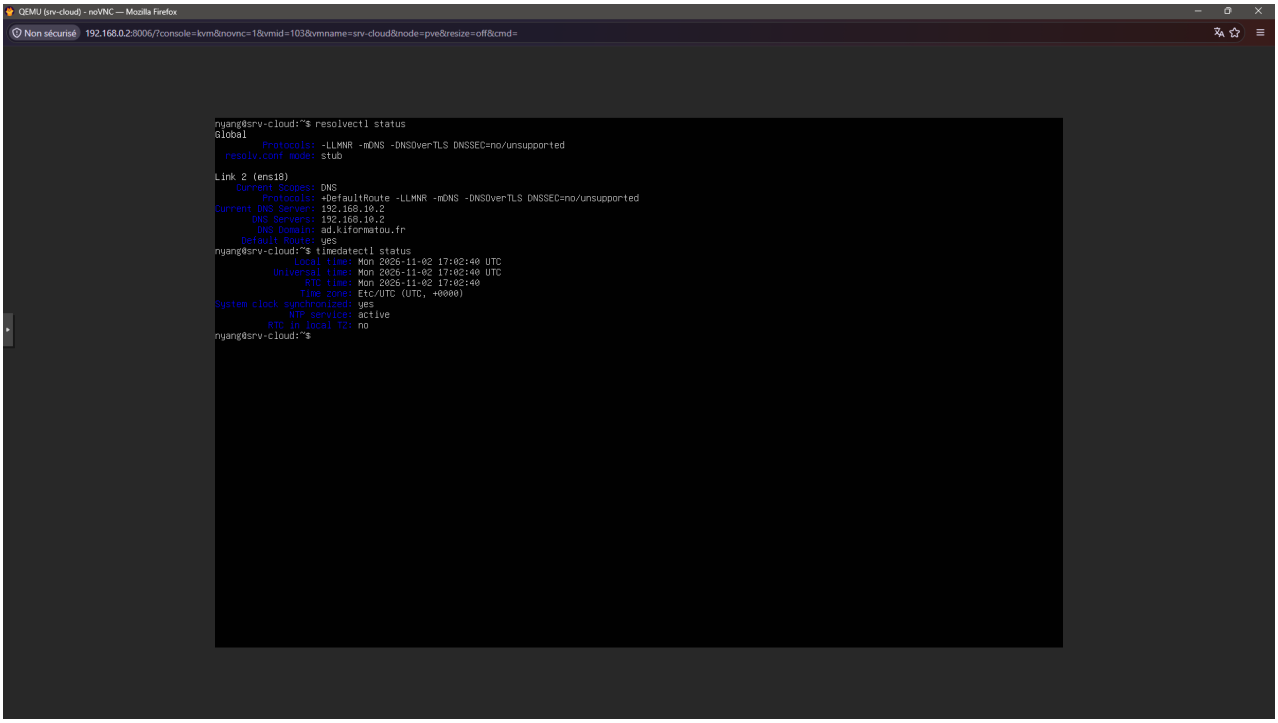

QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
Upgrade to Ubuntu Pro
[Help]
Upgrade this machine to Ubuntu Pro for security updates on a much wider range of packages, until 2036. Assists with FedRAMP, FIPS, STIG, HIPAA and other compliance or hardening requirements.
[About Ubuntu Pro ▸]
( ) Enable Ubuntu Pro
(X) Skip for now
You can always enable Ubuntu Pro later using the 'pro attach' command.
[Continue]
[Back]

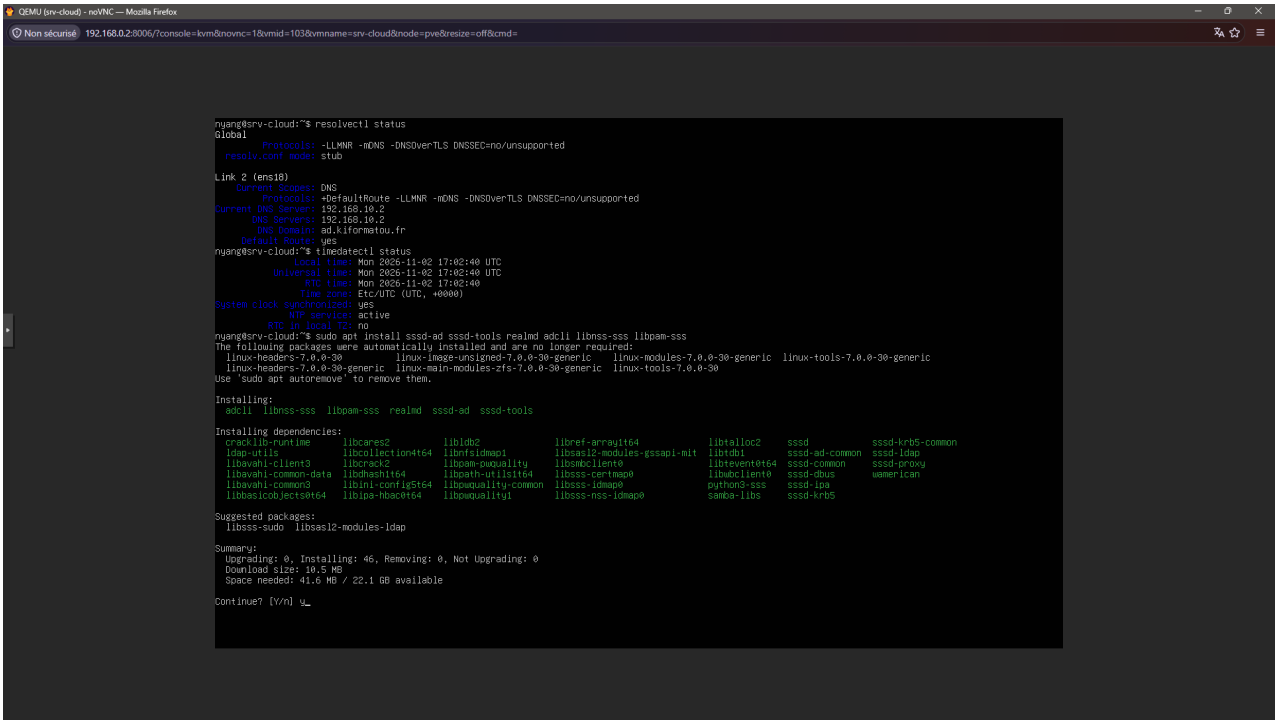

QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé
192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
SSH configuration
[Help]
You can choose to install the OpenSSH server package to enable secure remote access to your server.
[X] Install OpenSSH server
[X] Allow password authentication over SSH
[Import SSH key ▸]
AUTHORIZED KEYS
No authorized key
[Done]
[Back]

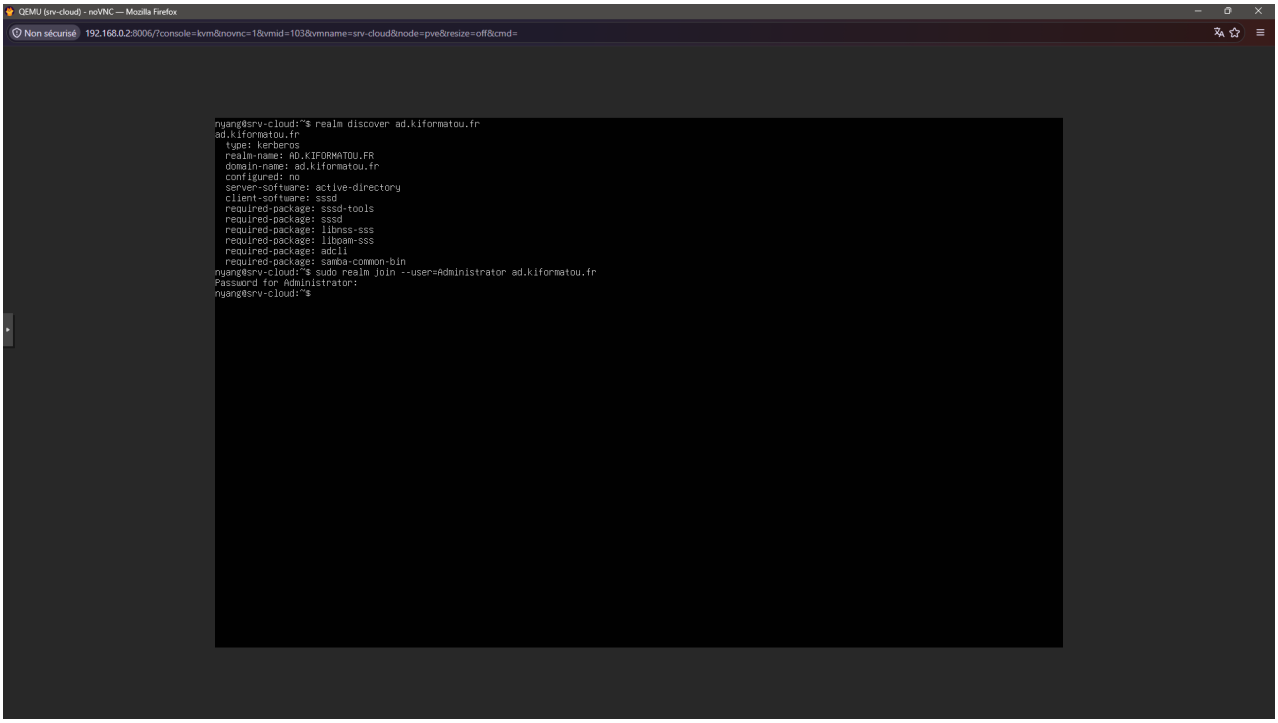

QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
Featured server snaps
[Help]
These are popular snaps in server environments. Select or deselect with SPACE, press ENTER to see more details of the package, publisher and versions available.
[ ] microk8s canonical✓ Kubernetes for workstations and appliances
[*] nextcloud nextcloud✓ Nextcloud Server - A safe home for all your data
[ ] wekan xet7 Open-Source kanban
[ ] canonical-livepatch canonical✓ Canonical Livepatch Client
[ ] rocketchat-server rocketchat✓ Rocket.Chat server
[ ] mosquitto mosquitto✓ Eclipse Mosquitto MQTT broker
[ ] etcd canonical✓ Resilient key-value store by CoreOS
[ ] powershell canonical✓ PowerShell for every system!
[ ] sabnzbd safihre SABnzbd - The automated Usenet download tool
[ ] wormhole snapcrafters get things from one computer to another, safely
[ ] aws-cli aws✓ Universal Command Line Interface for Amazon Web Services
[ ] slcli softlayer Python based SoftLayer API Tool.
[ ] doctl digitalocean✓ The official DigitalOcean command line interface
[ ] keepalived keepalived-project✓ High availability VRRP/BFD and load-balancing for Linux
[ ] prometheus canonical✓ The Prometheus monitoring system and time series database
[ ] lxd canonical✓ LXD - container and VM manager
[Done]
[Back]

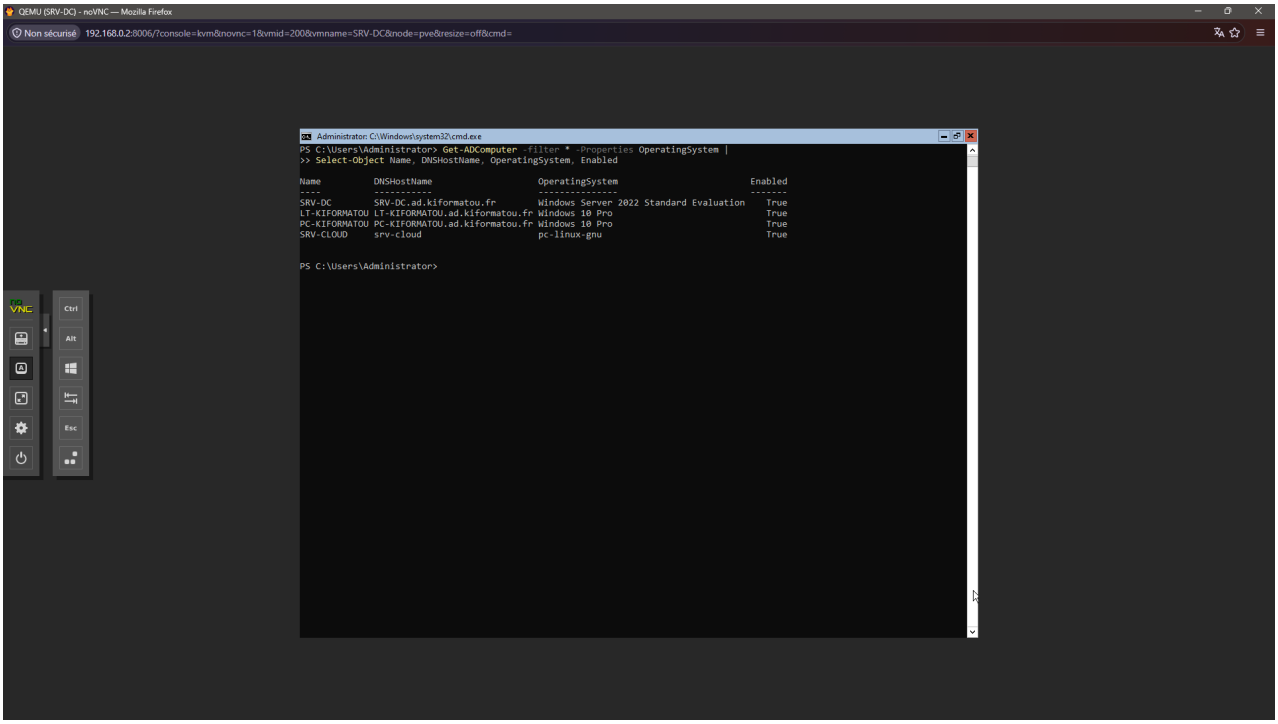

QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
Installation complete!
[Help]
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmp4zsbq_v8/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:
[View full log]
[Reboot Now]

2. Joindre le serveur au DC

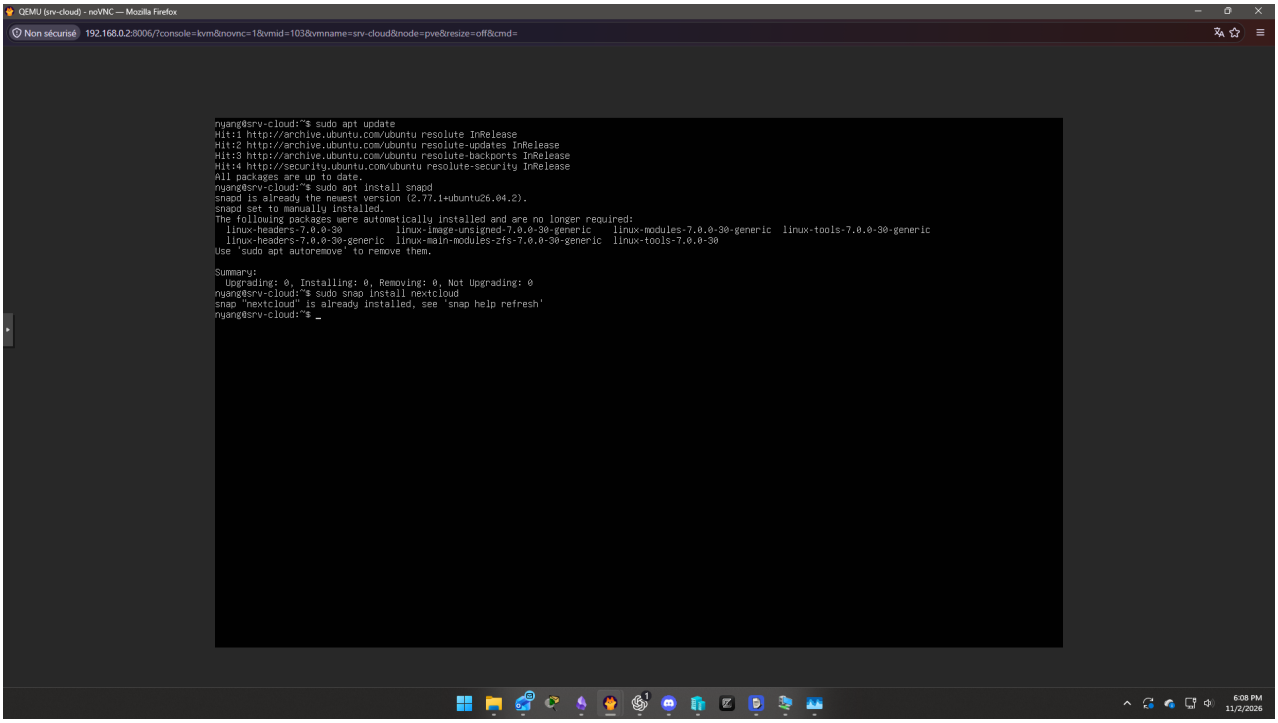


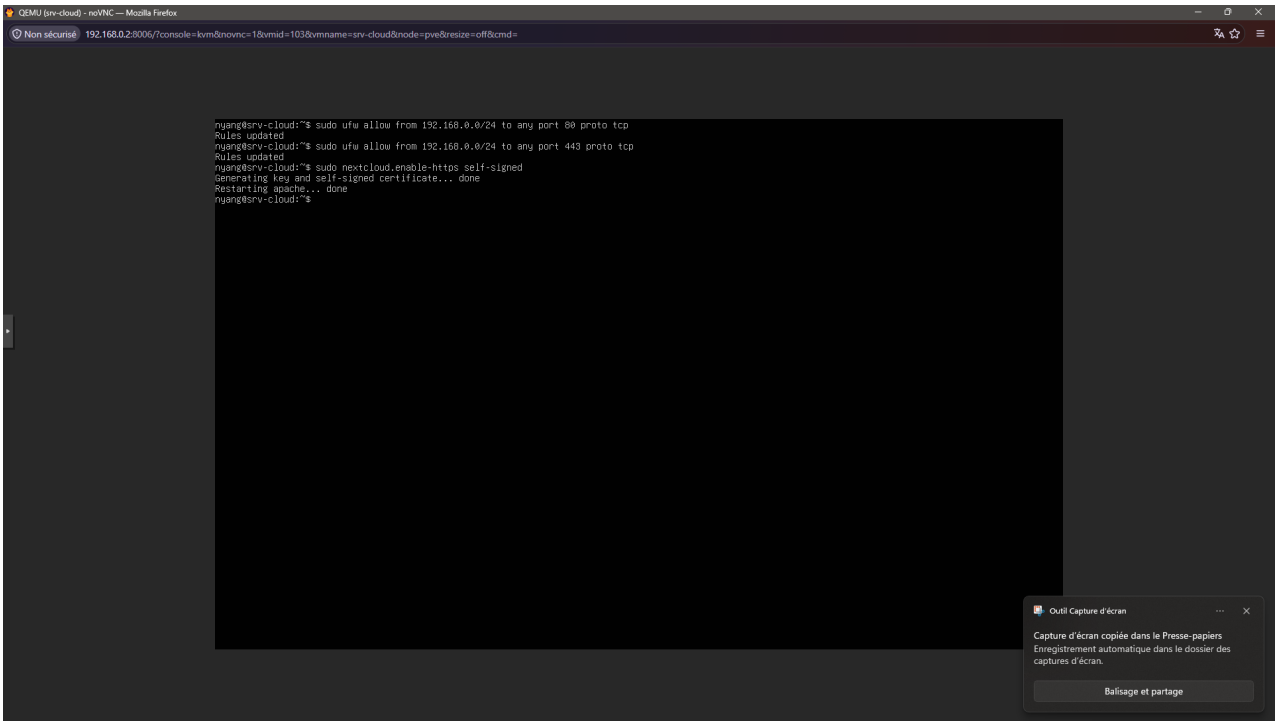



QEMU (srv-cloud) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=103&vmname=srv-cloud&node=pve&resize=off&cmd=
nyang@srv-cloud:~$ realm discover ad.kiformatou.fr
ad.kiformatou.fr
type: kerberos
realm-name: AD.KIFORMATOU.FR
domain-name: ad.kiformatou.fr
configured: no
server-software: active-directory
client-software: sssd
required-package: sssd-tools
required-package: sssd
required-package: libnss-sss
required-package: libpam-sss
required-package: adcli
required-package: samba-common-bin
nyang@srv-cloud:~$ sudo realm join --user=Administrator ad.kiformatou.fr
Password for Administrator:
nyang@srv-cloud:~$


QEMU (SRV-DC) - noVNC — Mozilla Firefox
Non sécurisé 192.168.0.2:8006/?console=kvm&novnc=1&vmid=200&vmname=SRV-DC&node=pve&resize=off&cmd=
Administrator: C:\Windows\system32\cmd.exe
PS C:\Users\Administrator> Get-ADComputer -filter * -Properties OperatingSystem |
>> Select-Object Name, DNSHostName, OperatingSystem, Enabled
Name DNSHostName OperatingSystem Enabled
SRV-DC SRV-DC.ad.kiformatou.fr Windows Server 2022 Standard Evaluation True
LT-KIFORMATOU LT-KIFORMATOU.ad.kiformatou.fr Windows 10 Pro True
PC-KIFORMATOU PC-KIFORMATOU.ad.kiformatou.fr Windows 10 Pro True
SRV-CLOUD srv-cloud pc-linux-gnu True
PS C:\Users\Administrator>
Ctrl
Alt
Esc

3. Installer Nextcloud





Création du serveur de sauvegarde (WIP)

6. Support et références

6.1. Diagnostic et support

Journaux [Emplacement ; accès ; rétention ; requête utile.]

Symptôme	Vérification	Action ou escalade
[Accès impossible]	[Réseau ; identité ; état du service]	[Action documentée ; contact]
[Erreur ou lenteur]	[Journaux ; ressources ; dépendances]	[Action documentée ; contact]

Support [Équipe ; canal ; horaires ; informations à fournir au ticket.]

6.2. Glossaire

- **VM** : Machine Virtuelle (Virtual Machine)
-

6.3. Références

Documentations

- Microsoft. Documentation de Hyper-V. 2026.
- Microsoft. Get-VMSwitch. 2026.
- Deciso. Virtual & Cloud-Based Installation. 2026.
- Microsoft. Configuration requise pour Windows 10. 2026.
- Microsoft. Configuration matétrielle requise pour Windows Server. 2026.
- https://learn.microsoft.com/fr-fr/windows-server/identity/ad-ds/deploy/install-active-directory-domain-services—level-100-

Ressources externes

- Installation d'OPNsense